

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/368690982>

INTERNET OF THINGS (IOT): SYSTEMS AND APPLICATIONS

Book · February 2023

CITATIONS

0

READS

527

1 author:



[Ashok John](#)

Eritrea Institute of Technology

30 PUBLICATIONS 25 CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:



Dynamic time series method by Artificial Neural Network [View project](#)

ABOUT THE AUTHORS



Dr. Monika Bhatnagar, B.E., M.Tech., Ph.D. from renowned University of India. Presently working as an Associate Professor in Electronics and Communication Engineering Department of Galgotias College of Engineering & Technology, Greater Noida, Uttar Pradesh. Dr. Bhatnagar having 21 yrs of academic experience from different Engineering College of India, she is published and grant national and international Patents in the field of Electronics and published so many Research paper in SCI journals and other national & international Scopus journals and conferences, Professor Bhatnagar awarded Star faculty award from renowned college of Engineering of U.P.



Dr. N. Rajendran is currently working as an Assistant Professor (Senior Grade) in the Department of Information Technology at B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, Tamil Nadu, India. He received the B.E. degree in Computer Science and Engineering from Bharathiar University, Coimbatore, India, in 2002, the M.E. degree in Software Engineering from Anna University, Chennai, India, in 2007 and Ph.D. degree in Information Technology from B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, India, in 2021. He has 17 years of teaching experience and 1 year of industry experience. His current research interests include Internet of Things, Cyber Security, Mobile Ad hoc Networks, and Big Data Analytics. He has 2 Patents granted, 5 Patents published and more than 40 Journal and Conference papers. He is a reviewer and co-editor for several reputed Journals and Conferences.



Dr. R. Priyadarshini is working as an Assistant Professor Senior Grade in the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai. She has 15+ years' experience teaching and research and 1+ years in industry. She has a Post-Doctoral fellowship in Faculty of Computer Science Engineering in Lincoln University Malaysia in the area IoT with ML and Healthcare. She completed her Ph.D. in Information Technology from UGC recognized B.S. Abdur Rahman Crescent Institute of Science and Technology and Completed M.Tech Information Technology in College of Engineering, Anna University, Chennai. She has a national and international patent grant to her credit in IoT Cloud and Data Analytics. She also has six patents published in IoT Cloud, Data Analytics, Cloud Security and Network Security. She has published 50+ papers in reputed international conferences and journals. She has 5 book chapters and 1 Book in amazon to her credit. More recently, she has contributed to interdisciplinary research in block chain, medical health and IoT. Being a passionate teacher & topper in all her academic records and continuous learner, she has 35+ certifications in her research areas to her credit from IBM, NPTEL, SWAYAM, Spoken Tutorials IIT Mumbai, Udemy etc.



Dr. Ashok J. received the B.E. Degree in Electrical and electronics engineering from University of Madras, in 2003, the M.E. Degree in Applied electronics engineering from Anna University, Chennai, in 2005, and the Ph.D. Degree in Applied Electronics engineering from Preet University, India, in 2016. He has teaching experience of more than 15 years in the areas of Microwave Engineering, Remote sensing and GIS, Digital Telephony, Networks, Micro-processor, Neural Networks, Electrical and Digital Circuits, Linear integrated circuits, Microprocessor and microcontroller, Medical Electronics, embedded system. Prof. Ashok J. currently serves as an Associate Professor of Department Electronics and communication Engineering, V.S.B. Engineering college karur, India. He was Associate professor at College of Engineering & Technology MAI NEFH, STATE OF ERITREA in year of 2019-2021. In the Department of Electronics and also Professor at Annasaheb Dange College of Engineering & Technology, sangli, Maharashtra, India. In the Department of Electronics and Telecommunication in the year of 2017-2019.



Dr. Dinesh Mavaluru is currently working as an Educator in the College of Computing and Informatics at Saudi Electronic University, Saudi Arabia. He received a doctorate from B.S. Abdur Rahman University, Chennai, India in 2014. His research interests span both data science and network science. Much of his research work has been on improving the understanding, design, and performance of parallel and networked computer systems, mainly through the application of data mining, statistics, and performance evaluation. Committed to helping researchers and students to identify and develop their own passions while becoming successful and confident scholars and learners.



NTL TECHNOLOGY
ntltechnology2020@gmail.com



ISBN 978-81-953929-2-6
9 788195 392926



INTERNET OF THINGS (IOT) SYSTEMS AND APPLICATIONS

ISBN: 978-81-953929-2-6

INTERNET OF THINGS (IOT) SYSTEMS AND APPLICATIONS

Dr. Monika Bhatnagar
Dr. N. Rajendran | Dr. R. Priyadarshini
Dr. Ashok J | Dr. Dinesh Mavaluru

INTERNET OF THINGS (IOT): SYSTEMS AND APPLICATIONS

Authors

Dr. Monika Bhatnagar

*Associate Professor,
Electronics and Communication Engineering Department,
Galgotias College of Engineering & Technology, 1,
Knowledge Park Phase -II, Greater Noida, Uttar Pradesh*

Dr. N. Rajendran

*Assistant Professor (Senior Grade)
Department of Information Technology at B.S. Abdur Rahman Crescent
Institute of Science and Technology, Chennai*

Dr. R. Priyadarshini

*Assistant Professor Senior Grade II,
School of Computer Science and Engineering,
Vellore Institute of Science and Technology, Chennai*

Dr. Ashok J

*Associate Professor,
Department of Electronic and Communication Engineering,
V. S. B. Engineering College, Karur*

Dr. Dinesh Mavaluru

*College of Computing and Informatics
Saudi Electronic University
Riyadh. Saudi Arabia*



Title: INTERNET OF THINGS (IOT):
SYSTEMS AND APPLICATIONS

Author's Name: Dr. Monika Bhatnagar
Dr. N. Rajendran
Dr. R. Priyadarshini
Dr. Ashok J
Dr. Dinesh Mavaluru

Published by: NTL Technology

Edition Details (I,II,III): I

Month & Year: February, 2023

ISBN: “978-81-953929-2-6”

Copyright @ NTL Technology

Pages: 198

Price: ₹490/-

CONTENTS

CHAPTER	TITLE	PAGE NO.
1	INTRODUCTION TO INTERNET OF THINGS 1-45	
1.1	INTRODUCTION	
1.2	TECHNOLOGICAL TRENDS AND FUTURE OPPORTUNITIES	
1.3	IOT AND BUSINESS SCOPE EVOLUTION	
1.4	BUSINESS PERSPECTIVES	
1.5	EMBEDDED SYSTEMS RELATIONSHIP	
1.6	CHALLENGES OF INTERNET OF THINGS (IOT)	
1.7	CHARACTERISTICS OF IOT	
1.8	IOT LIFE CYCLE	
1.9	PHYSICAL DESIGN OF IOT	
1.10	IOT CONCEPTUAL ARCHITECTURE	
1.11	IOT COMMUNICATION PROTOCOLS	
1.12	LEVELS OF THE INTERNET OF THINGS	
1.13	ARCHITECTURE OF THE INTERNET OF THINGS	
1.14	ADVANTAGES OF IOT	
1.15	IOT DISADVANTAGES	
2	SMART OBJECTS	
2.1	SMARTS OBJECTS IN IOT	
2.2	THE "THINGS" IN THE INTERNET OF THINGS (IOT)	
2.3	IOT FOR LOGISTICS: PACKAGES AND CONTAINERS AS "THINGS" CONNECTED TO THE INTERNET	
2.4	SENSORS AND ACTUATORS IN IOT ENABLING INDUSTRIAL AUTOMATION	
2.5	CLASSIFICATION OF SMART OBJECTS	
2.6	WIRELESS SENSOR NETWORK IN IOT	
2.7	COMMUNICATIONS CRITERIA	

3 IOT LAYERS

- 3.1 NETWORKING STANDARDS AND TECHNOLOGIES IN IOT
- 3.2 IOT NETWORKING PROTOCOLS
- 3.3 INTRODUCTION OF 5G NETWORKS IN IoT
- 3.4 IOT NETWORKING CONSIDERATIONS AND CHALLENGES
- 3.5 THE BUSINESS CASE FOR THE INTERNET OF THINGS
- 3.6 NETWORK OPTIMIZATION FOR IOT DEVICES
- 3.7 TRANSPORT LAYER PROTOCOLS
- 3.8 NETWORK LAYER PROTOCOLS
- 3.9 IOT COMMUNICATION CHALLENGES
- 3.10 PROVIDING SOLID IOT QUALITY OF EXPERIENCE
- 3.11 APPLICATION PROTOCOLS FOR IOT

4 DATA ANALYTICS AND SUPPORTING SERVICES

- 4.1 INTRODUCTION
- 4.2 IOT DATA ANALYTICS CHALLENGES
- 4.3 IOT NETWORK ANALYTICS APPLICATIONS
- 4.4 INTELLIGENCE IN BUSINESS
- 4.5 GATHERING SOFTWARE FOR EVENTS
- 4.6 COMPUTING FOR IOT/M2M
APPLICATIONS/SERVICES USING A CLOUD
PLATFORM
- 4.7 CLOUD COMPUTING TRENDS
- 4.8 CLOUD PLATFORM SERVICES
- 4.9 CLOUD DEPLOYMENT MODELS

5 IOT PHYSICAL DEVICES AND ENDPOINTS

- 5.1 INTRODUCTION
- 5.2 IOT BUILDING BLOCKS
- 5.3 IOT WORKS
- 5.4 IOT ARCHITECTURE LAYERS
- 5.5 EXEMPLARY DEVICE: RASPBERRY PI

6 IOT PHYSICAL SERVERS AND CLOUD OFFERINGS

- 6.1 INTRODUCTION TO CLOUD COMPUTING
- 6.2 CHARACTERISTICS
- 6.3 SERVICE AND DEPLOYMENT
- 6.4 CLOUD STORAGE API
- 6.5 IOT / CLOUD CONVERGENCE
- 6.6 WAMP FOR IOT
- 6.7 AMAZON EC2 - PYTHON EXAMPLE
- 6.8 PYTHON FOR MAPREDUCE
- 6.9 PYTHON WEB APPLICATION FRAMEWORK -
DJANGO

7 INDUSTRIAL INTERNET OF THINGS (IIOT)

- 7.1 INTRODUCTION
- 7.2 EVOLUTION OF IIOT
- 7.3 ADVANTAGES OF INDUSTRIAL IOT
- 7.4 DRIVERS OF IIOT
- 7.5 RISKS ASSOCIATED WITH IIOT SYSTEMS
- 7.6 BUSINESSES AND INDUSTRIES APPROACH IIOT
SECURITY
- 7.7 APPLICATIONS OF IIOT
- 7.8 WORK FLOW OF IIOT
- 7.9 SECURITY CONSIDERATIONS AND CHALLENGES
WHEN ADOPTING IIOT
- 7.10 COMPANIES CAN IMPLEMENT IIOT
INTELLIGENTLY
- 7.11 INDUSTRIAL INTERNET OF THINGS USE CASES

8 IOT APPLICATIONS & CASE STUDIES

- 8.1 IOT APPLICATIONS
- 8.2 IOT CONSUMER APPLICATIONS
- 8.3 IOT APPLICATIONS IN AGRICULTURE
- 8.4 IOT APPLICATIONS IN MEDIA, MARKETING, AND
ADVERTISING
- 8.5 APPLICATION OF IOT IN EDUCATION

- 8.6 GOVERNMENT APPLICATIONS
 - 8.7 IOT ENVIRONMENTAL MONITORING
 - 8.8 TRANSPORTATION AND THE INTERNET OF THINGS (IOT)
 - 8.9 IOT APPLICATIONS IN HEALTHCARE
 - 8.10 IOT LAW ENFORCEMENT APPLICATIONS
- ACRONYMS REFERENCES

CHAPTER 1

INTRODUCTION TO INTERNET OF THINGS

1.1 Introduction

It is a network of physical objects or things that send, receive, or communicate via the internet or other communication technologies. The intelligent link between the physical and digital worlds, often known as the Internet of Things (IoT), is expected to fundamentally alter every aspect of how we work and do business. By integrating apps with their integrated systems, businesses have the potential to significantly transform their industries. Currently, 90% of the data produced by tablets, smartphones, and linked devices is not utilised. Think about how you may be able to change it. No other technological platform has ever combined this level of complexity, global reach, and uniqueness, it is reasonable to claim. The Internet of Things (IoT) has created opportunities for directly integrating the real world with computer-based systems via sensors and the internet since it allows items to be handled remotely. Automation in practically every industry as well as the development of cutting-edge applications will result from the integration of these many embedded devices. As a result, accuracy, efficiency, and economic value have increased with reduced human influence. Among the technologies explored are smart grids, smart homes, intelligent transportation, and smart cities. Some of the main benefits of IoT include the ones listed below:

- ❖ Greater Customer Engagement - By automating the process, the Internet of Things (IoT) enhances the customer experience. For instance, the sensors on an automobile will detect any issue right away. Both the driver and the manufacturer will be communicated. Before the driver arrives the gas station, his or her time is not over.
- ❖ The manufacturer will make that the faulty component is accessible at the servicing facility.
- ❖ Technical Optimization - The Internet of Things has aided technological advancement. In order to improve the vehicle's design and efficiency, the manufacturer may make data from numerous automobile sensors and analyse it.

Reduced Waste – Our current understanding is limited, but real-time data from the Internet of Things (IoT) enables improved information management and decision-making. For instance, if a manufacturer finds a flaw in several engines, he may track the factory where the engines are produced and correct the issue with the production belt.

It is a network of physical objects or things that send, receive, or communicate via the internet or other communication technologies.

The intelligent link between the physical and digital worlds, often known as the Internet of Things (IoT), is expected to fundamentally alter every aspect of how we work and do business. By integrating apps with their integrated systems, businesses have the potential to significantly transform their industries. Currently, 90% of the data produced by tablets, smartphones, and linked devices is not utilised. Think about how you may be able to change it. No other technological platform has ever combined this level of complexity, global reach, and uniqueness, it is reasonable to claim. The Internet of Things (IoT) has created opportunities for directly integrating the real world with computer-based systems via sensors and the internet since it allows items to be handled remotely. Automation in practically every industry as well as the development of cutting-edge applications will result from the integration of these many embedded devices. As a result, accuracy, efficiency, and economic value have increased with reduced human influence. Among the technologies explored are smart grids, smart homes, intelligent transportation, and smart cities. Some of the main benefits of IoT include the ones listed below:

- ❖ Greater Customer Engagement - By automating the process, the Internet of Things (IoT) enhances the customer experience. For instance, the sensors on an automobile will detect any issue right away. Both the driver and the manufacturer will be communicated. Before the driver arrives the gas station, his or her time is not over.
- ❖ The manufacturer will make that the faulty component is accessible at the servicing facility.
- ❖ Technical Optimization - The Internet of Things has aided technological advancement. In order to improve the vehicle's design and efficiency, the manufacturer may make data from numerous automobile sensors and analyse it.

Reduced Waste – Our current understanding is limited, but real-time data from the Internet of Things (IoT) enables improved information management and decision-making. For instance, if a manufacturer finds a flaw in several engines, he may track the factory where the engines are produced and correct the issue with the production belt.

1.2 Technological Trends and Future Opportunities

Many companies see the Internet of Things (IoT) as a significant opportunity that has the potential to improve consumer interactions and

propel corporate development by increasing quality, productivity, and reliability on the one hand while reducing costs, risk, and theft on the other. IoT security breaches

Forrester asserts that the recent DDoS attack in the United States, which affected 1600 websites, was only the tip of the iceberg in terms of the threat posed by connected devices to the global world. That attack confirmed the concern about IoT device vulnerability with a significant distributed denial of service attack that crippled the servers of services including Twitter, NetFlix, NYTimes, and PayPal across the United States. It is the result of a significant internet assault that used malicious software and millions of IP addresses. All indications point to the virus gaining control of a sizable number of Internet of Things (IoT) devices, which power widely used technologies like closed-circuit cameras and smart home devices, and utilising them against the servers. a few IoT and mobile components

The Internet of Things is creating new possibilities and giving firms a competitive advantage in both developed and developing markets. It has an aspect on all facets of your data collecting, including the when, where, how, and why. In addition to changing the internet, the technologies that gave birth to the Internet of Things are also changing the things that are linked to it. More mobile moments (the times when a person pulls out a mobile device to get what he or she wants, quickly and in context) will display on the linked gadget, from home appliances to cars to smartwatches and virtual assistants. Each of these linked devices has the potential to provide a rich stream of data that product owners of goods and services may use to communicate with their clients.

Artificial intelligence (AI) and the Internet of Things (IoT) (AI)

In an IoT environment, AI can assist businesses narrow down the millions of data points to what matters most. The fundamental premise is the same as it is for retail applications: examine and analyse data to make trends or resemblances that can be utilised to inform and improve decisions.

Connectivity and the Internet of Things (IoT)

Various parts of the IoT may be connected to the sensors via Wi-Fi, Bluetooth, Low Power Wi-Fi, Wi-Max, traditional Ethernet, Long Term Evolution (LTE), and the new and promising Li-Fi technology (using light as a medium of communication between the different parts of a typical network including sensors).

There are new business models with the Internet of Things (IoT). Without strong and reliable business models for IoT, we risk experiencing another bubble. This model must satisfy all requirements for consumer markets, vertical markets, and horizontal markets. The inclusion of services with the product is essential; otherwise, devices like Amazon's Alexa would just be seen as another wireless speaker. Examples of these services include voice recognition, music streaming, and Uber booking. The Internet of applications has the potential to improve almost every aspect of our daily life. The list below includes a few of the examples.

1. Prediction of natural disasters: By integrating sensors with their autonomous coordination and modelling, it will be able to anticipate when landslides and other natural disasters will occur and take appropriate time in advance.
2. Applications for industry: For instance, managing a fleet of cars might be done using the Internet of applications in the industry. The processing of data to identify and choose the ones that need repair and monitoring of their environmental performance are made easier with the help of the Internet of Things.
3. Monitoring water scarcity: The Internet of Things may assist identify water shortages in different locations. When the networks of sensors are combined with the appropriate simulation exercises, they can be used to monitor long-term water interventions like catchment area management as well as warn stream users when an upstream event, like the unintentional release of sewage into the stream, may have potentially dangerous effects.
4. Smart home design: The Internet of Things (IoT) may assist with smart home design in a number of ways, including energy consumption management, appliance interaction, crisis detection, home safety and simple item location, home security, and more.
5. Medical applications: The Internet of applications (IoT) can be used in the medical field to save lives and improve quality of life. For instance, monitoring health parameters, monitoring activities, assisting with independent living, monitoring medication intake, and other similar things are some examples of how the IoT can be used in the medical field.
6. Agriculture application: A network of sensors may identify data, analyse it, and notify the farmer through communication infrastructure, such as a text message to a mobile phone, about data of land that need particular attention. Examples of this include smart seed packaging,

fertiliser, and pest control technologies that adjust to specific local conditions and provide recommendations. Intelligent farming systems will assist agronomists better understand plant development models and efficient farming practises by having information of the land conditions and climate variability. This will be advantageous.

A significant boost in agricultural output may be achieved by removing ineffective farming methods.

7. Intelligent transportation system architecture: The intelligent transportation system will provide effective transportation control and management by using contemporary sensor, information, and network technologies. Intelligent transportation may include features like
8. continuous electronic highway tolls, mobile emergency command and scheduling, transportation law enforcement, monitoring for vehicle rules violations, reducing environmental pollution, anti-theft system, avoiding traffic jams, reporting traffic incidents, smart beaconing, reducing arrival delays, and more.
9. Smart city design: The Internet of Things may assist with smart city design by, among other things, monitoring air quality, locating escape routes, providing efficient city lighting, and watering plants.
Smart metering and monitoring: Accurate automated metre reading and client invoice issuance will be made possible by the Internet of Things (IoT) design for smart metering and monitoring. The development of systems for gas, water, and environmental metering and monitoring, among other things, may be facilitated by the Internet of Things.
10. Smart Security: The Internet of Things (IoT) may be utilised in the security and surveillance sector for a variety of purposes, including space surveillance, asset and person monitoring, infrastructure and equipment maintenance, alerting, and more

1.3 IoT and Business Scope Evolution:

In 1999, British technology pioneer Kevin Ashton created the phrase "Internet of Things" (IoT) to describe a system in which real things may be linked to the Internet using sensors.

Ashton used the expression to illustrate the value of connecting RFID tags¹³ used in corporate supply chains to the Internet in order to count and track things without requiring human contact.

By the late 1970s, systems for remotely monitoring electricity grid metres through telephone lines, for instance, were already in use in the business world.

Due to developments in wireless technology, machine-to-machine (M2M) corporate and industrial solutions for equipment monitoring and operation became popular in the 1990s. Numerous early M2M systems, however, were dependent on closed, purpose-built networks and proprietary or sector-specific software.

From a broad technology, the cheap and simple interconnection of more and smaller devices is being made possible by the convergence of numerous technological and business trends²⁰:

Internet of Things -

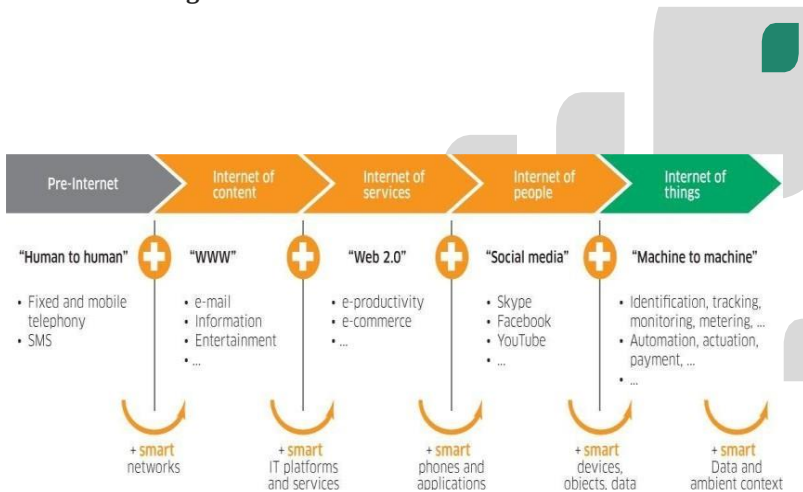


FIG 1.1 Iot Evolution Model.

- ❖ Ubiquitous Connectivity—Low-cost, high-speed, widespread network connectivity, particularly via licensed and unlicensed wireless services and technology, allows nearly everything to be connected. —connectable“.
- ❖ Widespread acceptance of IP-based networking—IP has established itself as the de facto worldwide networking standard, providing a well-defined and extensively deployed framework.
- ❖ Platform of software and tools that can be simply and cheaply incorporated into a wide variety of devices.
- ❖ Miniaturization— Manufacturing advancements have incorporated it possible to combine cutting-edge computer and communications technology into extremely small things. When combined with more computational power,

- ❖ This has driven the development of small, low-cost sensor devices, which are at the heart of many IoT applications.
- ❖ Data Analytics Advances— New algorithms and fast advances in computer power, data storage, and cloud services allow the aggregation, correlation, and analysis of enormous and dynamic datasets, opening up new avenues for extracting information and knowledge.
- ❖ The Rise of Cloud Computing– Cloud computing enables small and scattered devices to interact with sophisticated back-end analytic and control capabilities by using distant, networked computing resources to process, manage, and store data.

In this light, the Internet of Things reflects the coming together of a number of decades-old computer and connectivity developments. Currently, a broad variety of industry sectors – including automotive, healthcare, manufacturing, home and consumer electronics, and many more – are looking into the potential of implementing IoT technology into their products, services, and operations.

1.4 Business Perspectives

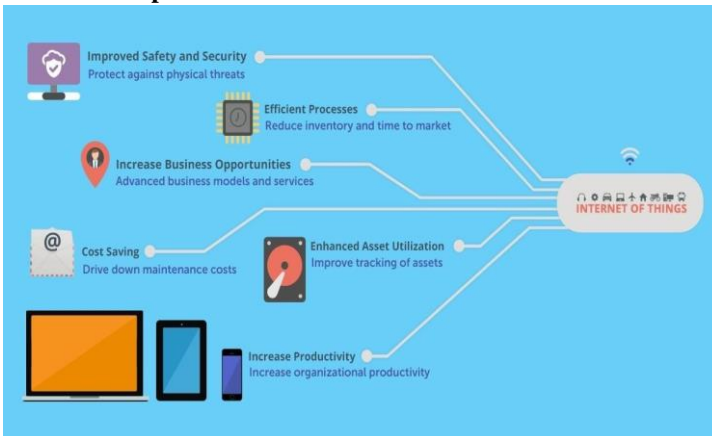


FIG 1.2 Scope of the Business.

Increase Business Possibilities Due to enhanced business models and services, the Internet of Things (IoT) opens up new business prospects and enables companies to capitalise on these chances.

Business cases are strengthened, time to market is sped up, and returns on investment are increased via IoT-driven innovations. The IoT has the potential to transform how customers and companies see the world by using its reach beyond connection.

Most Effective Asset Utilization

IoT will improve asset tracking (equipment, machinery, tools, and so forth) through sensors and connections, enabling businesses to get real-time information. Organizations may more easily identify asset flaws and do preventive maintenance to improve asset utilisation.

Efficacious Procedures

By tying a lot of devices to the internet, IoT enables organisations to become smarter with operational insights in real-time while lowering operating costs. Inventory, time to market, and maintenance downtime will all be decreased with the help of the data acquired from the logistics network, production floor, and supply chain.

Security and Safety have Increased.

IoT services are combined in conjunction with sensors, cameras, and video recording devices to monitor the workplace and prevent physical hazards. To resolve problems swiftly, many teams are coordinated utilising IoT connection.

Increased Productivity

The profitability of each company depends on how productive it is. The Internet of skills increases corporate productivity by enabling employees to get just-in-time training, increase labour productivity, and reduce skill mismatches.

Cost-Cutting

Utilization of resources, output, and process effectiveness You may perhaps all save money with 1iency. For instance, real-time diagnostics and predictive analytics save maintenance costs. The hype around emerging technology has culminated in the Internet of Things (IoT). Despite the potential benefits of IoT, businesses face significant obstacles to adoption, including issues with data and information management, a shortage of appropriate technologies, privacy and security concerns, and the skills necessary to handle IoT's growing complexity. On the other hand, a knowledgeable IoT service provider may assist you in overcoming these obstacles and increasing your return on investment.

Logistics

IoT sensors greatly improve order fulfilment and supply chain management processes to meet order demand. Managers can track their shipments and make sure they reach at their destination on time thanks to sensors on delivery containers and moving vehicles, for instance, which provide managers real-time status updates. an organised industry IoT provides automation potential for businesses that must manage and replenish their stockpiles. When data from IoT devices is connected to your enterprise resource planning (ERP) system, you can carefully monitor your inventory, evaluate purchase and consumption habits of a certain product, and automatically reorder products when IoT sensors show that supply is running low. There are fewer instances of being out of stock, and extra stock is avoided.

Fast Payment

The Internet of Things has the potential to transform how businesses conduct transactions given that the majority of payments are done electronically through point-of-sale systems or the internet. Examples of this may already be seen today, thanks to ApplePay, which enables customers to purchase products and services not only via smartphone applications but also through wearable technology.

Thanks to IoT devices, restaurants and businesses may soon be able to register or charge clients as soon as they walk through the door.

Market Research

Businesses will have a competitive edge if they can understand the data that IoT devices are collecting. Marketers, for instance, may discover a lot about how consumers use their products and which groups are most interested in them. Then, using this data, businesses can make better decisions about how to improve their products and services for consumers and direct future marketing initiatives. Businesses adopting the Internet of Things will likely face a number of challenges, but those that are successful will gain fully from this quickly developing technology.

1.5 Embedded Systems Relationship



FIG 1.3 Embedded Processing

Today's electrical components include embedded systems as a necessary component. Examples of low power consumption parts include remote controls, washing machines, microwaves, RFID tags, sensors, actuators, and thermostats used in a variety of applications, as well as networking equipment including switches, routers, modems, mobile phones, PDAs, and other devices. Embedded devices often function as a smaller component of a bigger device. For instance, embedded systems are used in home automation to automate and regulate lighting, security, audio/visual systems, detect temperature change, and monitoring, among other things. In HVAC systems, embedded systems are employed as networked thermostats. Embedded microcontrollers are present in practically all modern devices, including CT scanners, power tools, and DVD players. Compared to PCs, they are smaller and less powerful. Even though embedded systems are significantly smaller than their PC counterparts, they often feature a CPU, memory, and external interfaces. Frequently, the bulk of the electrical circuitry is housed on a single chip. A sensor detects (senses) changes in the information or the status of another device or system, and then sends or processes the data in a certain way.

- ❖ Analog sensors provide a continuous output signal or voltage that is proportionate to the quantity being measured in most cases.
- ❖ Temperature, speed, pressure, displacement, strain, and other physical variables are all analog values since they are continuous in nature.
- ❖ Digital sensors provide discrete digital output signals or voltages that indicate the quantity being measured in digital form.
- ❖ Digital sensors provide a binary output signal (ON or OFF) in the form of a logic 1 or a logic 0.

A component of a machine or system that moves or regulates the mechanism or system is known as an actuator. The mechanism through which a control system works on the environment is known as an actuator.

A control signal and a source of energy are required for an actuator. Power Savings

Until recently, one of the most frequent ways to preserve power in an embedded system was to execute as rapidly as possible before going into sleep mode. However, there are currently CPU core designs that require almost no power, but at a lower performance level. This is an appealing design for a WSN edge node.

C, C++, and Java are some of the programming languages utilized in deeply embedded systems. It's vital to note in mind that Java runs on top of an operating system. So the question isn't whether you'll use C/C++ or Java; it's whether you'll utilize both. Java is appealing for IoT devices because to the large number of Java developers globally, which means the industry has a potential of room for expansion. Java ME Embedded from Oracle is intended for small devices.

When cost isn't an issue, we may choose a single strong CPU to handle all of your device's functions. However, using two CPUs in the sensor/actuator device is a typical technical compromise. The physical-world interface is handled by one low-cost processor (8 or 16 bits), while the network interface is handled by a second 32-bit processor. This second processor is often housed in a separate module that has previously been approved by the FCC and has been certified for the protocol.

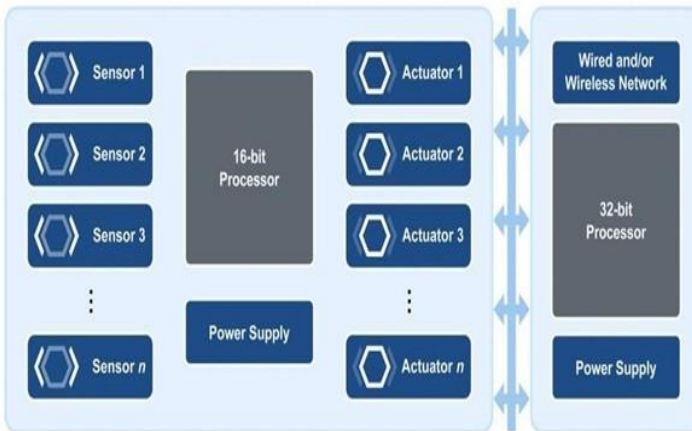


FIG.1. 4 IoT Devices With Two Processors

Designing a Design

A gateway is a device that joins two disparate networks to allow data to pass between them. This is usually a link between a private network and the Internet.

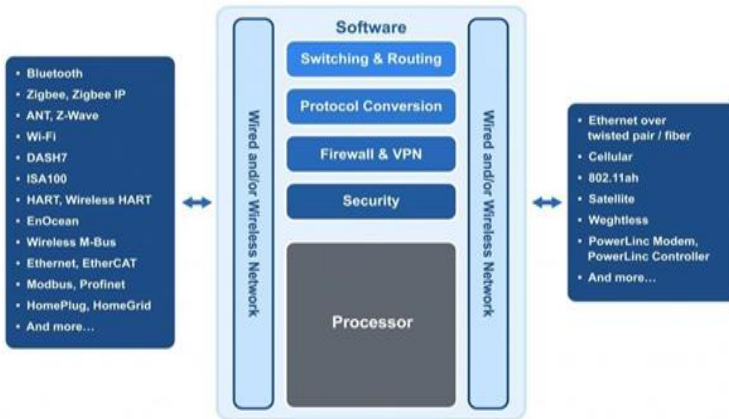


FIG. 5 Embedded Devices With Gateway

Bluetooth is a wireless technology standard for transmitting data between fixed and mobile devices over short distances and for establishing personal area networks. Zigbee wireless technology was created specifically for low-cost sensors and control devices, and it is extensively utilized in a variety of applications.

The Z-Wave wireless communication standard is primarily utilized for home automation. It's a mesh network that communicates from appliance to appliance using low-energy radio waves, allowing for wireless management of home appliances and other devices including lights, security systems, thermostats, and windows.

Wi-Fi is a popular wireless networking technology that employs radio waves to enable high-speed Internet and network connections across long distances. A widespread misunderstanding is that Wi-Fi stands for wireless fidelity "fidelity of wireless communication"

The International Society of Automation produced the ISA100.11a wireless networking technology standard (ISA). This is the official description: "Wireless (WiFi)"

Process Control and Related Applications in Industrial Automation Systems

EnOcean is a wireless energy harvesting technology that is primarily used in building automation systems, but it is also used in industry, transportation, logistics, and smart homes. EnOcean technology enables wireless communications between batteryless wireless sensors, switches, controllers, and gateways by combining micro energy converters with ultra low power electronics.

Different utility providers may install a number of IoT devices in your home, each with its own gateway, as part of home automation. Electricity or gas, water, phone, Internet, cable/satellite, alarm system, medical devices, and so on are examples of these. Additional features, such as local storage or a user interface, may be required by some of these gateways.

1.6 Challenges of Internet of Things (IOT)

At the moment, IoT faces several challenges, including:

- ❖ insufficient monitoring and updating
- ❖ concerns over data storage and privacy
- ❖ The difficulty of software
- ❖ Volumes of data and their interpretation
- ❖ AI and automation integration
- ❖ Devices necessitate a continuous power source, which is impossible to obtain.
- ❖ Interaction and close contact.

1.7 Characteristics of IoT

There are seven important IoT characteristics:

1. Interconnectivity. This does not need any clarification. There must be a link between different layers of everything in IoT devices and computers, with sensors and other electronics and related hardware and control systems.
2. Particulars. All that can be tagged or linked in the way that it is intended to be connected. Sensors, kitchen equipment, and tagged livestock are only a few examples. Sensors may be built into devices, or sensing materials may be added to devices and objects.
3. Details. Data is the glue that holds the Internet of Things together, and it is the first step toward action and intelligence.
4. Communicate. Devices are linked so that they can exchange data, which can then be analyzed. Communication may occur over short distances or medium to very long distances. Wi-Fi and LPWA network technology such as LoRa or NB-IoT is examples.

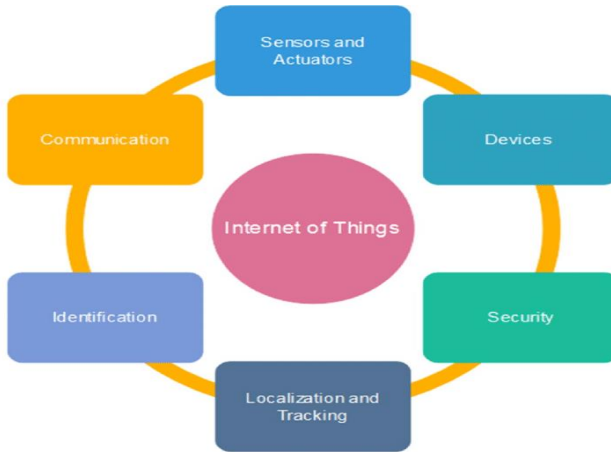


FIG 1.6: IoTCharacteristics

5. Intelligence The intelligence factor, such as sensing capabilities in IoT devices and intelligence derived from big data analytics (also artificial intelligence).
6. Take action. The product of intelligence. This can be a manual action dependent on disputes about phenomena (for example, in smart factory decisions) or automation, which is often the most critical piece.
7. The ecosystem. The Internet of Things' position on other technology, cultures, priorities, and the overall image into which the Internet of Things fits. The Internet of Everything dimension, the network dimension, and the need for strong relationships are all important considerations.

1.8 IoT Life Cycle

The lifetime of IoT development is pretty basic.

Monitoring, maintenance, and management come after deployment, then regular upgrades and decommissioning.

IoT Product Lifecycle is described in the below diagram.

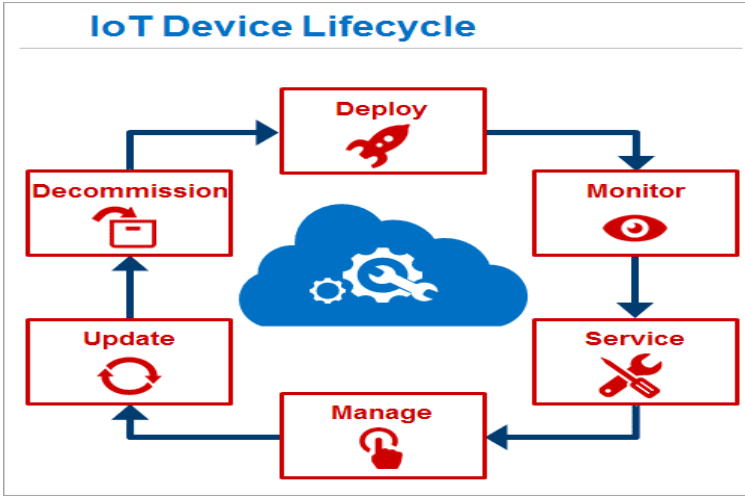


FIG 1.7 IoT Life Cycle

Aside from these facts, there are certain benefits and drawbacks of Internet of Things devices that can significantly affect humans' present and future generations.

1.9 Physical Design Of IoT

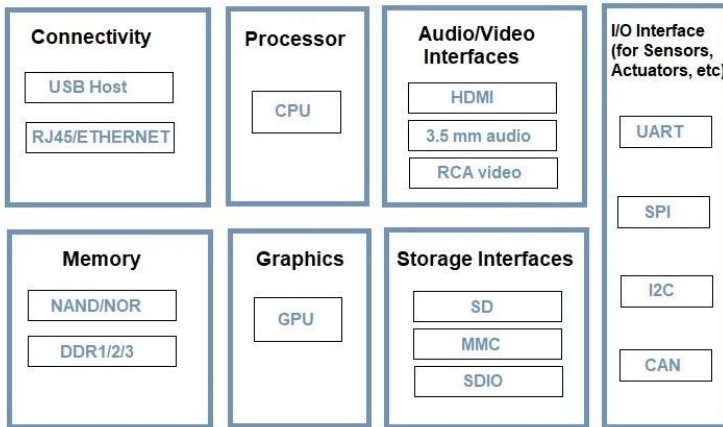
1.9.1 Things

Things are IoT Devices with special personalities that can do remote sensing, actuation, and monitoring. Things are the most critical component of an IoT program. Sensing Devices, Smart Watches, Smart Electronics Appliances, Wearable Sensors, Automobiles, and Automotive Robots are all examples of IoT Devices. These devices collect data in various ways. Data analytics tools' analysis yields valuable information to direct subsequent activities locally or remotely.

Temperature data produced by a Temperature Sensor in the home or other location, for example, when analyzed, may aid in determining temperature and taking action based on user input. The diagram above represents a typical block diagram of an IoT device. It can provide multiple interfaces for connecting to other devices. Sensors have I/O interfaces, Internet connectivity, storage, and audio/video. IoT devices gather data from onboard or connected sensors. The data is transmitted to another device or a cloud-based server. There are numerous cloud servers available today, especially for IoT systems. These platforms are known as IoT

Platforms. These clouds are specifically designed for IoT purposes. However, we can quickly review and process data here.

For example, suppose a relay switch attached to an IoT device will turn on/off an appliance based on commands sent to the IoT device over the Internet.



Generic Block Diagram of IoT Devices

FIG 1.8 Block Diagram of IoT

Temperature data produced by a Temperature Sensor in the home or other location, for example, when analyzed, may aid in determining temperature and taking action based on user input. The diagram above depicts a typical block diagram of an IoT device. It can provide multiple interfaces for connecting to other devices. Sensors have I/O interfaces, Internet connectivity, storage, and audio/video. IoT devices gather data from onboard or connected sensors. The data is transmitted to another device or a cloud-based server. There are various cloud servers available today, especially for IoT systems. These platforms are known as IoT Platforms. These clouds are specifically designed for IoT purposes. However, we can quickly analyze and process data here.

For example, suppose a relay switch attached to an IoT device will turn on/off an appliance based on commands sent to the IoT device over the Internet.

1.9.2 IoT Protocols

IoT protocols aid in establishing Internet-based communication between IoT devices (Node Devices) and cloud-based servers. It aids in

sending commands to IoT devices and receiving data from IoT devices through the Internet. An illustration is shown below. You can tell which protocols were used by looking at this graphic.

1.9.2.1 Link Layer

Link-layer protocols govern how data is physically transmitted through the network's physical layer or medium (Coaxial cable or another radio wave). This Layer specifies how packets are encoded and signaled by the hardware device over the medium to which the host is connected (e.g., coaxial cable).

802.3 – Ethernet: Ethernet is a collection of technologies and protocols mainly found in local area networks (LANs). IEEE 802.3 specification was the first to standardize in the 1980s. IEEE 802.3 specifies the physical and medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks. Ethernet is divided into two types: traditional Ethernet and switched Ethernet.

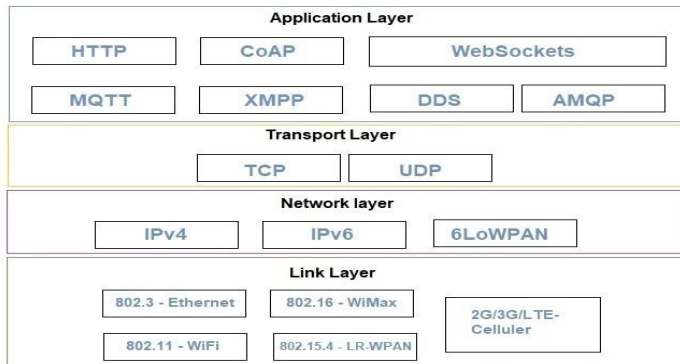


FIG 1.9 Layers of IoT

Ethernet is a collection of applications and protocols mainly used in local area networks (LANs). IEEE 802.3 standard was the first to standardize in the 1980s. IEEE 802.3 specifies the physical and medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks. Ethernet is divided into two types: traditional Ethernet and switched Ethernet.

Classic Ethernet is the original version of Ethernet, with data speeds ranging from 3 to 10 Mbps. The variants are widely known as

10BASE-X. In this case, 10 denotes the highest throughput, i.e., 10

Mbps, BASE denotes the use of baseband communication, and X denotes the medium form. In today's communication setting, most types of traditional Ethernet have become outdated.

Switches are used to link LAN stations in a switched Ethernet network. It replaces the traditional Ethernet repeaters and allows for maximum bandwidth use. IEEE 802.3 Popular Versions

There are several versions of the IEEE 802.3 protocol. The most popular ones are -

IEEE 802.3: This was the initial 10BASE-5 standard. It used a dense single coaxial cable through which a link could be tapped by drilling into the cable to the heart. In this case, 10 represents the highest throughput, i.e., 10 Mbps, BASE represents the use of baseband transmission, and 5 represents the maximum segment length of 500m.

- ❖ IEEE 802.3a: This provided the standard for thin coax (10BASE-2), a thinner version in which BNC connectors join coaxial cable parts. The number 2 corresponds to the overall section length of roughly 200m (185m, precisely).
- ❖ IEEE 802.3i: This standard established the twisted pair (10BASE-T) standard, which employs unshielded twisted pair (UTP) copper wires as the physical layer medium. IEEE 802.3u included additional variants for 100BASE-TX, 100BASE-T4, and 100BASE-FX.
- ❖ IEEE 802.3j: This is the standard for Ethernet over Fiber (10BASE-F), which uses fiber optic cables as the communication medium.

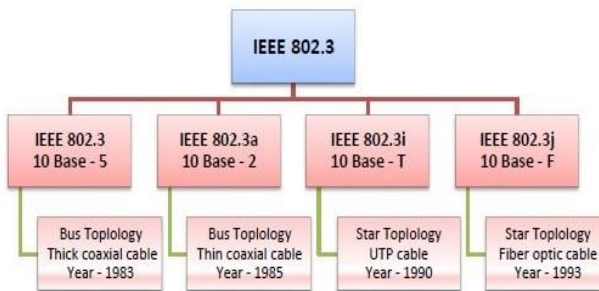


FIG 1.10 IEEE 802.3 Popular Versions

Classic Ethernet Frame Format and IEEE 802.3

The primary fields of a traditional Ethernet frame are as follows:

- ❖ Preamble: This is the initial area that contains the alert and timing pulse for transmission. In the case of traditional Ethernet, it is an 8-byte field, while in the case of IEEE 802.3, it is a 7-byte field.

- ❖ **Frame Delimiter Start:** It is a one-byte field in an IEEE 802.3 frame involving alternating patterns of ones and zeros ending with two ones.
- ❖ **Destination Address:** This is a 6-byte sector that contains the physical addresses of destination stations.
- ❖ **Source Address:** This is a 6-byte field that includes the sending station's physical address.
- ❖ **Length:** This is a 7-byte field that keeps track of the number of bytes in the data field.
- ❖ **Data:** This variable-sized area contains data from the upper layers. The actual data field size is 1500 bytes.
- ❖ **Padding:** This is applied to the data to make it the minimum length of 46 bytes.
- ❖ **CRC:** CRC is an abbreviation for cyclic redundancy search. It includes details about error detection.

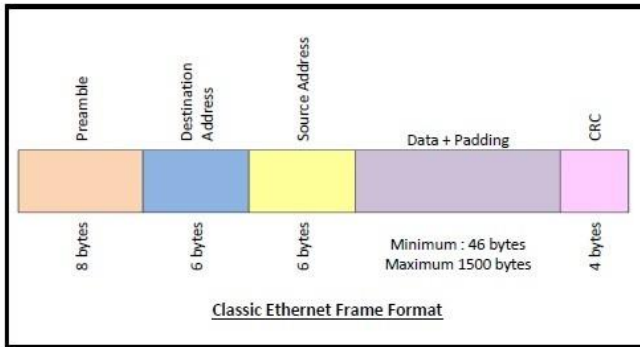


FIG 1.11 Classic Ethernet Frame Format

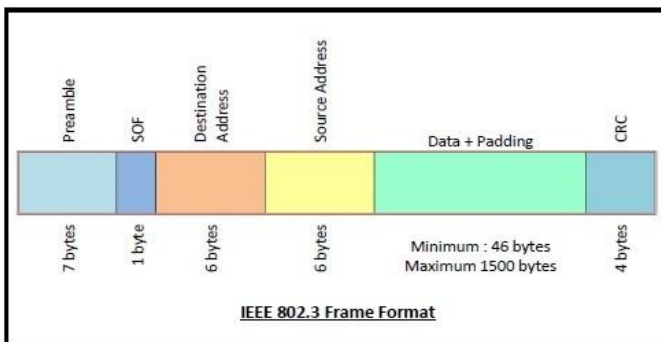


FIG 1.12 IEEE 802.3 Frame Format

IEEE 802.11 – WiFi: Part of the IEEE 802 family of LAN protocols, IEEE 802.11 defines a set of media access control (MAC) and physical layer (PHY) protocols for integrating wireless local area network (WLAN) Wi-Fi device communication at different frequencies, including but not limited to 2.4 GHz, 5 GHz, and 60 GHz.

IEEE 802.11 is a series of medium access control (MAC) and physical layer (PHY) protocols for incorporating wireless local area network (WLAN) device communication that is part of the IEEE 802 set of local area network (LAN) technical standards. The standard and its revisions serve as the foundation for wireless network devices bearing the Wi-Fi name. They are the most commonly used wireless computer networking protocols in the world. IEEE 802.11 is used in most home and office networks to allow computers, printers, smartphones, and other devices to communicate and link to the Internet without wires.

The LAN/MAN Standards Committee of the Institute of Electrical and Electronics Engineers (IEEE) develops and maintains the standards (IEEE 802). The standard's base version was published in 1997, and it has since been amended. Although each amendment is legally repealed when introduced into the most recent edition of the standard, the business community continues to market to amendments because they concisely denote the capabilities of their goods. As a result, each revision becomes its standard in the marketplace.

IEEE 802.11 operates at various frequencies, including, but not limited to, 2.4 GHz, 5 GHz, 6 GHz, and 60 GHz. While the IEEE 802.11 requirements list potential networks, the radio frequency

spectrum availability allowed differs substantially by regulatory domain.

The protocols, which are usually used in combination with IEEE 802.2, are intended to interoperate easily with Ethernet and often carry Internet Protocol traffic.

802.16 – Wi-Max: The WiMAX infrastructure standard is a standard for Wireless Metropolitan Area Networks (WMANs) established by IEEE 802 working group number 16, specializing in point-to-multipoint broadband wireless networking.

WiMAX is a wireless broadband networking technology that uses the IEEE 802.16 standard to provide high-speed data over a wide region. WiMAX stands for Worldwide Interoperability for Microwave Access (AXess), and it is a point-to-multipoint wireless networking technology.

WiMAX technology may address the demands of a wide range of consumers, from those in developing countries who wish to build a modern high-speed data network quite cheaply without the expense and time taken to install a wired network to those in remote areas who need quick connectivity where wired options might not be possible due to distances and costs involved - simply delivering WiMAX broadband. It is also used by smartphone apps, which provide high-speed data to consumers on the go.

The Fundamentals of WiMAX Technology

The WiMAX technology standard is a wireless metropolitan area network (WMAN) established by IEEE 802 working group number 16, specializing in point-to-multipoint broadband wireless connectivity. Initially, 802.16a was developed and published, but it has been refined. The 802.16d standard, also known as the 802.16-2004 standard, was introduced as a streamlined version of the 802.16a standard targeted at fixed implementations. Another variant of the standard, 802.16e or 802.16-2005, targeted at the roaming and smartphone markets, was also announced. WiMAX broadband infrastructure relies on many main innovations to provide high-speed data rates:

- ❖ OFDM (Orthogonal Frequency Division Multiplex): OFDM has been integrated into WiMAX technology to provide high-speed data without the selective fading and other problems that other signal formats provide.
- ❖ MIMO (Multiple Input Multiple Output): WiMAX technology employs MIMO for multipath propagation. MIMO allows for lower signal intensity levels or higher data speeds by using the various signal paths.

WiMAX Discussion Forum

The WiMAX Forum is a wireless industry consortium composed of a rising number of industry leaders. It was founded to promote and grow WiMAX technology globally and bring universal standards around the globe to allow it to become a globally established technology.

One of the forum's objectives is to facilitate the implementation of a standard that will allow complete interoperability between goods. The WiMAX Forum seeks to avoid the issues associated with previous wireless standards' low interoperability and the effect on adoption. Eventually, vendors will be able to have their products approved under the Forum's auspices and then market their products as "Forum Certified."

Even though WiMAX technology can accept traffic dependent on communication technologies such as Ethernet, Internet Protocol (IP), and Asynchronous Transfer Mode (ATM), the Forum will only validate the IP-related elements of 802.16 items. The focus is on IP operations since this is still the most commonly used protocol.

The Evolution of WiMAX

The origins of WiMAX can be traced back to the 1990s when it was realized that there would be a major rise in data traffic over telecommunications networks. Since wired telecommunications networks are very costly, especially in rural areas, and are not widely available in many countries, methods of delivering wireless broadband were investigated.

WiMAX history began with these inquiries into what was known as "last mile connectivity" - means of providing high-speed data to a vast number of people who do not have a wired link.

The promise of low-cost last-mile connectivity and technology capable of handling backhaul over a cellular link proved to be a convincing case for developing a modern wireless data link system.

The IEEE's standards development was the next big milestone in WiMAX history.

The IEEE 802 LAN/MAN Standards Committee formed the 802.16 standards working group in 1999. In December 2001, the first 802.16 standards were accepted, followed by two revisions to the original 802.16 standards. These revisions, defined as 802.16a and 802.16c, discussed radio spectrum and interoperability problems.

A new revision project was launched in September 2003. This was to match the standard with the European / ETSI HIPERMAN standard. It was also planned to have conformance evaluation requirements as the overall standard. The project was finished in 2004, and the standard was introduced as 802.16d. However, due to the implementation date, it is sometimes referred to as 802.16-2004. The previous 802.16 manuals, including the a, b, and c amendments, discontinued the introduction of the 802.16-2004 standard.

WiMAX variants

New WiMAX applications have been developed since its introduction. As a result, there are two "flavors" of WiMAX technology available:

The two flavors of WiMAX broadband technologies are used for various implementations. Although they are based on the same standard,

each implementation has been optimized for its specific use. 802.16d - DSL substitute The 802.16d version is also known as 802.16-2004, and it is closer to the original version of WiMAX described by 802.16a. It is targeted at fixed applications and provides a wireless counterpart to DSL broadband data, commonly referred to as WiMAX broadband. In reality, the WiMAX Forum defines the technology as "a standards-based technology that enables the distribution of last-mile wireless broadband service as an alternative to cable and DSL."

Because 802.16d can have data speeds of up to 75 Mbps, it is suitable for fixed, DSL replacement applications such as WiMAX broadband. It can also be used for backhaul, transmitting the final data to individual users. Cell radii can range up to 75 kilometers.

802.16e - Mobile / Nomadic Although 802.16 / WiMAX was initially intended to be a fixed-only technology, people on the move demand high-speed data at a lower cost than wireless networks, created a need for a mobile version, and 802.16e was established. This standard is often referred to as 802.16-2005. It currently allows users to bind to a WiMAX cell from some locations, with potential updates to include cell handover.

Data speeds of up to 15 Mbps are possible with 802.16e, and cell radius lengths are usually between 2 and 4 km. Competition

There are some implementations of the IEEE 802.3 protocol. The most well-known are -

The rivalry between WiMAX and 802.16 is dependent on the form or variant used. While it was originally believed that WiMAX would be a major competitor to Wi-Fi, there are other places where WiMAX is a challenge.

- ❖ Cable DSL bars WiMAX is capable of providing customers with high-speed data connections. As such, it can pose a challenge to DSL cable operators.
- ❖ Cell phone service providers Cell phone providers viewed the smartphone version of WiMAX as a potential challenge as LTE was being established and the first roll-outs began. It was also proposed as the IMT 4G standard. However, LTE was chosen as the standard, leaving WiMAX for fixed WiMAX broadband, last-mile connections, and other point-to-point applications.

WiMAX technology has been used in a variety of applications. Although it was originally seen as a contender for 4G, its use is dwindling. However, it is still used for WiMAX broadband and last-mile connections.

802.15.4 -LR-WPAN: A collection of Low-rate wireless personal area network standards. The IEEE 802.15.4 standard specifies the MAC and

PHY layer used for networking protocols such as Zigbee®, 6LoWPAN, Thread, WiSUN, and MiWiTM. The standards allow low-cost, low-speed communication for devices with limited power.

2G/3G/4G- Cell Communication: There are separate generations of telecommunications. These specifications allow IoT devices to connect over cellular networks.

1.9.2.2 Networking Layer In charge of transmitting IP datagrams from the source network to the destination network. The Network layer handles host addressing and packet routing. For Host recognition, we used IPv4 and IPv6. IPv4 and IPv6 are also hierarchical IP address allocation systems. IPv4: Internet Protocol Version 4.

An Internet Protocol address (IP address) is a numerical mark assigned to every device attached to a computer network that communicates using the Internet Protocol. An IP address performs two primary functions: defining a host or network interface and addressing location. An IP address is specified as a 32-bit number in Internet Protocol version 4 (IPv4). However, due to the proliferation of the Internet and the shortage of valid IPv4 addresses, a new variant of IP (IPv6) was standardized in 1998, using 128 bits for the IP address. Since the mid-2000s, IPv6 implementation has been continuing.

An Internet Protocol address (IP address) is a numerical mark assigned to every device attached to a computer network that communicates using the Internet Protocol.

[1][2] Formalized paraphrase An IP address performs two primary functions: defining a host or network interface and addressing a particular location.

An IP address is specified as a 32-bit number in Internet Protocol version 4 (IPv4).

[2] Formalized paraphrase, However, due to the proliferation of the Internet and the shortage of valid IPv4 addresses, a new variant of IP (IPv6) was standardized in 1998, using 128 bits for the IP address. (3rd) [4]. Formal paraphrase [5] Formal paraphrase Since the mid-2000s, IPv6 implementation has continued.

In IPv4, IP addresses are written and displayed in human-readable notation, such as 172.16.254.1, and in IPv6, 2001:db8:0:1234:0:567:8:1. In CIDR notation, the size of the routing prefix of the address is specified by suffixing the address with the number of significant bits, for example,

192.168.1.15/24, which is equal to the traditionally used subnet mask 255.255.255.0.

The Internet Assigned Numbers Authority (IANA) and five regional Internet registries (RIRs) are responsible in their designated regions for assigning IP addresses to local Internet registries, such as Internet service providers (ISPs) and other end users. IANA allocated IPv4 addresses to RIRs in blocks of approximately 16.8 million addresses each. However, these have been depleted at the IANA stage since 2011. Just one of the RIRs has a supply of local assignments in Africa left. [6] Formalized paraphrase Some IPv4 addresses are reserved for private networks and are thus not internationally exclusive.

Network administrators give each device connecting to a network an IP address. Such assignments can be static (fixed or permanent) or dynamically based on network policies and software features. An IPv4 address is 32 bits long, limiting the address space to 4294967296 (2³²) addresses. These addresses are reserved for specific uses, such as private networks (18 million) and multicast addressing (270 million).

IPv4 addresses are typically expressed in dot-decimal notation, consisting of four decimal numbers separated by dots ranging from 0 to 255, for example, 172.16.254.1. Each component represents an 8-bit (octet) address group. IPv4 addresses can be provided in different hexadecimal, octal, or binary representations in some specialized writing.

History of Subnetting

During the early stages of Internet Protocol development, the network number was always the highest order octet (most significant eight bits). Since this approach only allowed for 256 networks. It quickly proved insufficient when new networks emerged independently of the existing networks already designated by a network number. With the implementation of classful network architecture in 1981, the addressing specification was updated. [2] Formalized paraphrase

Classful network architecture allowed more individual network assignments and fine-grained subnetwork design. The class of an IP address was specified as the first three bits of the most important octet. Three classes (A, B, and C) were specified for universal unicast addressing. Depending on the class, the network recognition depended on the entire address's octet boundary segments. Each class used an increasing number of octets in the network identifier, decreasing the available number of hosts in higher-order classes (B and C). The table below provides an outline of this now-defunct scheme.

Historical Classful Network Architecture

Class	Leading bits	Size of network number bit field	Size of rest bit field	Number of networks	Number of addresses per network	Start address	End address
A	0	8	24	128(2^7)	16777216 (2^{24})	0.0.0.0	127.255.255.255
B	10	16	16	16384 (2^{14})	65536 (2^{16})	128.0.0.0	191.255.255.255

Classy network architecture served its function during the Internet's early days. However, it lacked scalability in the face of the 1990s' exponential proliferation of networking. In 1993, the address space's hierarchy structure was replaced by Classless Inter-Domain Routing (CIDR). CIDR is based on variable-length subnet masking (VLSM) to make allocation and routing based on arbitrary-length prefixes. Today, fragments of classful network principles serve mainly in a small capacity as the default configuration parameters of certain network software and hardware components (e.g., netmask) and technical terms used in network administrator discussions.

Private Addresses

When universal end-to-end connectivity was envisioned for communications with all Internet hosts, IP addresses were meant to be internationally exclusive. However, as private networks grew and public address space needed to be conserved, it was discovered that this was not always possible.

Computers that are not wired to the Internet, such as factory machines that communicate only through TCP/IP, do not need globally specific IP addresses. Today, such private networks are common, and they usually link to the Internet through network address translation (NAT) when necessary.

Three non-overlapping IPv4 address ranges are reserved for private networks. [8] Formalized paraphrase Since these addresses are not accessible on the Internet; their use would not require coordination with an

IP address registry. Any person is free to use the reserved blocks. A network administrator will typically split a block into subnets; for example, many home routers have a default address range of 192.168.0.0 to 192.168.0.255 (192.168.0.0/24). Internet Protocol version 6 (IPv6) is the successor to IPv4. The Internet Engineering Task Force (IETF) created IPv6 to solve the long-awaited issue of IPv4 address depletion. IPv6 was designated as a Draft Standard by the Internet Engineering Task Force (IETF) in December 1998. It was adopted as an Internet Standard on July 14, 2017. IPv6 employs a 128-bit address, allowing for 2128, or roughly 3.41038 addresses.

The address size in IPv6 was expanded from 32 bits in IPv4 to 128 bits, allowing for up to 2128 (approximately 3.4031038) addresses. This is thought to be adequate for the near future.

The goal of the new design was not only to have a sufficient number of addresses but also to reinvent Internet routing by enabling a more effective aggregation of subnetwork routing prefixes. This slowed the development of routing tables in routers. The smallest individual allocation available is a subnet for 264 hosts, which is the square of the whole IPv4 Internet. Real address consumption ratios on any IPv6 network segment would be poor at these stages. The new architecture also allows for separating a network segment is addressing infrastructure, i.e., the local management of the segment's usable capacity, from the addressing prefix used to redirect traffic to and from external networks. If the global connectivity or routing strategy changes, IPv6 has facilities that instantly adjust the routing prefix of whole networks without requiring internal overhaul or manual renumbering. Because of the vast IPv6 addresses available, large blocks may be allocated for specific purposes and aggregated for efficient routing. There is no need for complicated address conservation approaches like CIDR with a huge address space.

Private Addresses

IPv6 addresses are reserved in the same way as IPv4 addresses for private networks. In IPv6, these are known as special local addresses (ULAs). This block has the routing prefix fc00::/7 reserved for it,[9] and is separated into two /8 blocks of separate implicit policies. The addresses contain a 40-bit pseudorandom integer, which reduces the possibility of address collisions if sites merge or packets are misrouted.

For this reason, early practices used a separate block (fec0::), which was called site-local addresses.

[10] Formalized paraphrase; however, the concept of what constituted a location remained vague. The ill-defined addressing strategy generated routing ambiguities. This address form has been deprecated and can not be used in modern systems. [11] Formalized paraphrase

Link-local addresses begin with fe80::, are allocated to interfaces for communication over the attached link. The operating system produces addresses for each network interface automatically. This allows for immediate and automated communication between all IPv6 hosts on a link. This functionality is used in IPv6 network administration's lower layers, such as the Neighbor Discovery Protocol.

Private and link-local Address Prefixes are not Permitted to be Routed on the Public Internet.

IPv6 over Low-Power Wireless Personal Area Networks is abbreviated to 6LoWPAN. 6LoWPAN is a completed IETF working group in the Internet region. This protocol enables even the smallest devices with minimal computing power to transfer data wirelessly using an internet protocol. 6LoWPAN can connect with 802.15.4 devices and other types of devices through an IP network connection, such as WiFi. IPv6 over Low-Power Wireless Personal Area Networks is abbreviated to 6LoWPAN.

The principle of 6LoWPAN arose from the belief that “the Internet Protocol might and should be extended even to the smallest devices” and that low- power devices with minimal computing capacities should be able to engage in the Internet of Things.

The 6LoWPAN group has established encapsulation and header compression mechanisms that allow IPv6 packets to be sent and received over IEEE 802.15.4 networks. IPv4 and IPv6 are the workhorses for data transmission in local-area and wide-area networks like the Internet. Similarly, IEEE 802.15.4 devices provide sensing communication in the wireless domain. The two networks' underlying characteristics, however, vary.

The IPv6 offers a simple transport mechanism to generate complex control systems and cost-effectively communicate with devices through a low- power wireless network, making the 6LoWPAN suitable for home or building automation. The following are the most popular applications for it:

It is possible to achieve distinct advantages over other IoT platforms by connecting smart home devices via IPv6. The Thread initiative was developed to standardize a protocol that runs over 6LoWPAN to allow

home automation. Moreover, OpenThread makes it easier for developers to start with Thread and build smart home solutions.

Smart Agriculture: allowing all types of sensors used in agriculture and farming by linking devices far apart in remote areas. This protocol's possibilities for creating mesh networks make it just the right one for this kind of use. Industrial IoT: Automated factories and industrial plants provide a tremendous opportunity for 6LoWPAN, and using automation will result in considerable savings. The potential of 6LoWPAN to bind to the cloud opens up a wide range of possibilities for data monitoring, analysis, and predictive maintenance. 6LoWPAN is the newest ZigBee competitor. What exactly is ZigBee? Like 6LoWPAN, ZigBee is intended for low-data-rate and battery-powered applications. ZigBee is currently the most common, low-cost, low-power wireless mesh networking standard on the market and the more advanced technology of the two (ZigBee, 6LoWPAN). It is commonly used in personal or home-area networks or wireless mesh networks for networks that run over longer distances.

The ZigBee IP is based on the IEEE 802.15.4 standard, but, unlike 6LoWPAN, it cannot interface with other protocols easily. However, one advantage of ZigBee is that nodes can remain in sleep mode most of the time, significantly extending battery life. M2M/IoT Implementations for ZigBee Wireless Light Switches are a kind of switch that allows you to turn on and off the lights Meters for electricity (Smart grid, demand response, etc.) Monitoring of Industrial Equipment

The name is derived from the wiggly dance performed by honey bees on their way to deposit honey. Data in a ZigBee network, like those wild bees, "hops" through a mesh of transceivers before a route to the host (usually the internet) is discovered. It runs at a fixed data rate of 250 kbit/s and is based on the IEEE 802.15 protocol. Because of this speed and poor transmit capacity, ZigBee has a limited range. Repeaters and a dense network of nodes are often needed to achieve the desired coverage.

1.9.2.3 Transport Layer

This layer handles error management, segmentation, flow control, and congestion control. As a result, these layer protocols allow end- to-end message transmission independent of the underlying network.

TCP (Transmission Control Protocol): TCP (Transmission Control Protocol) is a network protocol that specifies how to create and manage a network conversation in which application programs can share data. TCP communicates with the Internet Protocol (IP), which determines how computers transmit data packets to one another. TCP and IP are the

fundamental principles that define the Internet. TCP is defined by the Internet Engineering Task Force (IETF) in Request for Comment (RFC) standards document number 793.

UDP is a Transport Layer protocol that stands for User Datagram Protocol. UDP is a component of the Internet Protocol suite, also known as the UDP/IP suite. It is an insecure and connectionless protocol, as opposed to TCP. As a result, there is no need to communicate prior to data transmission.

1.9.2.4 Application Layer

Application layer protocols describe how systems communicate with lower layer protocols to transfer data over a network.

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used to distribute hypermedia documents such as HTML. It was created to facilitate communication between web browsers and web servers. However, it can also be used for other purposes. HTTP follows the traditional client-server model. A client opens a connection to make a request and then waits for an answer. HTTP is a stateless protocol, which means that the server does not store any data (state) between requests.

RFC 7252 defines CoAP-Constrained Application Protocol as a specialized Internet Application Protocol for constrained devices. It allows devices to connect over the Internet. The protocol is designed specifically for restricted hardware such as 8-bit microcontrollers, low-power sensors, and other devices that cannot run HTTP or TLS. WebSocket Protocol: The WebSocket Protocol allows two-way communication between a client running untrusted code in a managed environment and a remote host who has agreed to receive messages from that code. The origin-based protection paradigm, widely used for web browsers, was used.

MQTT is a machine-to-machine (M2M)/Internet of Things (IoT) connectivity protocol. It was created as an incredibly lightweight publish/subscribe messaging transport that can be used for communications with remote locations where a minimal code footprint is needed, and there is limited network capacity.

Extensible Messaging and Presence Protocol (XMPP) is an XML-based communication protocol for message-oriented middleware (Extensible Markup Language). It facilitates the sharing of organized and extensible data between two or more network entities in near- real-time.

The Object Management Group® (OMG®) Data Distribution Service (DDSTM) is a middleware protocol and API standard for data-centric

connectivity. It connects device elements, offering low-latency data connectivity, extreme stability, and a modular infrastructure needed by enterprise and mission-critical Internet of Things (IoT) applications.

AMQP: The AMQP – IoT protocols comprise a collection of components that route and save messages within a broker carrier and a set of policies for connecting the components. The AMQP protocol allows patron systems to communicate with the dealer and interact with the AMQP model.

1.10 IoT Conceptual Architecture

In this post, we will look at the logical nature of the Internet of Things. The conceptual architecture of an IoT framework refers to an abstract representation of the entities and processes without delving into the low-level implementation requirements. We define the words below to help you visualize the Logical Design of IoT.

Functional Blocks for IoT Models of IoT Communication APIs for IoT Communication

1.10.1 Internet of Things Functional Blocks

An IoT device comprises various functional blocks that enable the system to perform functions such as recognition, sensing, actuation, communication, and management.

The following are practical blocks:

An IoT system comprises devices that perform sensing, actuation, monitoring, and control.

Communication: Manages the IoT system's communication. Resources provide device monitoring services, device regulation services, data publication services, and device exploration services.

Management: This block includes several roles for governing the IoT system. Protection: This block protects the IoT device by providing authentication, authorization, message and information integrity, and data security.

Application: An GUI from which users can access and manage different facets of the IoT framework. Users may also monitor the device status and view or evaluate the generated data using the application.

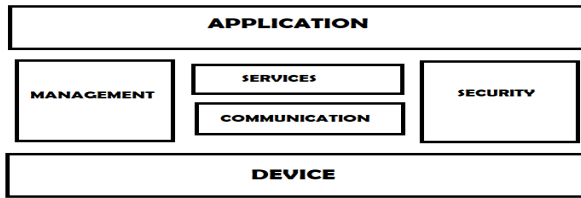


FIG1.13 IoT Functional Blocks

1.10.2 IoT Communication Models

Model of Request-Response

The request-response model is a communication model in which the client sends requests to the server, and the server replies. When a request is sent, the server determines how to respond, retrieves the data, retrieves the resource representation, prepares the response, and then sends the response to the client. Request-response communication is a stateless model in which each request-response pair is independent of the others.

HTTP is a request-response protocol that links a client and a server. The client could be a web browser, and the server could be an application running on a device that hosts a web.

Example: A client (browser) sends an HTTP request to the server, and the server responds with a response. The response includes request status information as well as the requested material.

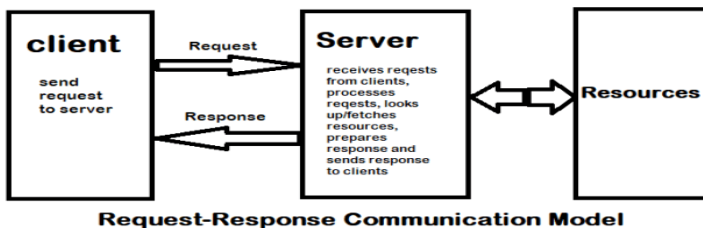


FIG 1.14 Request –Response Communication Model

Publish-Subscribe Model

Publish-Subscribe is a communication model involving authors, brokers, and customers. Data were gathered from publishers. The publishers send the data to the topics handled by the broker. Publishers are ignorant of their clients. Consumers subscribe to the issues that the broker manages. When the broker receives data for an issue from the publisher, it distributes it to all subscribers.

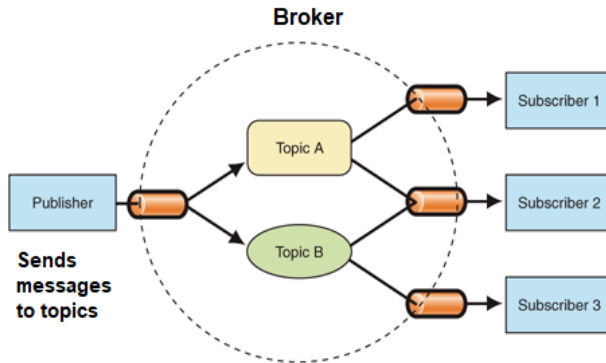


FIG 1.15 Publish-Subscribe Model Push-Pull Model

Push-Pull is a communication model in which data suppliers push data to queues, and users pull data. Take data from the Queues. Producers are not expected to be aware of their clients. Queues aid in the separation of messaging between producers and consumers. Queues often act as a buffer, which is useful for a difference between the rate suppliers push data and the rate at which consumers pull data.

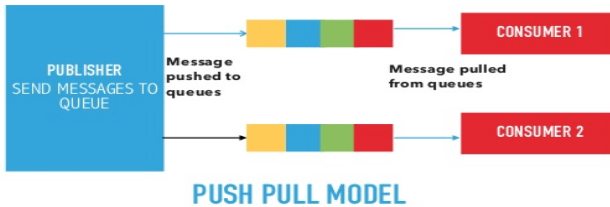


FIG 1.16 Push-Pull Model

Exclusive Pair Model

Exclusive Pair is a bidirectional, entirely duplex communication model that relies on a continuous link between client and server. When a link is created, it stays open until the client sends a request to close the connection. After the connection is established, the Client and server can send messages. Exclusive pair is a stateful communication model in which the server is aware of all open links.

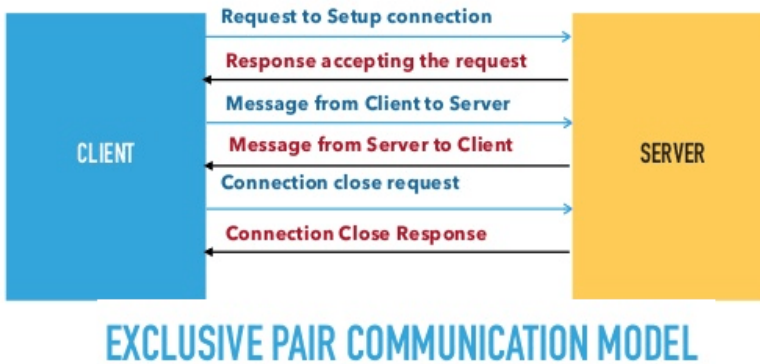


FIG 1.17 Exclusive Pair Model

1.10.3 IoT Communication APIs

For IoT communication, we usually use two APIs. These are the IoT Communication APIs: REST-based Communication APIs
APIs for WebSocket-based Communication
Communication APIs based on REST

Representational state transition (REST) is a collection of architecture standards that can be used to design Web services and Web APIs that rely on the properties of networks and how resource states are handled and transmitted. The rest of the architectural restriction applies to the modules, connectors, and data elements within a distributed hypermedia system that uses REST APIs to implement the request-response communication model. Below are the remaining architectural constraints:

Client-server architecture – The division of interests is the driving theory behind the client-server constraint. Clients, for example, need not be concerned with data management, which is the provider's responsibility. Similarly, the server should not be concerned with the user interface, which is the client's responsibility. Separation enables client and server to be created and updated separately.

Stateless – Each request from the client to the server must contain all of the information required to interpret the request and cannot make use of any server-stored context. The session state is completely stored on the client.

Cache-able – Cache constraints demand that the data in response to a request be leveled as cache-able or non-cache-able, either implicitly or

directly. If a response is cache-able, a client cache is allowed permission to reuse the response data for future, similar requests. Caching will reduce or delete certain orders, improving performance and scalability.

Layered system constraints – layered system constraints limit the action of components. They cannot see past the immediate layer in which they are communicating. E.g., the client does not know if it is linked directly to the end server or by an intermediary. Allowing intermediaries to respond to requests instead of the end server improves system scalability without requiring the client to do anything different.

Uniform interface – Uniform interface constraints demand that the mode of communication between client and server be uniform. Resources are defined in requests (via URIs in web-based systems). They are distinct from the representations of the data returned to the client. When a client owns a resource representation, it includes all of the information needed to update or uninstall the resource (provided the client has the required permissions). Each communication contains sufficient information to explain how to process the message.

Code on demand – Servers may have executable code or scripts for clients to run in their context. This is the only restriction that is discretionary.

A RESTful web service is a “Web API” built with HTTP and REST concepts. REST is one of the most commonly used IoT communication APIs.

Uniform Resource Identifier (URI)	GET	PUT	PATCH	POST	DELETE
Collection, such as https://api.example.com/resources/	<i>List the URIs</i>	<i>Replace the</i>	<i>Not gender</i>	<i>Create a new</i>	<i>Delete the</i>
	and perhaps other details of the collection's members.	entire collection with another collection.	ally used	entry in the collection. The new entry's URI is assigned automatically and is usually returned by the operation.	entire collection

	Retrieve a representation of the addressed collection member, expressed in an appropriate Internet media type.	Replace the addressed member of the collection, or create it if it does not exist.	Update the addressed member of the collection.	Not generally used. Treat the addressed member as a collection in its own right and create a new entry within	Delete the addressed member of the collection.
--	--	--	--	---	--

Http Methods

Websocket Based Communication API

Websocket APIs allow full-duplex bidirectional communication between clients and servers. Websocket APIs use the exclusive pair communication model. Unlike the request-response model used by REST, WebSocket APIs support complete duplex communication and do not require a new link to be formed for each message sent. The client sends a link configuration request to the server, initiating websocket communication. The request (known as a websocket handshake) is sent over HTTP and is treated by the server as an update request. The server responds to the websocket handshake answer by whether it accepts the websocket protocol. After the link is established, the client and server can send data and messages in full duplex mode. Since there is no overhead for link setup and termination demands for each packet, the Websocket API eliminates network traffic and letency. Websockets are ideal for IoT applications that require low latency or high throughput. As a result, Web sockets are the perfect IoT communication APIs for IoT systems.

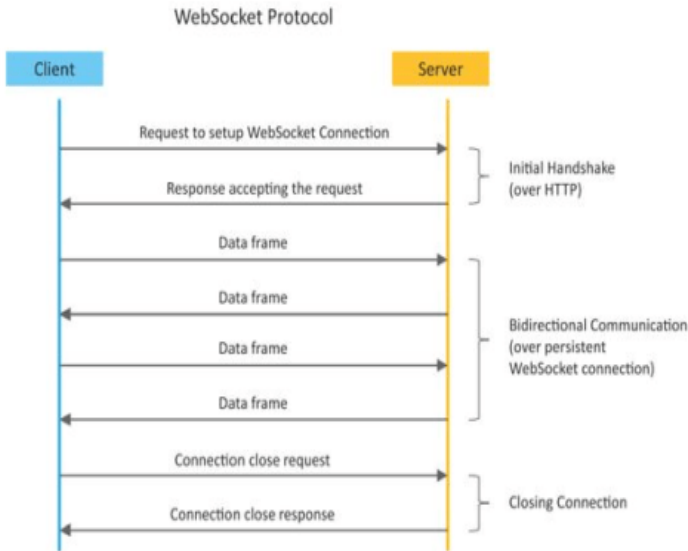


FIG 1.15 Websocket Protocol

1.11 IOT Communication Protocols

Various communication protocols and technologies are used on the Internet of Things. Bluetooth, USB, Radio Protocols, LTE-A, and WiFi-Direct are some of IoT innovations and protocols (IoT Communication Protocols). These IoT communication protocols are designed to cater to and satisfy the basic functional requirements of an IoT framework.



FIG 1.16 IoT Technologies and Protocols

A Bluetooth a.

A vital short-range IoT communication protocol/technology, Bluetooth, which has grown in popularity in the computer and consumer goods industries, is supposed to be critical for wearable devices in particular, which would once again be bound to the IoT, but most likely from a smartphone in many situations. The new Bluetooth Low-Energy (BLE) – or Bluetooth Smart, as it is now called –is critical for IoT applications. Importantly, although it has a similar range to Bluetooth, it has been built to use less power.

Zigbee Wireless Technology.

ZigBee, similar to Bluetooth, is mainly found in commercial environments. It has major advantages in complex systems, such as low-power operation, high security, robustness, and high efficiency. It is well placed to take advantage of wireless control and sensor networks in IoT applications. The most recent version of ZigBee is 3.0, which effectively unifies the different ZigBee wireless protocols into a single standard.

Z-Wave

Z-Wave is a low-power RF communications IoT technology designed specifically for home automation products such as lamp controllers and sensors, among many other applications. Z-Wave uses a simpler protocol than most, making for quicker and smoother development. However, the only producer of chips is Sigma Designs, instead of numerous sources for other wireless technologies such as ZigBee and others.

Cellular Internet (Wi-Fi)

WiFi connectivity is a common IoT communication protocol that is always an obvious option for many developers, particularly given the availability of WiFi within the home environment and within LANs. Conclusion: there is a strong current system and fast data transfer and the capacity to process large volumes of data. The most popular WiFi protocol currently found in households and many enterprises is 802.11n, which provides a spectrum of hundreds of megabits per second, which is good for file transfers but could be too power-consuming for many IoT applications.

Molecular

Any IoT program that needs service over longer distances can use GSM/3G/4G cellular communication features. Although cellular technology can transmit significant data, especially for 4G, the cost and power usage would be prohibitively expensive for many applications. However, it could be suitable for sensor-based low-bandwidth data ventures that transmit very small volumes of data across the Internet.

NFC

NFC (Near Field Communication) is an Internet of Things (IoT) technology. It enables easy and secure communication between electronic devices, especially smartphones, enabling consumers to conduct transactions without physically being present. It allows the user to attach mobile devices and view digital information. It expands the capability of contactless card technologies by allowing devices to exchange information at less than 4 cm.

LoRaWAN LoRaWAN is a popular IoT technology aimed at wide-area network (WAN) applications. LoRaWAN was created to provide low-power WANs with features directly needed to enable low-cost mobile secure communication in IoT, smart city, and industrial applications. It meets low- power consumption standards and serves vast networks of millions of users, with data speeds ranging from 0.3 kbps to 50 kbps.

Embedded Devices (System) in the IoT

It is important to understand embedded devices when learning about the IoT or developing IoT projects. The objects that make up the one-of-a-kind computer machine are embedded systems. This system may or may not provide Internet access.

In most instances, an embedded device framework operates as a single program. On the other hand, these devices can connect to the internet and communicate with other network devices.

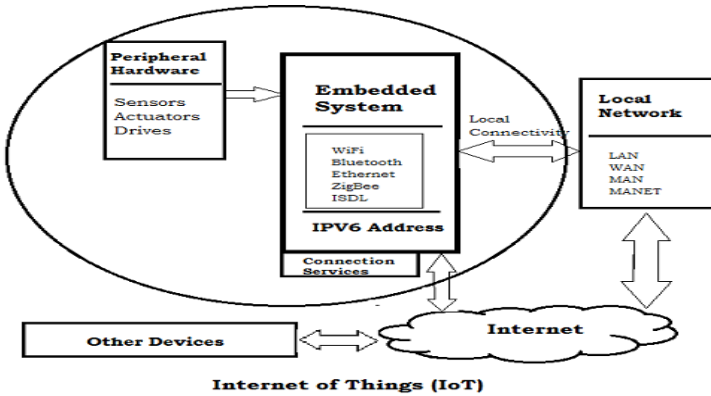


FIG 1.17 Embedded System Hardware

The embedded device may be of the microcontroller or microprocessor variety. An integrated circuit is present in each of these forms (IC).

A RISC family microcontroller, such as the Motorola 68HC11, PIC 16F84, Atmel 8051, and others, is a vital part of an embedded device. The most critical aspect that separates these microcontrollers from microprocessors such as the 8085 is their internal read and write memory. The following parts identify the vital embedded computer modules and system design.

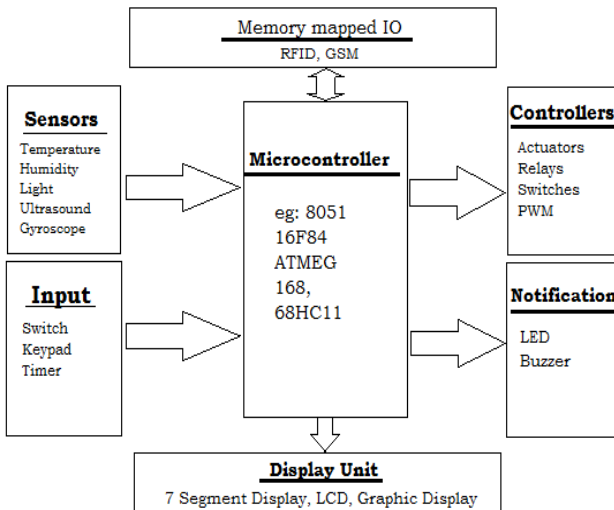


FIG:1.18 Basic Embedded System

Software for Embedded Systems

The embedded architecture that uses devices for the operating system is based on the language platform and is essentially where real-time operations are conducted. Manufacturers create embedded applications for electronic devices such as automobiles, telephones, modems, appliances, etc. Lighting controls run on an 8-bit microcontroller can be used as embedded system applications. There may also be sophisticated applications for missiles, process control systems, and aircraft, among other things.

1.12 Levels of the Internet of Things

The elements of the IoT architecture differ depending on the programme. Based on this fact, different levels of the IoT framework have been identified. Let's take a look at these IoT levels and their modules and several explanations of how they can be used. Consider an air conditioner, which must be kept at a certain temperature to understand how the Internet of Things works.

IoT at the first level

This level has an air conditioner, a temperature sensor, a data collection and interpretation app, and an app that can be used to control and track things. The sensed data is saved locally.

- ❖ The data collection is carried out at a territorial level.
- ❖ Monitoring and control were accomplished by using a mobile app or a web app.
- ❖ The amount of data produced by this level of application is not large. All control activities are carried out via the internet.
- ❖ For example, a temperature sensor is used to measure the temperature of a room, and the data is stored and analysed locally. Smartphone apps may be used to start a control operation, or they may keep track of how things are going.
- ❖ IoT at Level 2
- ❖ This level includes an air conditioner, a temperature sensor, big data (larger than level-1, with data processing performed here), a cloud, and a control and monitoring app.
- ❖ This level-2 is more complicated than level 1. Furthermore, the rate of sensing is higher than at level 1.
- ❖ The data at this level is very large. As a consequence, cloud computing is used.

- ❖ Data collection is done at a local level. The cloud is mainly used for computing.
- ❖ A control operation is initiated using a web interface or a smartphone app. Based on data analysis.

IoT at the Third Level

The figure shows that this level contains an air conditioner, a temperature sensor, large data storage (larger than level 1), a cloud (for data analysis), and a control and tracking app.

The data in this case is massive, i.e., huge data. The data sensing frequency is huge, and the gathered sensed data is processed on the cloud since it is massive.

A smartphone app or a web app can be used to start control actions based on the results of an analysis. The data processing takes place in the cloud.

Examples cover agricultural systems, odour sensor-based space freshening technologies, etc. IoT at the fourth level

This level has a lot of sensors, data processing, and an app that lets you control and track things.

At this level, multiple sensors independent of one another are used.

The data obtained by these sensors is uploaded to the cloud in its own right. Cloud storage is used at this level instead of the need for large amounts of data computing.

A web interface or a smartphone app can be used to control the actions taken based on the data collected.

Level 5, IoT

This level has sensors, a coordinator node, data collection and interpretation, and an app that lets you control and track things.

This level is close to level 4, which also has many data. As a result, it is sensed by many different sensors at a much faster rate and all at the same time.

Data processing and analysis were carried out in the cloud.

The control operation is carried out through a smartphone app or a web app based on the study.

1.13 Architecture of the Internet of Things

There is no commonly established special or normative agreement on the Internet of Things (IoT) architecture. The IoT architecture differs in terms of functional areas and solutions. The IoT architecture technology, on the other hand, is mostly made up of four main parts:

Elements of IoT Architecture

Sensors and Devices Networks and Gateways Cloud/Management Service Layer The Application Layer

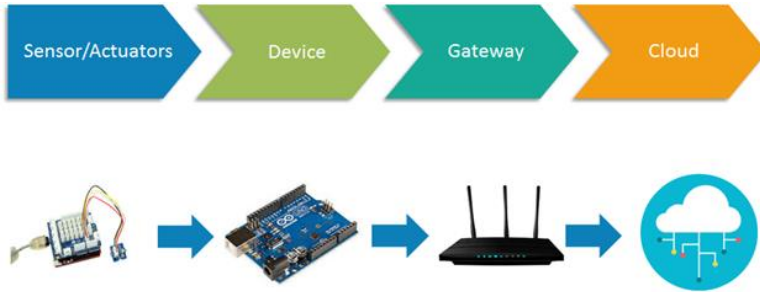


FIG 1.19 IOT ARCHITECTURE

The architecture of IoT Solutions in Stages

Multiple layers of IoT are based on the capability and performance of IoT components to deliver the right approach to business organizations and end-users. The IoT architecture is a foundational approach for constructing the different components of IoT so that it can distribute services over networks to satisfy future needs.

The primary stages (layers) of IoT that provide IoT architecture solutions are as follows.

1. **Sensors/Actuators:** Sensors and actuators are devices that can transmit, retrieve, and process data over a network. These sensors or actuators can be linked through wired or wireless connections. GPS, electrochemical, gyroscope, RFID, and other technologies are included. The majority of sensors need communication through sensors gateways. Sensors and actuators may be connected using a Local Area Network (LAN) or a Personal Area Network (PAN).
2. **Gateways and Data Acquisition:** Since these sensors and actuators generate a vast amount of data, high-speed Gateways and Networks are needed to process the data. This network may be of the Local Area Network (LAN) variety (such as WiFi, Ethernet, and so on), or it may be of the Wide Area Network variety (WAN such as GSM, 5G, etc.).
3. **Edge IT:** The hardware and software gateways that evaluate and pre-process data before moving it to the cloud are referred to as the edge in the IoT Architecture. If the data read from the sensors and gateways is not changed from its previous reading value, it does not pass to the cloud, saving the data.

4. Data center/Cloud: The Data Center or Cloud is a Management Service that processes information through analytics, system management, and security measures. Aside from security controls and system management, the cloud transfers data to end-user applications such as Retail, Healthcare, Emergency, Environment, and Energy.

The 4 Stage IoT Solutions Architecture

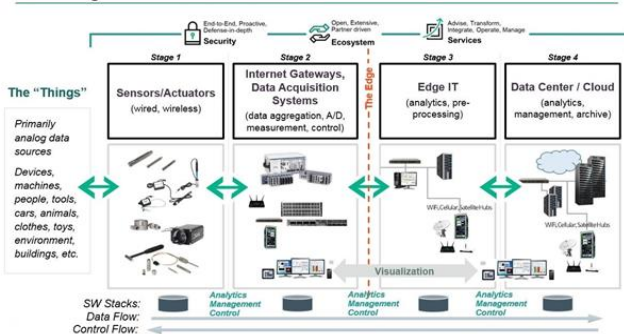


FIG 1.20 IoT Solutions Architecture

1.14 Advantages of IoT

Below are the key advantages of IoT technology:

- ❖ **Technological Optimization:** IoT technology leads greatly to the advancement and development of technologies. For example, a manufacturer may collect data from different car sensors using IoT. The manufacturer analyses them to enhance their design and make them more effective.
- ❖ **Enhanced Data Collection:** Conventional data collection has drawbacks and is structured for passive use. IoT allows for quick action on data.
- ❖ **Less Waste:** IoT provides real-time information, allowing for more efficient decision-making and resource management. For example, suppose a manufacturer discovers a problem in several car engines. In that case, he will monitor the production plan of such engines and resolve the problem using the manufacturing belt.
- ❖ **Improved Customer Engagement:** IoT helps you enhance customer service
- ❖ by identifying challenges and enhancing processes.

1.15 IoT Disadvantages

Now, in this Internet of Things guide, let us look at some of the drawbacks of IoT:

- ❖ Security: IoT technology produces a network of interconnected devices. However, the device can have little authentication control despite adequate security precautions during this phase.
- ❖ Privacy: IoT reveals a large volume of sensitive data in extreme detail
- ❖ without the user's direct consent. This raises a slew of privacy concerns.
- ❖ Flexibility: The flexibility of an IoT system is a big problem. It is mostly concerned with integrating with another system since several systems are involved.
- ❖ Complexity: The IoT system's architecture is often very difficult. Furthermore, deployment and servicing are difficult.
- ❖ Compliance: The Internet of Things has its collection of rules and regulations. However, owing to its complexity, the task of compliance is very difficult.

CHAPTER 2

SMART OBJECTS

2.1 Smarts Objects In IoT

IoT uses the term "smart" to describe physical objects that are interactive, digital, networked, somewhat autonomous, reconfigurable, and have local resource control. Energy, data storage, etc. are necessities for smart objects.

An object that enhances interaction with both people and other smart objects is referred to as a smart object. The world of Things (IoT) is a network of uniquely addressable, standard communication protocol-based heterogeneous objects (such as smart devices, smart objects, sensors, actuators, RFID, embedded computers, etc.).

People regularly use a variety of devices with internet, wireless, or wired connections. like as

- ❖ Smartphone
- ❖ Tablets
- ❖ TV, computer

No matter the circumstance, home, accessibility to a sensor, size, scenario, or hazard risk, these objects may be connected to one another and make our everyday lives easier (smart homes, smart cities).



Fig 2.1 Smart Objects

The Internet of Things (IoT) technologies are extensively used to transform the physical environment around us into a digital one. A smart object is equipped with application logic building elements that are sensible for their environment and communicate with users. A smart object can detect, occurrence, interpret, interact with other smart objects, and share information with people. They can also record events that happen to them and in their environment.

The work of the smart object has been focused on technical elements (such as software infrastructure, hardware platforms, etc.) and application scenarios. Supply-chain management, corporate applications (for the home and hospital), healthcare, and workplace assistance in industry are just a few examples of application fields. The environment is only beginning to pay attention to smart object technologies' human interface features.

Smartphone

Up to 10 sensors may be found in a contemporary smartphone, which makes it possible to record anything from location to device orientation to lighting conditions. Together, these sensors generate a considerable quantity of data, both organised (like GPS or acceleration data) and unstructured (like images or movies). With the popularity of wearables like the Apple Watch and Android Wear, your smartphone plays an even bigger part. Given the storage and communication capabilities of the smart phone, this new role may be referred to as the "brain" of your Body Area Network (BAN). Last but not least, thanks to technologies like Near Field Communications (NFC), a growing number of smartphones may function as actuators, triggering things like payments or controlling other things like TVs and cars, in addition to being sensors.

Here are the four IoT application areas and where smartphones may or have been utilised in an IoT setup:

We see an increase in the number of fitness and health-related services in the Personal IoT. Then there are a number of devices designed to help with jobs and issues encountered every day, such as Tile, which lets you use your phone to locate things, and NFC rings, which can be used to unlock your phone.

Smart phones are useful for a number of Group IoT applications, including connected automotive applications that use them to control or check system status. The same is true for smart homes, a subject we shall go into more depth about in a later piece.

Smart phones are essential for the Community IoT; for additional role, check the prior blog article on smart cities. For instance, residents may contribute to a smart city using crowd-sourcing applications.

Smart phones are utilised for B2C reasons in the context of Industrial IoT applications, such as giving contextualised and tailored suggestions in real-time and vouchers. Additionally, solutions for client 360-degree views are emerging in the retail sector, and the insurance services sector may profit from employing smart phones and Internet of Things applications

(think: adaptive policies that take place, time or type of travel into account).

I recommend reading through the following sites if you want to learn more about the IoT applications available or maybe want to get ready with regard to tooling:

IoT applications, both current and planned, are listed on [iotlist](#).

An good sandbox environment is provided by the Eclipse Foundation IoT.

It is possible to use a number of development frameworks, including AllJoyn, thethingsystem, or smarthings.

Web APIs and services such as [freeboard.io](#), [wit.ai](#), and [dweet.io](#)

Televisions and the Internet of Things Since they are connected devices, TVs—at least smart TVs, which account for the majority of those sold today—are already a component of the Internet of Things. A wide number of practical applications currently exist for smart connected TVs. For instance, a Smart TV may now be used as a video calling device, allowing families to converse while seeing one other in all their splendour.

Up until now, the major uses of Smart TVs have been for Internet content streaming or showing private images and movies. In internet, more and more people are watching TV online via streaming. Nowadays, a lot of Smart TVs will pick up on your favourite programmes and suggest new programmes you should watch out based on that information. There is no reason why a Smart TV can't also be utilised as a control platform, even if the majority of IoT devices normally use Smartphones and tablets for their controls since such applications are very simple to install.

The Internet of Things in the home has just recently begun, particularly when compared to its potential. It is yet unclear what role the TV will eventually serve in comparison to a computer or a mobile device, but it is possible that it will play a significantly larger role. For instance, all of your home's sensors may be connected and shown on a large screen, saving you from having to goggle at your smartphone. While you are eating breakfast, the TV may converse with you using speech recognition, informing you what it thinks is the best course of action for the day based on the weather, your calendar, your body monitor, etc.

Then there is software that you might utilise on a Smart TV, such as IFTTT (If this, then that). You want to unwind and watch your favourite programmes, but you need to wait for a call or an email from your parents, your work, or someone else. Using IFTTT, you can automatically pause your programme (and start recording it on your PVR) and show the email or connect the important phone call you are waiting for while still allowing

you to continue waiting your favourite shows.

Given the enormous prospects, the connected TV will surely play a significant role in the future of the IoT. Future TVs could have a lot more storage space so they can manage your music and movie collection and/or gather information from your home's sensors. It could also have more powerful processors so that it can control your lighting and temperature systems, monitor your security system, and alert you to any unusual behaviour. Keep an eye out as the Internet of Things develops every day.

The Internet of Things (IoT) technologies are extensively used to transform the physical environment around us into a digital one.

A smart object is equipped with application logic building elements that are sensible for their environment and communicate with users. A smart object can detect, occurrence, interpret, interact with other smart objects, and share information with people. They can also record events that happen to them and in their environment.

The work of the smart object has been focused on technical elements (such as software infrastructure, hardware platforms, etc.) and application scenarios. Supply-chain management, corporate applications (for the home and hospital), healthcare, and workplace assistance in industry are just a few examples of application fields.

The environment is only beginning to pay attention to smart object technologies' human interface features



FIG 2.2 IoT Process With Smart Devices

2.2 The "Things" in the Internet of Things (IOT)

Tractors, thermostats, and shipping containers are just a few examples of the "things" that make up the burgeoning "Internet of Things." The introduction of computers to the internet and the subsequent arrival of smartphones both revolutionised world. Imagine the possibilities once the other 99 percent of objects that are present in our daily lives are connected.



FIG 2.3 IoT for all

You could be knowledgeable about how the internet operates. Additionally, you've definitely heard about how the "Internet of Things," sometimes known as "IoT," is sparking the subsequent industrial revolution. However, you may be wondering what "things" in the Internet of Things really mean. What distinguishes these "things" from, say, an internet-connected computer or smartphone?

What Do the "Things" in the Internet of Things Stand for?

Here's your response: the "things" in the Internet of Things can be anything from farming equipment (a tractor, an irrigation system, a drone) to the thermostat on your wall, to your "connected car," to a freight container with a connected temperature sensor inside it that transmits data to a logistics coordinator.

Agriculture

Healthcare IoT: Our Bodies as Internet-connected "Things"

IoT "things" may also be "wearables" with medical applications. Devices that continuously track your body's systems and functions may serve as an early warning system for health problems and a tool for you to hold yourself responsible for achieving your health objectives.



FIG 2.4 Sensors Magazine

Preventive care may considerably lower hospitalisation and acute care costs using streaming information, according to Lalit Panda in *What Is the Internet of Medical Things (IoMT)?* The major risk of mishaps and misuse that comes with connecting our bodies to the internet is, however, highlighted by both Panda and Kristina Podnar in her article. Our physical bodies would be a "thing" in the Internet of Things for healthcare.

2.3 Iot For Logistics: Packages and Containers as “Things” Connected To The Internet

Moving things from point A to point B is the main focus of supply chain management and logistics. In a perfect world, keeping track of those things isn't really difficult. You know everything will be OK when you transfer cargo X from warehouse Y to client Z, right? Wrong! The world is chaotic, fluid, and dynamic. There are things. Global maritime lines may be impacted by everything from weather catastrophes to piracy, which can lead to delays and monetary losses. Additionally, sometimes losses and delays are just the result of innately present inefficiencies.

A report described the supply chain inefficiencies that contributed to losses of over \$2B in the UK alone in 2018 as "a game of Chinese whispers" as parcels zigzagged across the world. "Supply chain actors are connected—just not to each other or at the correct time," said Ericsson in a recent article on IoT For All.

The Internet of Things can help with that. Cargo containers, vehicles, port utility equipment, and, most significantly, data pipelines and digital ledgers for securely monitoring shipments as they travel the world and change hands are "things" in IoT for logistics and supply chain applications.



FIG 2.5 All Things Supply Chain

IoT for Agriculture: “Things” as Tractors, Soil Sensors and Irrigation

Systems

Tractors, soil sensors, and irrigation systems: IoT for Agriculture We have been modifying how we produce and collect food for thousands of years, whether it is via irrigation, crop rotation, fertilisation, or genetic manipulation. Farms have evolved from being tiny, family- or community-run businesses to huge conglomerates spanning thousands or perhaps millions of acres. In order to satisfy rising needs, farmers have resorted to technology as populations have increased over the 20th century. Farmers continue to come up with creative solutions to satisfy the world's growing supply for food in an interconnected and unpredictable environment in the twenty- first century.

A wide range of agricultural improvements are made possible via IoT. Farmers are beginning to employ IoT, drones, and AI to "improve the quantity and quality of products while optimising the human labour needed by manufacturing," according to SciForce's article on IoT For All.

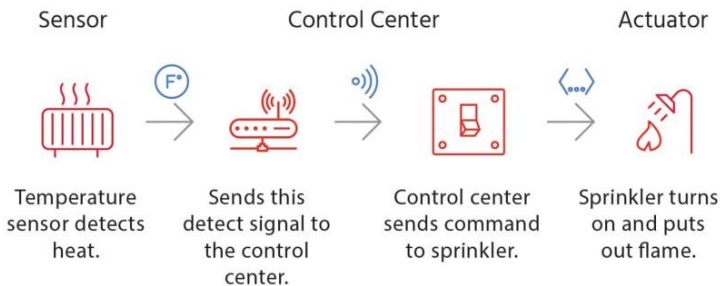
The image below demonstrates how a remote, large-scale soil monitoring system called "CropX" is being used to assist farmers better understand which fields require irrigation when and how soil conditions change over time.



FIG 2.6 Smar zero

Everything was unconnected in the distant past. Then the telephone appeared, allowing us to link people and spaces across distances of hundreds, then thousands. Then, in addition to our voices, we began transmitting papers, photographs, movies, and other types of data across the line. But we couldn't stop with merely connecting things to the internet, such as laptops and smartphones. So why end there? Our current emphasis is on connecting the many additional items and locations we use for daily living and commercial operations to the internet so that we may get information about our surroundings and communicate with our "things" remotely and on a grand scale.

2.4 Sensors and Actuators in Iot | Enabling Industrial\ Automation



Sensor to **Actuator** Flow

FIG 2.7 Sensor Actuator Flow

The new Data Economy greatly benefits from the Internet of Things. When it comes to automation, for instance, the value of an IoT system expands beyond its initial use case. This is due to the fact that an IoT system's intelligence has additional value. IoT data is generated by sensors. Additionally, IoT sensors and actuators may work to allow automation on an industrial scale. Finally, over time, analysis of the data generated by these sensors and actuators might provide insightful business information.

Sensor technology is developing at an ability never before seen, driven by advances in materials science and nanotechnology. As a result, it is becoming more accurate, smaller and less expensive, and able to measure or detect things that weren't before feasible. In fact, in a few years we'll see a trillion new sensors deployed yearly since sensing technology is developing so quickly.

Sensors

A transducer is a more appropriate name for a sensor. Any physical device that converts one kind of energy into another is referred to as a transducer. As a case, for a sensor, the transducer converts a physical phenomena into an electrical impulse that determines the reading. A microphone is a sensor that converts sound waves from vibrational energy to electrical energy in an usable fashion so that other parts of the system may link the electrical energy back to the original sound.

Actuators

Actuator is another type of transducer that is used extensively in IoT systems. Simply said, an actuator works the opposite way from a sensor. It transforms an electrical input into physical movement. Actuators come in a variety of forms, such as electric motors, hydraulic systems, and pneumatic systems.

Controller

A sensor may gather data and send it to a control centre in a conventional Internet of Things system. There, the judgement is governed by previously established rationale. As a result, the perceived input causes a matching instruction to control an actuator. As a result, with the IoT, sensors and actuators collaborate from opposing ends. We'll talk more about the location of the control centre inside the larger IoT system later.

IoT Variety Is Key

In an IoT system, there are several types of sensors. The list of sensors includes flow sensors, temperature sensors, voltage sensors, humidity sensors, and more. There are other methods to measure the same thing as well. One instance to measure airflow is using a little propeller, like the one on a weather station. This approach, however, wouldn't work in a moving car. Vehicles may also measure airflow by heating a tiny element and measuring how quickly it cools.

Different approaches of measuring the same thing are required for different applications. Several actions might be started by the same variable at the same time. As a result, the IoT's sensors and actuators need to work well together.

The Value of Reliable Sensors

Consider that you are the owner of a bar and that you want to measure the amount of beer from one of your taps. Installing a sensor along the line connecting the beer keg to the tap is one line of doing this. Most likely, this sensor would have a little impeller. Like the propeller on a weather station, the impeller would spin as the beer passed by the sensor.

A computer will receive a stream of electrical signals while the impeller rotates. To determine how much beer is pouring through, the computer will interpret the impulses. Sounds easy, doesn't it?

Sensors become interesting at this point. If you read our explanation again, you'll see that we never calculated the amount of beer that was passing through the sensor; instead, we inferred it from a stream of electrical impulses. That implies that we first need to determine how to interpret it.

Sensor Calibration

We would need to use a container with a known carrying capacity, such as a pint glass, to calibrate the sensor. Then, in order to determine what the electrical pulse signal looked like, we would have to fill the container under a number of circumstances. Then, keep an eye on the actuator that controls the other end's flow by turning it on and off.

For instance, the initial pour from a fresh keg could have a tendency to have more froth, which would affect how it reads in comparison to a pour from the centre of the keg that had just beer. We can only be certain that we can interpret the data after many trials and a large amount of data. IoT sensors and actuators can work to automate tasks like filling bottles.

Why Accurate Calibration Is Important

A methodology can guarantee the sensor is reading accurately at all times after this correlation has been found. This is a calibrating procedure. Reputable producers will provide properly calibrated devices and include instructions on how to recalibrate to check sensor accuracy.

Since you will rely mission-critical judgments on subsequent analysis of the data, which will be of little value if the data is incorrect, the quality of sensed data is crucial.

2.5 Classification of Smart Objects

A smart object is one that facilitates communication not just with people but also with other smart objects. They are products, assets, and other things embedded with processors, sensors, software, and connectivity that help in enabling information to be exchanged between the product and its environment, as well as other products and systems. These products are also known as smart connected products or smart connected things (SCoT).

Additionally, connectivity enables part of the product's capabilities to be seen outside of the physical device, in what is known as the product cloud. The information acquired from these products' records may then be examined to help with decision-making, facilitate operational efficiency, and continuously improve the product's overall performance.

Grouping of intelligent objects:

These are the types of "smart" objects: Static or Mobile? Whether the "thing" should move or remain in the same place at all times determines this classification. An additional fact that a sensor is a cell is when it is transferred from one object to another (for fact, a viscosity sensor in a chemical plant that is moved from batch to batch) or when it is connected to an object that is moving (Example, an area sensor on transferring items in a warehouse or manufacturing unit floor).

Additionally, the movement's frequency might change, ranging from infrequent to constant. The potential power source is often determined by the mobility range (from a few inches to miles distant).

High or Low Reporting Frequency -

This classification is mostly based upon how often the object must submit monitored metrics. Additionally, a rust sensor has a reporting time of up to one month. Several hundred times per second, a motion sensor may also detect acceleration.

Higher frequencies need more object consumption, which might limit the available power supply (and hence the mobility of the objects) as well as the transmission range.

Power-Connected or Battery-Powered -

Whether or whether an object includes its own energy source or continuously draws electricity from an external source determines how it is classified. Objects powered by batteries can be transported more easily than things powered by lines.

Riding transmission fluctuates in frequency because batteries limit the amount of power and longevity that the object may use.

Rich or Simple Data?

This classification is entirely based on the volume of records transferred throughout each reporting cycle. While an engine sensor may also record many characteristics, such as temperature to pressure, compression speed, carbon index, etc., a humidity sensor in a specific area may also produce an easy daily index value (on a binary scale from zero to 255). Richer records often result in higher energy use.

The object information throughput is often determined by combining this classification with the previous (low throughput to excessive throughput). A medium-throughput object may also convey successful information at a low frequency instead of a high frequency (in which case the glide shape seems continuous) (in which case the flow shape appears bursty).

Object of Objects Per Cell

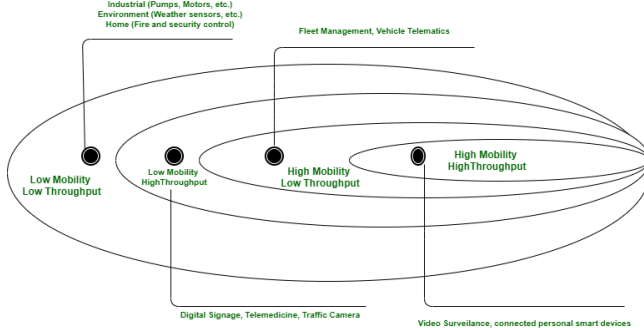
This classification is entirely based on the number of smart objects connected to the same gateway in a specific area that have a similar demand for communication. A single sensor placed strategically every few miles along an oil pipeline is another option.

In contrast, a tiny area is covered by hundreds or even thousands of mirrors in a telescope like the SETI Colossus at the Whipple Observatory, each of which has many gyroscopes, gravity sensors, and vibration sensors.

Reporting Range

This classification is mostly based upon the distance from the gateway. For instance, your fitness band has to be no more than a few metres from your phone in order to speak with it. The idea is that you must be able to view your smartphone in order to consult the records shown on the screen for guidance. If the phone is far away, you probably won't use it, and it's no

longer required to report band data to the phone. On the other hand, a moisture sensor embedded in a street's asphalt may likewise want to speak with its reader hundreds of metres or even miles distant.



An Example of Sensor applications based on mobility and throughput

FIG 2.8 Reporting Range

2.6 Wireless Sensor Network In IoT

An infrastructure-less wireless network called a "wireless sensor network" is used in the Internet of Things to install a large number of wireless sensors that monitor system, physical, and environmental factors. If you're wanting to integrate WSN into your business, our highly driven and experienced engineers can provide you an all-encompassing solution.

Networks Connecting Wireless Sensors

A communication protocol is used to link Sensors that are built into IoT devices. These IoT devices can communicate over great distances thanks to a type of wireless network called an LPWAN, or low-power wide-area network. It is common to utilise a wireless sensor network based on Lora. Data gathered from this sensor network may be delivered to the cloud through cellular networks such as NBIoT, LTE-M or wifi, among others, and sub-1 GHz, Zigbee, Thread, etc. are also used to link sensor networks and gateway.

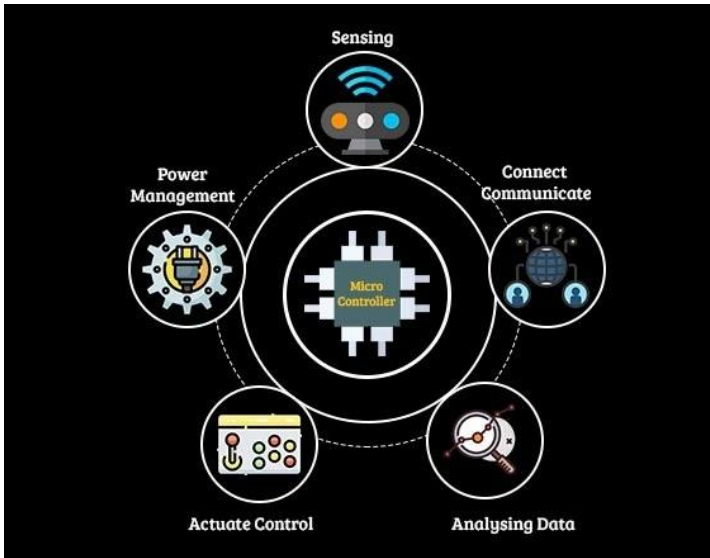


FIG 2.9 Networks Connecting Wireless Sensors

2.7 Communications Criteria

In the present market research trend, executives from a variety of IoT industries are establishing patterns for how and when businesses might begin adopting IoT. IBM, PTC, GE, Microsoft, and Amazon Web Services are among the industry leaders in the Internet of Things platform space at the moment. How do you choose the right platform for your business? Think about these standards:

The ability to build and provide administration for communication from a device to the platform is known as connectivity. Consider the categories of communication that the IoT platforms offer, which may be categorised into the following groups:

Long Range: Devices might communicate over the air using low power radio waves, cell networks, or satellites.

Close Range: WiFi, Bluetooth, Z-Wave, etc.

MQTT, CoAP, and XMPP are specialised protocols.

Security: IoT is always under strain due to security issues and leaks from all directions. There have been countless posts describing how communications are being recorded or how devices are being hacked. A top concern is protecting the lines of communication between the device and the backend. IoT systems are combining security elements such as network

connectivity, authentication, identity management, data loss prevention, and device management to assure protection.

Manageability: Platforms take into account an IoT device's ability to be operated, provisioned, and maintained. A platform will need to have the ability to manage and scale to these devices when there are hundreds or even thousands of devices to manage the connection and incoming data. A crucial component of IoT platform solutions is the ability to monitor, use testing, upgrades, and troubleshooting as a single solution.

Analytics: The ability to analyse and transform sensor data that has been gathered in real time is a rare commodity on today's market. IoT Analytics should provide the tools necessary for users to quickly collect and view this data. Suppliers like IBM Watson and Amazon provide predictive analysis for data being recorded, offering even more knowledge about consumers or the status of assets like machines that have been acquired.

Development: As an IoT platform gets more diverse, the ability to integrate all of its features into a single application for internal or external usage becomes a key factor in determining how to offer analytics and data. A major component of any offering for IoT platforms is the ability to make it simple for developers to create code, create business rules, and manage data for the criteria we reviewed today. A complete solution for IoT may be achieved by combining development tools and API management tools to connect to your internal enterprise applications, such as CA, Mulesoft, Apigee, IBM, or Amazon, via a single platform.

There are several radio technologies including licenced and unlicensed spectrum that are used for IoT access technology.

These access may be divided into two groups at high: Alternative Technologies

Cellular Technologies

An overview of these technologies is provided below.

Parameter	Non-Cellular IoT					Cellular Technologies				
	Bluetooth	Zigbee	LoRa	SigFox	Wifi	GSM	E-GSM IOT Rel-13	LTE-Rel.8	LTE-M Rel. 12	NB-IoT Rel.13
LTE User Equipment cat	NA	NA	NA	NA	NA	NA	NA	Cat. 1	Cat. M1	Cat. NB
Coverage and MAPL	less than 10 m	10 ~ 20 m	Few Km	Few Km	100 m	< 35 Km	< 35 Km	< 100 Km	< 100 Km	< 35 Km
Spectrum	Unlicensed					Licensed				
System Bandwidth	1 MHz	2 MHz	<500 KHz	100 KHz	20 MHz	200 KHz	200 KHz	20 MHz	1.04 MHz	180 KHz
Max Data Rate	20 Mbps	250 Kbps	50 Kbps	100 Kbps	100 MHz	< 500 Kbps	< 140 Kbps	<10 Mbps < 5 Mbps	< 1 Mbps	DL < 170 Kbps UL < 250 Kbps

There are benefits and drawbacks to each of the IoT connectivity technologies that are now available. However, because to the vast range of

technical and commercial IoT connectivity needs, cellular technologies may provide obvious advantages in a wide range of applications. The following requirements should be considered into account while picking technology.

It should be Global in Scope Developed Ecosystem Various and Secure
scalable and supporting QoS
Affordable Total Cost of Ownership (TOC)



Considered all above requirements regarding access network the best option is the Cellular Network Technologies. In terms of global reach, cellular networks already cover 90 percent of the world's population. WCDMA and LTE are catching up, but GSM will offer superior coverage across the Globe. Cellular networks have been developed and deployed over three decades, and they will be around for the foreseeable future. The cellular mobile industry represents a huge and mature ecosystem, incorporating chipset, device and network equipment vendors, operators, application providers and others.

The global cellular ecosystem is governed by the 3GPP standardization forum, which guarantees broad industry support for future development. When it comes to scalability, cellular networks are built to handle massive volumes of mobile broadband traffic; the traffic from most IoT applications will be relatively small and easily absorbed. Operators are able to offer connectivity for IoT applications from the start-up phase and grow this business with low TCO and only limited additional investment and effort.

Cellular connectivity offers the diversity to serve a wide range of applications with varying requirements within a network. QoS mechanisms is essential for many IoT applications and cellular systems have mature QoS functionality, and this enables critical IoT applications to be handled together with traffic from sensors, voice and mobile-broadband traffic on the same carrier.

Traditionally, the security mechanisms of cellular networks have been based on a physical SIM attached to the device, referred to as a Universal Integrated Circuit Card (UICC). This has also enabled roaming between operators, which has been one of the main factors behind the huge success of mobile networks. The SIM will also be essential in future IoT

applications, with SIM functionality embedded in the chipset (eUICC) or handled as a soft-SIM solution running in a trusted run-time environment of the module. One network connecting the whole diversifying IoT market will guarantee the lowest possible TCO as well as fast time to market.

•Consumers' market adoption:•The main consumer devices range from smart phones to tablets and PCs. •The common protocol that links IoT in the consumer space to these devices is IP. •The innovation factor:•IP is the underlying protocol for applications ranging from file transfer and e-mail to the World Wide Web, e-commerce, social networking, mobility, and more Cellular networks currently reach 90% of the world's population on a worldwide scale. Although WCDMA and LTE are coming up, GSM will have better global coverage. Over the last three decades, cellular networks have been created and implemented, and they will continue to exist for some future. The cellular mobile sector is made up of a vast and developed ecosystem that includes operators, application providers, and manufacturers of chipset, device, and network equipment.

The 3GPP standards forum oversees the whole cellular ecosystem, ensuring wide industry support for future development. The traffic from the majority of IoT applications will be very little and readily absorbed since cellular networks are built to manage enormous amounts of mobile broadband traffic. Operators are able to provide connectivity for IoT applications right from the start and expand this business with no extra investment or work and a low TCO.

Within a network, cellular connectivity provides the diversity needed to support a range of applications with various needs. For many Internet of sensors (IoT) applications, quality of traffic (QoS) techniques are essential. Cellular systems have handled QoS capabilities, allowing for the management of key IoT applications alongside phone, mobile broadband, and sensor traffic on the same carrier.

The security measures of cellular networks have historically been based on a physical SIM that is connected to the device, also known as a Universal Integrated Circuit Card (UICC). Additionally, this has made it possible for users to roam between carriers, which has been a major contributor to the enormous success of mobile networks. Future IoT applications will also depend on the SIM, which will either be handled as a soft-SIM solution operating in an essential run-time environment of the module or as SIM functionality incorporated in the chipset (eUICC). The lowest TCO and quickest time to market are guaranteed by a single network that connects the whole, diversifying IoT industry.

Adoption by consumers: The primary consumer electronics are tablets, PCs, and smart phones. IP is the standard protocol that connects consumer IoT to these devices. • IP is the foundational protocol for a wide range of applications, including file transmission and email as well as the World Wide Web, e-commerce, social networking, mobility, and more.

2.9 Connecting all the Things in the Internet of Things

IoT devices may communicate with other IoT devices, cloud-based applications, and services thanks to networking technologies. The internet depends on defined protocols to provide safe and dependable communication between diverse devices. Standard protocols define the networks and formats that devices must follow in order to create, maintain, and transfer data across networks.

A "stack" of technologies is how networks are built. At the bottom of the stack is a technology called Bluetooth LE. Others, such as IPv6 technologies, are located higher up the stack and are in charge of logical device addressing and network traffic routing. Applications that are executing on top of such layers leverage technologies at the top of the stack, such as message queuing technologies.

Additionally, it offers recommendations for selecting one network protocol over another. The following section examines the important factors and difficulties associated with IoT networking, including range, bandwidth, power consumption, sporadic connectivity, interoperability, and security.

CHAPTER 3

IOT LAYERS

3.1 Networking Standards and Technologies In Iot

The Open Systems Interconnection (OSI) model is a stack of seven protocol layers that is an ISO-standard abstract model. They are application, presentation, session, transport, network, data connection, and physical, listed in order of decreasing top. The foundation of the internet is TCP/IP, or the Internet Protocol suite, which offers an abridged concrete implementation of these layers in the OSI model.

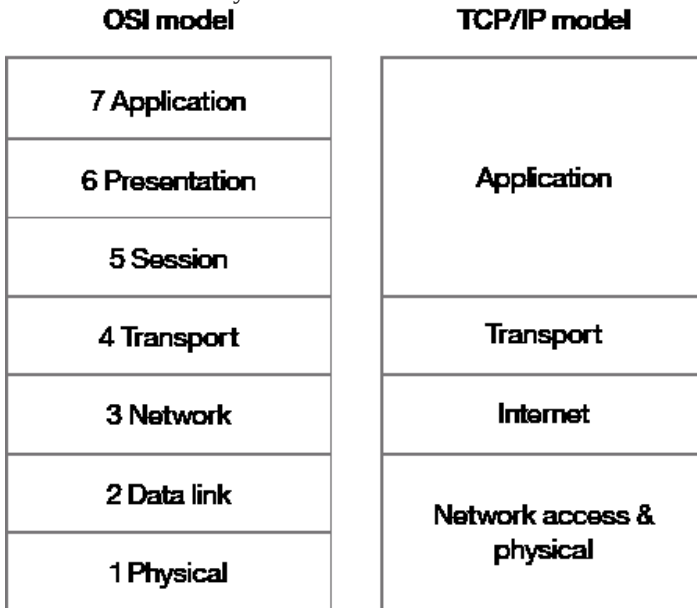


Figure 3.1. Osi And Tcp/Ip Networking Models

Only four layers make up the TCP/IP model, which combines certain OSI model layers:

Physical and Network Access Layer

The OSI layers 1 and 2 are combined into one TCP/IP layer. The physical (PHY) layer (Layer 1 of the OSI model) specifies how each device is physically connected to the network via hardware (such as an optical cable, cables, or radio for wireless networks like IEEE 802.11 a/b/g/n). Devices are identified by MAC addresses at the link layer (Layer 2 of the OSI level), where protocols are concerned with physical addressing, such as

how switches transmit messages to devices on the network. Internet Layer

This layer corresponds to OSI Layer 3. (network layer). The third OSI layer deals with logical addressing. This layer's protocols specify how routers transmit data packets between IP-identified hosts at the source and destination. Commonly used for IoT device addressing is IPv6.

Transport Layer End-to-end communication is the primary emphasis of the transport layer (Layer 4 in the OSI model), which offers characteristics like reliability, congestion avoidance, and the assurance that packets will be delivered in the same order as they were received. Due to its performance, UDP (User Datagram protocol) is frequently used for IoT transmission.

application App Layer

Application-level communications is covered by the application layer (Layers 5, 6, and 7 in the OSI model). An example of an application layer protocol that is often used on the internet is HTTP/S.

Despite the fact that the TCP/IP and OSI models implement you valuable abstractions for talking about networking protocols and the particular technologies that each protocol uses, certain protocols don't fit nicely into these layered models and are unsuitable. For instance, the Transport Layer Security (TLS) protocol, which employs encryption to operate network traffic's privacy and data integrity, may be considered of as operating across OSI layers 4, 5, and 6.

3.2 IoT Networking Protocols

Some of the networking protocols that are widely adopted within IoT and where they fit within the TCP/IP layers are shown in Figure 2.

Figure 2. IoT network protocols mapped to the TCP/IP model

The Internet of Things (IoT) market is seeing the adoption of several new and competing networking technologies. Numerous technologies from various manufacturers provide different implementations of the same standard protocols and are aimed at various vertical markets, such as industrial IoT, healthcare, or home automation. For instance, IEEE 802.15.4, which is implemented by a number of competing technologies like as ZigBee, Z-Wave, EnOcean, SNAP, and

6LoWPAN, specifies the operation of low-rate wireless personal area networks (LR-WPANs).

111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040 1041 1042 1043 1044 1045 1046 1047 1048 1049 1050 1051 1052 1053 1054 1055 1056 1057 1058 1059 1060 1061 1062 1063 1064 1065 1066 1067 1068 1069 1070 1071 1072 1073 1074 1075 1076 1077 1078 1079 1080 1081 1082 1083 1084 1085 1086 1087 1088 1089 1090 1091 1092 1093 1094 1095 1096 1097 1098 1099 1100 1101 1102 1103 1104 1105 1106 110

El 6 6 1 : 11 C 1 : 1

Aims of the study and objectives

A. L. ...

The network technologies listed below each have a short description.

LPWAN

A group of technologies designed (Low Power Wide Area Network) are made for low-power, long-distance wireless communication. They are ideal for widespread IoT deployments of low-power devices like wireless sensors. LPWAN technologies include LTE-M, NB-IoT, Haystack, SigFox, and LoRa (LongRange Physical Layer Protocol) (Narrow-Band IoT).

Cellular

Low-power, low-cost IoT communication methods using current cellular networks are addressed by the LPWAN NB-IoT and LTE-M standards. The newest of these standards, NB-IoT, is focused on long-range communication between a significant number of devices, mostly indoor ones. Although current cellular technologies are frequently used for long-range wireless communication, LTE-M and NB-IoT were adopted expressly for IoT. In addition to 2G (GSM), which is now being phased out in older devices, and CDMA, which is also being retired or phased out, 3G is also being phased out quickly, with numerous network operators retiring all 3G devices. Up until the time when 5G is completely implemented and available, 4G will continue to be in use.

Low Energy Bluetooth (BLE)

The well-known Bluetooth 2.4 GHz wireless communication protocol has a low-power variant called BLE. It is designed for short-range communication (no more than 100 metres), typically in a star arrangement, with one major device controlling several subsidiary devices. Both layers 1 (PHY) and 2 (MAC) of the OSI model are supported by Bluetooth. Devices that transmit brief bursts of small amounts of data are best suited for BLE. When they aren't sending data, devices are designed to sleep and save power. BLE is frequently used by personal IoT devices, such as monitors for fitness and health.

ZigBee

On the 2.4GHz wireless communication frequency, ZigBee runs. Its range is up to 100 metres longer than BLE's. Additionally, compared to BLE, it offers a little lower maximum data rate (250 kbps as opposed to 270 kbps). A mesh network protocol is zigbee. Not all devices can sleep in between bursts, unlike BLE. Their location inside the mesh and whether

they must act as controllers or routers within the mesh will both have a significant impact. ZigBee was designed for applications in home and building automation. The Z- Wave technology, which is likewise based on IEEE 802.15.4, is a near relative of ZigBee. For home automation, Z-Wave was designed.

NFC

In order to hold an NFC card or tag adjacent to a reader, for example, the near field communication (NFC) protocol is used for extremely short range communication (up to 4 cm). NFC is frequently utilised in payment systems, but it is also beneficial in check-in systems and asset monitoring using smart labels.

RFID

Radio Frequency Identification is referred to as RFID. Identifiers and data are stored on RFID tags. An RFID reader reads the tags after they have been affixed to devices. RFID typically has a range of less than one metre. Active, passive, or assisted passive RFID tags are all possible. Since the reader reads the ID passively, passive tags are ideal for battery-free devices. While aided passive tags become active when an RFID reader is available, active tags periodically broadcast their ID. Dash7 is an active RFID-based communication protocol designed for Industrial IoT applications that need secure long-range

communication. Similar to NFC, monitoring inventory goods in retail and industrial IoT applications is a popular use case for RFID.

Wifi

Based on IEEE 802.11a/b/g/n standards, WiFi is a standard for wireless networking. IoT devices could only utilise 802.11b or g in order to save power as 802.11n has the best data speed but also the highest power consumption. Although many prototype and current- generation IoT devices use wifi, it is expected that lower-power alternatives may eventually replace wifi as longer-range and less- powerful options become more accessible.

Ethernet

Ethernet implements the IEEE 802.3 standard for wired connectivity inside local area networks. Not all Internet of Things (IoT) devices have to be wireless. For instance, wired networking technologies like Ethernet may

be used by sensor units that are installed into a building automation system. An alternative hard-wired method called power line communication (PLC) substitutes pre-existing electrical wiring for specialised network lines.

3.3 Introduction of 5g Networks in IoT

The next generation of wireless networks is called 5G. The infrastructure for 4G Long-Term Evolution (LTE) is built. The bandwidth has increased noticeably. But so too are the wireless service's capacity and reliability.

In order to meet the increased data and communication needs brought on by the billions of connected devices that will make up the Internet of Things, 5G is ideal (IoT). Additionally, it helps with the very low latency needed for real-time communications. It is a device-new data pipeline that acts as the plumbing for all signals coming from all Internet-using devices. Dense data can be handled.

The user may download a film, for example, in only 17 seconds thanks to its speed, which is nearly twenty times quicker than 4G. It is a connected development for the world of Things (IoT) and for what it can do in the future.

The introduction and complete implementation of 5G are still in the early stages; shortly, they will be outdated. Rest certain that as data travels in bigger quantities and more quickly than ever, it will improve our world, our lives, and make the benefits of technology to shine with a little bit more brilliance than before.

IoT network technologies at the Internet layer

Technologies in the OSI Layer 3 of the Internet Layer are used to locate and route data packets. This layer contains several commonly used IoT technologies, including IPv6, 6LoWPAN, and RPL.

IPv6

Devices are identified by IP addresses at the Internet layer. As opposed to IPv4, IPv6 is typically used for IoT applications. Compared to the present number of connected IoT devices, IPv4 is restricted to 32-bit addresses, which only supply around 4.3 billion addresses overall. By contrast, IPv6 utilises 128 bits and offers 2¹²⁸ addresses, which amounts to over 3.4 10

38 or 340 billion billion billion billion addresses. IoT devices don't always all need public addresses in real practise. Many of the tens of billions of IoT-connected devices that are anticipated over the next years will be installed in private networks with private address ranges and will

only be able to communicate with other devices or services on external networks through gateways.

6LoWPAN

Using IPv6 over 802.15.4 wireless networks is possible thanks to the IPv6

Low Power Wireless Personal Area Network (6LoWPAN) standard. The Thread protocol for home automation devices also utilises 6LoWPAN, which is frequently used for wireless sensor networks.

RPL

Routing is covered by the Internet Layer. The IPv6 Routing Protocol for Low- Power and Lossy Networks (RPL) is designed to route IPv6 traffic across networks implemented by low power sources, such as 6LoWPAN networks. RPL, which is pronounced "ripple," is designed for packet routing in restricted networks like wireless sensor networks, where not all devices are always available and there is significant or unexpected packet loss. By building a graph of the network's nodes based on dynamic measurements and constraints, such as reducing latency or energy usage, RPL may determine the best route.

IoT Network Technologies at the Application Layer

RESTful HTTP and HTTPS interfaces are commonly used in IoT, while HTTP and HTTPS are widely used throughout internet applications. CoAP (Constrained Application Protocol), which resembles a simplified version of HTTP, is frequently used with 6LoWPAN over UDP. MQTT, AMQP, and XMPP are three messaging protocols that are frequently used in Internet of Things applications:

MQTT

The publish/subscribe-based messaging protocol Message Queue Telemetry Transport (MQTT) was designed for use in low bandwidth scenarios, notably for sensors and mobile devices on unreliable networks.

AMQP

Message-oriented middleware uses the Advanced Message Queuing Protocol (AMQP), an open standard messaging protocol. In particular, RabbitMQ implements AMQP.

XMPP

Extensible Messaging and Presence Protocol (XMPP) was first designed for instant messaging and other forms of real-time human-to-human communication. To implement lightweight middleware and to route XML data for machine-to-machine (M2M) communication, this protocol has been adapted. Smart appliances are where XMPP is most often utilised.

The exact application needs of your IoT project will determine the technologies you use for this layer. For instance, MQTT would be a wise choice for a low-cost home automation system with several sensors since the protocol is easy to implement and lightweight, making it ideal for messaging on devices with limited storage or processing power.

3.4 IoT Networking Considerations and Challenges

Adopt the following limitations in mind when you decide which networking technologies to use for your IoT application:

Range Bandwidth Power use

Sporadic connectivity Interoperability Security

Range

The distances that IoT devices transmitted to a network typically transmit data across may be used to characterise networks:

PAN (Personal Area Network)

PAN has a limited range and allows for measurements in metres, such as a wearable fitness tracker that uses BLE to connect to an app on a mobile device.

LAN (Local Area Network)

LAN refers to short- to medium-range networks, where the distances between devices may reach hundreds of metres or more. Examples include home automation systems or sensors installed within a factory production line that communicate through wifi with a gateway device installed inside the same building.

MAN (Metropolitan Area Network)

MAN is long-range (city-wide), where distances are measured up to a few kilometres, such as smart parking sensors installed all over a city and connected in a mesh network architecture.

WAN (Wide Area Network)

Long-range networks, or WANs, may monitor across distances that can be measured in kilometres. An example of this would be agricultural sensors installed throughout a large farm or ranch to track localised climatic conditions known as microclimates.

The IoT devices should provide data to your network, which should then transmit it to the correct location. An appropriate network protocol must be required for the range. For a WAN application, for instance, that has to operate over a range of many kilometres, avoid using BLE. Consider edge computing if sending data across the required range is difficult. Instead of using data from a far-off data centre or somewhere else, edge computing examines data that is straight from the devices.

Bandwidth

The volume of data that may be sent during a transmitted amount of time is known as the bandwidth. It restricts the amount of data that may be transmitted from IoT devices and sent upstream. There are several factors that impact bandwidth, including:

The volume of data that each device collects and sends

The number of deployed devices

Whether data is being transferred continuously or in short bursts, and whether there are any particularly busy times.

The networking protocol's packet size should correspond to the volume of data that is typically transmitted. Sending packets padded with useless data is inefficient. In contrast, there are costs associated with dividing bigger data amounts across too many little packets. Not all data transmission speeds are symmetrical (that is, upload rates might be slower than download rates). Data transmission must thus be taken into consideration if there is two-way communication between the devices. Because cellular and wireless networks typically have little bandwidth, think carefully before deciding if a wireless technology is the best choice for high-volume applications.

Think about whether or not all raw data has to be transmitted. Reduce the amount of data that is collected by sampling less frequently as one option. As a result, you'll record fewer variables and could filter the device's data to eliminate irrelevant information. The volume of data transmitted is decreased if the data is aggregated before being transmitted. However, the granularity and flexibility of the upstream analysis are impacted by this procedure. Time-sensitive or latency-sensitive data may not always be suited for aggregation or bursting. All of these methods raise the IoT device's storage and data processing needs.

Power use

Power is used when a device transmits data. In comparison to small ranges, long range data transmission demands greater power. A device's power source, such as a battery, solar cell, or capacitor, as well as its whole

lifespan, must be taken into account. A lengthy and robust lifespan will increase reliability and save operating costs. Longer power supply lifecycles may be achieved with the aid of actions. For instance, you may put the device in sleep mode while it is inactive to increase battery life. To make sure that the device's power supply and storage capacity correspond with the power required to transmit the necessary data using the networking technologies you adopted, it's also a recommended practise to model the energy consumption of the device under various loads and network circumstances.

Sporadic connectivity

The Internet of Things (IoT) devices are not always connected. Devices may sometimes be designed with periodic connections in mind. However, sometimes a faulty network may result in devices ceasing to function as a result of connectivity problems. On a wireless network using shared spectrum, there may sometimes be quality of service problems, such as dealing with interference or channel congestion. If intermittent connectivity is a key consideration in the design of the IoT landscape, designs should take that into account and look for any solutions that can guarantee continuous service.

Interoperability

Devices are interoperable; they can operate with other devices, machinery, systems, and technology. Interoperability may be difficult with the IoT since there are so many different types of connected devices. The usual method for preserving interoperability on the Internet has been to adopt standard protocols. Industry participants have established standards that prevent the use of several unique designs and approaches. Interoperability problems may be avoided with appropriate standards and people who agree to them.

However, standardisation procedures often find it difficult to keep up with innovation and change in the IoT. They are created and based in accordance with standards that are currently under development and may change. Take into account the technological ecosystem: Are they often used? Are they proprietary or open? How many different implementations are there?

These planning questions might help you create IoT networks with improved interoperability and greater strength.

Security

Priority one is security. It is essential to use networking technologies that implement end-to-end security, including open port protection,

authentication, and encryption. Access control, message integrity, message secrecy, and replay protection are among the security characteristics offered by the IEEE 802.15.4 security model, which is implemented by technologies based on this standard like ZigBee.

To build a secure and reliable IoT network, take into account the following factors:

Authentication

Adopt authentication for devices, gateways, users, services, and applications by implementing secure protocols. For device authentication, take into account implementing the X.509 standard.

Encryption

Use Wireless Protected Access 2 (WPA2) for wireless network encryption if you're utilising wifi.

A Private Pre-Shared Key (PPSK) strategy is another option. Adopt TLS or Datagram Transport-Layer Security (DTLS), which is based on TLS but adapted for unstable connections that operate over UDP, to secure privacy and data integrity for communication between applications.

TLS protects the integrity of application data by encrypting it.

Port security

Only the ports necessary for communication with the gateway or upstream applications or services are required accessible to external connections thanks to port protection. All other ports need to be closed or firewalled off. Exploiting Universal Plug and Play (UPnP) vulnerabilities may expose device ports. So, the router's UPnP function has to be disabled.

3.5 The Business Case for the Internet of Things



Much of the technology needed for the Internet of Things (IoT) has been available for some time, but connecting a world's worth of devices to

th...The IoT; intelligent systems; surround computing. Whatever you choose to call it, the big buzz in embedded over the last couple of years has been around ubiquitous computing and how connecting edge devices to the cloud will change our perceptions of data. However, once you get beyond the buzzwords, the potential, and even the technology, the IoT is nothing more than a concept unless it addresses the business models and use cases of the enterprise and carrier networks.

In the context of the IoT, both carriers and the enterprise face significant deployment challenges, with businesses having to

consider how to integrate new and existing devices into a “system of systems” approach, and network operators needing to respond to the data traffic produced by billions of sensors. Both must also determine how to move data northbound to the cloud for analysis and then back southbound to execute on gained intelligence, and how to manage it along the way (Figure 1). Until recently, these setbacks were the result of a lack of technology. But with the advent of Software-Defined Networking (SDN), Network Intelligence (NI), and manageability tools, the discussion turns to how companies can incorporate these technologies to use the IoT to their advantage.

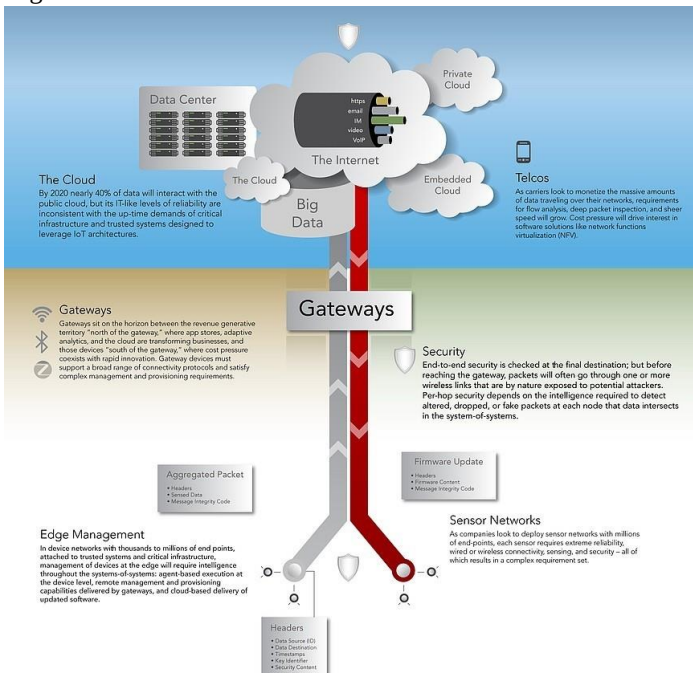


Figure 3.3 : The IoT relies on a North/South concept with multiple components to ensure that data moves from the device to the cloud and back again in a secure and efficient manner. Graphic courtesy Wind River.

“The technology ingredients are there today [for the IoT], but not necessarily the business models and supply chains,” says Gareth

Noyes, Chief Strategy Officer, Wind River. “We are going to need to deploy infrastructure at multiple levels – whether it is at the edge, gateway, sensor hubs, or the cloud – that is scalable, both in terms of performance, price, and other things. One of the reasons why the embedded space is so fragmented is that we are dealing with price points, different business models, different technology demands. That leads to us having to be very scalable in that approach.”

“You have to identify where the pain points are on the carrier side and the enterprise side,” says Jim Douglas, Chief Marketing Officer, Wind River. “The starting point is, ‘what value is the IoT going to create?’ The IoT is a big concept, but if you break it down in its simplest form, it is all about trying to unlock the value of your assets and business processes; trying to make them perform better; trying to make sure they are more sustainable; trying to ensure there is no disruption.”

The business case for connecting “everything”

As mentioned, the business case for the IoT is that it allows carriers and the enterprise to make better use of existing infrastructure. For carrier networks, the IoT presents the opportunity to provide tiered services for data that is more or less critical. This can be achieved through software solutions that enable new revenue streams and also limit additional overhead.

“From the carrier side, the question is not how will the networks be able to handle the IoT, but how the carriers will get paid,” says Douglas. “How do they quickly drive down Operating Expenditures (OPEX) but make money at the same time? It is an issue of effectively accelerating information through data pipes, which is a lot more effective with data acceleration engines. Content inspection also allows you to look for bad actors, as well as implement some flow analysis and traffic shaping. This enables you to see what is going through the pipes, then manage, analyze, and monetize it.”

“If you do not have enough physical bandwidth in your pipe, or the data volume is too high on your device, you are going to need some intelligence to throttle that in an intelligent way,” Noyes explains. “Unless you own all of the infrastructure, you do not have control over the latency

and data paths. Today you are leveraging either public infrastructure or leased infrastructure. That does not necessarily give you the underlining performance that you need.

That is probably going to be a business opportunity for service providers – to provide different classes of infrastructure. You may pay more for something that has low latency or high determinism, and there are services that have adapted to the ‘good-enough’ model – video streaming as an example. On the other hand, in a lot of end- user applications, where you have maybe security or safety-critical applications, ‘good-enough’ is not good enough.

“The concept here is if you are providing different types of traffic over a single IP pipe – if you are transmitting video, monitoring data, tweets, e-mails, and a bunch of things all over one pipe, you are probably going to want to have a better quality of service with protocol data,” Noyes continues. “What the Intelligent Network Platform allows you to do is look at different flows and on a packet- by-packet basis, determine what you want to do with the packet (Figure 2). You can make it high priority or more critical.”

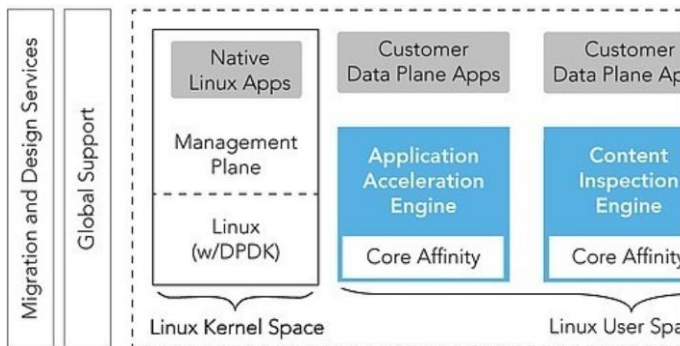


Figure 3.4: The Wind River Intelligent Network Platform integrates application acceleration engines, content inspection, and flow analysis that give Network Service Providers a business incentive for the IoT.

Intelligent networking enables the IoT business case for service providers and, when combined with SDN technology, helps facilitate the northbound/southbound flow of information. On the device side, the IoT will provide enterprise businesses with actionable intelligence about their resources, allowing data to be extracted from the edge, analyzed, and

returned. However, executing on this process requires that intelligence is applied at the bottom layers of the cloud so that information can be retrieved from disparate devices in a secure and digestible fashion. As many existing devices were deployed without regard for the IoT, gateway architectures can be layered in to preclude extensive field upgrades on legacy systems.

“We have devices everywhere, and a lot of these were purpose built to be safe and secure,” Douglas says. “The way they accomplished that was by being isolated from things like enterprise networks where nasty things happen. A lot of high-reliability systems in Aerospace and Defense and carrier networks have taken the network out of the question, and many apps have tried to stay off the Internet. This results in no leverageable data. This data can be used for predictive maintenance for remote repairs, to get information from a lot of devices to reduce OPEX, and for adaptive analytics so you can adjust elements to make everything more optimum.”

“You cannot have a closed, proprietary, non-API for this infrastructure,” Noyes explains. “It is going to be based on interoperable building blocks, standards, compatibility, well-defined structures, and APIs. We have a platform called the Intelligent Device Platform that is architected specifically around an open operative environment that caters for different management protocols, whether it is OM, OSGI, OMDA, and other protocols on the management side (Figure 3). On the communications side, there are things like ZigBee, Bluetooth, and other things. We are already architecting programs like the Intelligent Device Platform that have the building blocks, and we can fully expect to incorporate others. That is on the edge side facing down to the devices.”

3.6 Network Optimization For Iot Devices

IoT is experiencing rapid development, which is leading to challenges for network operators. Learn how IoT device intelligence can help overcome these challenges.

The internet of things (IoT) is developing at a rapid pace. Gartner forecasts that, in the enterprise and automotive IoT market, there will be 5.8 billion

IoT endpoints in use by the end of 2020. This represents a 21% increase from 2019, which was already up 21.5% compared to 2018.

Additionally, Statista notes that there will be an estimated 50 billion IoT connected devices in use around the world by 2030.

The vast number of devices and rapid adoption rate demonstrates that businesses are increasingly leveraging the opportunities enabled by IoT. This is resulting in increased challenges for network operators and service providers.

Network Optimization Challenge

With the rate of growth and diversity of connected devices, the challenge for operators is to understand the implications for their networks to help ensure optimal performance. Operators need deeper insight into the nature and capabilities of the terminals (e.g. cameras, data collection terminals, wireless hotspots, etc.).

Many of the new classes of terminal have very different behaviours on the network compared with the traditional voice and data capable cell phone. As a result, operators need to be able to reliably segment traffic to understand the patterns and trends of network usage. This is important for many reasons, including overcoming the following specific challenges.

Example problems that need to be solved:

1. Capacity has to be reserved at a cell tower level for IoT devices. Operators must be able to measure IoT activity levels on a per-cell basis, to assess whether the capacity reservations are optimal, as both under-reserving and over-reserving cause inefficiencies;
2. Rate plan misuse. Since different data rate plans are provided for IoT devices and consumer devices, the operator needs to be able to monitor whether the device in use is aligned with the plan; and
3. Limited understanding of device capabilities on a network. Limited insight into subscriber device capabilities, and their changes over time, makes planning network enhancements, to meet demand, more difficult. Enabling Network Optimization With Enriched Device Intelligence

Device Map, developed in partnership with the GSMA, is the authoritative source of device intelligence for the mobile ecosystem.

Device Map combines the GSMA issued TAC with detailed device capabilities to deliver rich IoT and non-IoT device insights, that enable mobile network operators and related service providers to make more informed business decisions.

IoT Device Intelligence

As mentioned, operators need deeper insight into the nature and capabilities of terminals. Device Map provides this.

There are now over 20 device classifications in Device Map, providing full granularity into diverse traffic sources such as geolocation trackers, embedded network modules, digital home assistants, meter hubs and security hubs, along with the above-stated cameras, data collection terminals, and wireless hotspots.

Additionally, Device Map includes a set of properties to classify IoT devices. These property details are as follows:

Property Name	Property Description	Example
IoT Endpoint	Sensor and/or actuator, with cellular connectivity	
Camera	Data Collection Terminal, Geolocation tracker	
IoT Enabler	Provides cellular connectivity to otherwise unconnected devices	Embedded network module, Modem
IoT Controller	Data receiver and aggregator, command/control of remote and/or local IoT devices	Digital home assistant

To further support the optimization of networks for IoT devices, Device Map also includes the following protocol properties:

Property Name	Property Description
NB-IoT	The device is capable of connecting to the specific type of narrow- band Low Power Wide Area Network (LPWAN) using radio technology standard specified in 3GPP Release 13.
LTE-M	The device is capable of machine – to –machine communication through connecting to the specific type of Low Power Wide Area Network (LPWAN) using radio technology standard specified in 3GPP Release 13.
NB-IoT and LTE-M	are communication protocols designed to support IoT applications. Their lower data bandwidth and low power consumption are more appropriate for IoT applications. NB-IoT tends to be used for very low data volume applications, such as simple sensors that may make hourly or daily updates. Whereas, LTE-M supports larger data volumes, appropriate for continuous monitoring or surveillance.

To leverage and enable IoT opportunities, operators need to make NB-IoT and/or LTE-M support decisions. Many factors can impact support decisions, such as if an operator already has LTE networks - LTE-M may be lower cost to roll out, but is less efficient for very low data volume applications. Additionally, device capabilities on a network (specifically, the level of devices that are NB-IoT and LTE-M capable) and the change over time are considerable factors.

Device Map enables operators to understand whether devices on their network support these technologies, which can have significant benefits.

Benefits Of These Properties

Operators and relevant service providers can leverage the above Device Map properties to overcome many network optimization challenges. Specific benefits of the properties include:

- ❖ Simplified analysis and comparison of traffic types, to permit accurate capacity allocation in the network;
- ❖ Facilitates measurement of trends of IoT traffic, including levels of rate plan misuse;
- ❖ Understand whether devices on a network support specific technologies and, based on the scale and trends, plan network enhancements accordingly; and
- ❖ Permits distinguishing between traffic which is 100% IoT (IoT Endpoints) and traffic which may represent a mix of IoT and non-IoT traffic (IoT Enablers).

Distinguish Between Consumer Devices And M2M Devices

Consumer behaviour on a network is very different to IoT device behaviour. Therefore, how operators choose to optimize their networks for each type of behaviour will likely differ.

For cases where the operator need is to distinguish between consumer devices and M2M devices, additional Device Map properties can be leveraged. Specifically, “Primary Hardware Type” can be used. Examples of Primary Hardware Type include device types such as Mobile Phone, Desktop and Games Console—typically used by consumers—in addition to Data Collection Terminal and Embedded Network Module—common in M2M systems.

3.7 Transport Layer Protocols

Transport is the OSI Level 4 layer and is recognized by the same name in the TCP-IP model. Like physical layer, data link layer and network layer, this layer is also part of the infrastructure layer in IOT reference architecture. In the previous tutorials, physical and data link layer protocols and network layer protocols were discussed.

To learn about physical and data link layer protocols commonly used for computer and mobile devices networking, check out the following tutorial – Physical and Data Link Layer Protocols for Computers and Mobile Devices – Ethernet BLE Wi-Fi Wi-Fi Direct WPA

To learn about physical and data link layer protocols developed for LPWAN, check out the following tutorial –

<https://www.engineersgarage.com/articles/part-5physical-and-data-link-layer-protocols-lpwan>

Physical and Data Link Layer Protocols for LPWAN

To learn about physical and data link layer protocols developed for PAN, HAN and LAN, check out the following tutorial –

Physical and Data Link Layer Protocols for LAN HAN and PAN

To learn about RFID and Mobile standards with applications in IOT, check out the following tutorial –

Physical and Data Link Layer Protocols – RFID and Mobile Standards

To learn about various network layer protocols, check out the following tutorial –

3.8 Network Layer Protocols

In this tutorial, transport layer protocols will be discussed. Transport layer which is also known as routing layer is responsible for routing of data packets over a network. At this layer, ordering of packets, error detection and correction in delivery of data packets is performed. The commonly used transport layer protocols include the following –

- ❖ TCP
- ❖ UDP
- ❖ DCCP
- ❖ SCTP
- ❖ RSVP
- ❖ DTLS
- ❖ TLS
- ❖ RPL
- ❖ CARP
- ❖ CORPL
- ❖ QUIC
- ❖ uIP
- ❖ ROLL
- ❖ Aeron
- ❖ CCN (Content Centric Networking)
- ❖ NanoIP
- ❖ TSMP (Time Synchronized Mesh Protocol)

TCP – Transmission Control Protocol (TCP) is connection oriented and heavyweight protocol. It is suitable for reliable communication because

in this protocol acknowledgment is received when the client sends the packet to the server via TCP protocol. The data must be guaranteed sent at the other end if the packet is sent via TCP protocol.

The protocol operates in three phases – Connection establishment, data transfer and connection close. A TCP connection is managed by an internet socket which lying at the end point (physical) undergoes various state changes.

The packet overhead is very large in this protocol. TCP consumes more power from the devices and has large overhead so it is not suitable for low power devices with constrained environment. Therefore, UDP is preferred as a connection less protocol for low overhead.

UDP – User Datagram Protocol is a connection less protocol and is not reliable for guaranteed transmission of data. However, the UDP protocol is a best protocol to send data to the server when packet loss during transmission of the data can be afforded. UDP protocol is a lightweight protocol and is suitable for wireless sensor network communication. UDP is often used in applications specially tuned for real-time performance.

DCCP – Datagram Congestion Control Protocol (DCCP) is a message-oriented transport layer protocol. This protocol is more secure than TCP protocol. Compared to TCP which has a single byte long ID for each packet, the packet ID is 48-bit long (6-bytes) in DCCP. This makes it hard for any attacker to hack data packets. This protocol is generally used for time critical data transfers like media streaming and VoIP.

SCTP – Stream Control Transmission Protocol (SCTP) protocol is a message oriented transport layer protocol and it uses congestion control to reliably transfer data over a network. The data transferred along with a 12-byte header is secured using 4-way handshake. Due to multi-homing, data is reliably sent to the destination.

RSVP – Resource Reservation Protocol (RSVP) is a control protocol mainly used for unicast and multicast data transmission. It actually does not transport data but is used to ensure quality of service (QoS) for data streams.

DTLS – Based on Transport Layer Security (TLS) protocol, Datagram Transport Layer Security (DTLS) is a stream oriented transport layer protocol. This is a security protocol designed against message forgery, tampering and eavesdropping. However, large packet size, packet reordering and loss of datagram are some of the major drawbacks of this protocol stack.

TLS – Transport Layer Security (TLS) is a security protocol which uses symmetric cryptography to secure data. This protocol has been now prohibited from use due to security considerations.

RPL – IPv6 Routing Protocol for Low Power and Lossy Networks is a transport layer protocol developed for 6LoWPAN networks. This protocol stack has been specially designed for constraint networks. The protocol has been designed handle unknown packet losses and minimize energy consumption by the network. The protocol uses a graph of nodes (Destination Oriented Directed Acyclic Graph, DODAG) to route data through the most optimum path, so that data is delivered with minimum latency. It is also designed to tackle situations where the destination end-point is not reachable or not available the time packet is routed. Despite that when the end-point wakes up, it manages to deliver the data packets.

A node can join the network by sending DODAG Information Solicitation (DIS) request to the root device which responds by a DAO Acknowledgment (DAO-ACK) confirming the inclusion of the node in the network. It is the root which determines how to route data through various nodes and deliver it to the destination node or end point.

Not only for 6LoWPAN, the RPL (or Ripple) can be used along with various other low power data link layer protocols. It is based on IPv6 standard, so it is quite suitable for any IOT application.

CARP – Channel-Aware Routing Protocol (CARP) is a transport layer protocol developed for underwater networks. This protocol is quite suitable for IOT implementations as header size in this protocol is very small. The protocol keeps track of data communication history to select nodes for data transfer. The focus in this protocol remains on building quality links through which data packets can be reliably and securely delivered. The data transfer happens in two stages – Network Initialization and Data Forwarding. In network initialization phase, a sink node (node requesting sensor data) sends a HELLO packet to all the nodes of the network. Then, in the data forwarding phase, the data from the sensors is routed to the sink node from one hop to another hop where each hop route data independently. This protocol is designed especially for sensor networks.

CORPL – Cognitive RPL (CORPL) is an extension of RPL protocol developed for cognitive networks. It is also based on Destination Oriented Directed Acyclic Graph (DODAG) topology with few modifications to suit the cognitive networks. In CORPL, there is opportunistic forwarding. There are multiple forwarders (called forwarder set) from which best node to forward data can be selected. Not only the parent node, instead each node

maintains a forwarder set. The nodes communicate changes in their forwarder set by sending Destination Advertisement Object (DAO) messages.

QUIC – Quick UDP Interconnections (QUIC) is a protocol stack based on User Datagram Protocol with additional features that make this protocol as secured as TLS protocol. It uses a set of multiplexed connections between the source and destination such that there is minimum latency and data losses.

uIP – It is an open-source TCP-IP protocol stack which can be used with tiny 8-bit and 16-bit microcontrollers.

ROLL – Routing Over Low power and Lossy networks (ROLL) is an IPv6 routing protocol based on RPL designed for Low power and Lossy Networks.

Aeron – Aeron is a protocol stack designed for UDP unicast and UDP multicast. It is a high throughput and low latency communication protocol generally used for streaming data.

CCN – Content Centric Networking (CCN) or Information Centric Networking (ICN) is a network architecture and protocol stack designed for content distribution. In this protocol stack, the data is delivered as named packets which are saved in memories wherever available in the network. By automatic and application neutral caching of content or information, the protocol removes the need of any application layer protocol or service to deliver data.

NanoIP – Based on TCP-IP, NanoIP is an underdevelopment protocol stack for sensors and embedded devices.

TSMP – Time Synchronized Mesh Protocol (TSMP) is a protocol stack developed for motes or sensor nodes. It is used for networking of wireless sensors so that sensors could communicate data to each other in timeslots. It has an approach similar to time division multiplexing.

In the next tutorial, service delivery protocols will be discussed. Common application layer protocols in IoT explained

IoT doesn't exist without connectivity, but those connections would not be possible without the right application layer protocols in IoT.

IoT is simple in theory: Use the vast connectivity of the internet to send data from the tiniest devices -- such as switches and sensors -- to the cloud, on-premises server farms or large-scale data platforms to model the world more accurately. That model forms the foundation for entire new classes of applications, analyses and previously unimagined efficiencies and optimizations.

The global growth in IoT-based applications makes the cloud look tame, as far as technological innovations and social importance go. IoT has given the world automated industrial operations, personal healthcare monitoring, traffic control and smart homes. The list is very long and growing.

However, for all the available connectivity and the technological revolution IoT promotes, it presents communication challenges rarely seen in more conventional networking scenarios.

3.9 IoT Communication Challenges

Application layer protocols in IoT must address some major challenges in IoT's network communications. The data in IoT are more complex than in conventional networking. In a simple scenario, a sensor reports a temperature to a local network, which then passes the data to a cloud through the internet. Complex scenarios can get as tricky as dialog between multiple machines on a factory floor that selectively power down when environmental sensors report the detection of fire, while a cloud AI system monitors the shutdown and reports the event to multiple external systems. The simple scenario includes a single data type and a single message moving in one direction. The complex one includes many data types in many messages among many devices that must efficiently exchange the data. The hardware itself can vary wildly.

The conventional protocol standard, HTTP, isn't necessarily appropriate for small devices. Some IoT devices are HTTP-capable and benefit from the features of HTTP transport, but many, if not most, are not and cannot. HTTP is designed for very heavy, rich communications, while most IoT communications are necessarily minimal, given the small size and sparse hardware and firmware of most devices. An HTTP header can contain hundreds of bytes; that's nothing in most internet communications, but it's huge for most IoT devices. Protocols that are leaner and able to ride HTTP but not be HTTP are necessary.

How the Application Layer Works

Some IoT devices have network capability directly built in, but most connect to the internet through a gateway.

IoT makes use of the Open Systems Interconnection model, which employs seven interactive layers to handle each phase of message transport. The IoT application layer is one of seven layers, three of which are of particular importance to IoT application protocols.

The physical network layer is the actual infrastructure that is moving the data, such as Ethernet, Wi-Fi or Bluetooth.

The transport layer is the protocol supporting the movement of the data, such as Transmission Control Protocol (TCP), HTTP or User Datagram Protocol (UDP).

The application layer is the interface between the IoT device and the network with which it will communicate. It handles data formatting and presentation and serves as the bridge between what the IoT device is doing and the network handoff of the data it produces. Top application layer protocols in IoT

Engineers have many IoT application layer protocols to choose from, covering a broad range of functionality. The right protocol for a particular

IoT application depends on a number of factors based on the type of device involved and the function it will perform:

- ❖ Data latency. How fast does data transport need to be? How much time can reasonably pass for a data packet to move from one point to another?
- ❖ Reliability. How consequential is data loss in the IoT application? How redundant must device communication be?
- ❖ Bandwidth. What's the volume of data that needs to be accommodated?
- ❖ Transport. What's the best transport protocol for the IoT application? TCP, UDP and HTTP each offer features that can be exploited by application layer protocols with which they're compatible.

Engineers must answer these questions before considering specific protocols to pin down which features will be most important.

The application layer handles data formatting and presentation and serves as the bridge between what the IoT device is doing and the network handoff of the data it produces.

Five of the top protocols and their features that matter most to IoT are: Constrained Application Protocol. Organizations use CoAP with limited

Hardware that has a low transmission rate because it's lightweight. The protocol is HTTP-friendly, using two basic message types: request and response. Messages can be confirmable or non-confirmable. Data packets are small, so message losses are few. The disadvantage is the protocol lacks security, which engineers normally can remedy with datagram transport layer security, but DTLS is of limited use in IoT.

Message Queue Telemetry Transport. MQTT is a publish /subscribe protocol, which is effective for lightweight machine-to-machine (M2M) connectivity over TCP and minimizes data loss. Publish/subscribe for IoT means clients don't have to call for updates, which lessens network traffic and processing load. The protocol also accommodates a range of quality enforcement levels, from single handshake delivery to acknowledgment requirement.

Extensible Message and Presence Protocol. XMPP is based on XML, the document-encoding markup language popular for its human-friendly readability. As an HTML extension, XMPP is useful for real-time communication, including presence, content syndication and instant messaging. The protocol is highly scalable and offers devices data-bearing nodes that can connect with other nodes by request to create complex local networking and data-sharing.

Advanced Message Queuing Protocol. AMQP is an asynchronous protocol. Like MQTT, it uses a publish/subscribe approach. Engineers use the protocol primarily over TCP, but it is also capable of other transport methods. The protocol is quality-flexible and offers an optional one-or-more delivery guarantee. AMQP implements security via Transport Layer Security and Secure Sockets Layer.

Representational State Transfer. REST is the most ubiquitous protocol and gives IoT synchronous request-response via HTTP. HTTP makes it feature- rich, as well as capable of authentication and caching, both of which are useful in complex environments, although difficult to implement in IoT. The protocol is both XML- and JSON-compatible, which is useful for M2M and communication with tablets and smartphones, a boon for IoT.

Accelerating IoT Transformation

IoT is the primary driving force of digital transformation and it has complex technology requirements. It includes a cascade network architecture, each network consists of massive number of electronic devices, and each device has a different design objective. Optimizing your network and delivering the required capacity and coverage for each IoT use case is a challenging task. P.I. Works solution helps you tackle this challenge.

Understanding IoT Use Cases and Changing Network Requirements

Analysys Mason predicts that the total number of IoT connections worldwide will grow at a CAGR of 22% between the end of 2018 and 2028, and will reach 5.3 billion in 2028. Consequently, the number of

challenges will mount proportionally, outpacing humans' abilities to deal with them and reinforcing the need for more software and automation solutions. While traditional wireless networks have been built into a range of one-size-fits-all options utilizing 2G, 3G, LTE and 5G with limited cross-system integration, IoT networks are being designed around a broad spectrum of use cases.

Each IoT use case has its own requirements. In some cases you need high bandwidth for applications such as real-time surveillance. In some other cases you need low bandwidth with many handovers like in vehicle tracking. For example, while remote measurement devices require lower bandwidth, strong indoor coverage and battery lifetime, auto guided vehicles look for ultra low latency and high bandwidth connection.

P.I. Works' service-aware EXA and SmartPlan products drive network efficiency by delivering conformal coverage, mobility robustness and signaling along with the latency reduction and self-healing functions in IoT enabled mobile networks.

3.10 Providing Solid Iot Quality of Experience

Increasing radio network complexity and emerging new services bring new network design and optimization challenges. Massive number of simultaneously connected IoT devices used in remote measurement, smart houses, medical, transportation industries increase signaling loads and radio interference levels. This requires extensive and continuous QoS monitoring and optimization for the specific services delivered. P.I. Works service-aware network monitoring and optimization platform helps operators reach IoT SLAs by ensuring solid service quality as expected from each service type like including ultra-low latency, wide-band throughput and deep indoor coverage.

P.I. Works EXA addresses IoT challenges through signaling load mitigation, interference reduction, mobility performance improvement, and power consumption optimization with coverage boosting activities. P.I. Works EXA offers the following capabilities.

Battery Life Time Improvement and Coverage Continuity Assurance: Besides real time optimization of defined parameters, EXA configures the capacity and coverage optimization policies and monitors the network performance.

Recommended modules include CCO (Coverage and Capacity Optimization), RSI (Rach Index Optimization), ACO (Area Code Optimization), AIC (Automated Inconsistency Correction), CEM enabled VantageNQ for performance monitoring and SLA alarms.

Signaling Reduction and Interference Mitigation for CapEx Savings: The massive increase in the number of connections along with the bursty traffic behavior of IoT devices cause more packet collisions in uplink during the network access. Cell Random access channel optimization with RSI and PCI optimizers, coverage overlap reduction with CCO, location update traffic reduction with ACO (Area Code Optimization) brings not only infrastructure efficiencies by reducing the need for new carriers in radio network and minimizing CPU loads on core network nodes but also allows operators to achieve improved IoT service quality (e.g., battery life time, solid accessibility).

Recommended Modules: CCO (Coverage and Capacity Optimization), RSI (Rach Index Optimization), PCI (Physical Cell Identity), ACO (Area Code Optimization).

3.11 Application Protocols For Iot

One of the major aspect of the IoT is the data exchange between Sensors, Servers, Databases, Business Engines and Content Consumers. While there are many protocols available to enable this communication between different classes of endpoints, there are some protocols which are designed keeping IoT specific requirements in mind. Let's explore some of such protocols which are becoming more and more popular in IoT Applications.

1. Message Queuing Telemetry Transport (MQTT):

MQTT or Message Queuing Telemetry Transport is a publish-subscribe based lightweight messaging protocol. It is best suited for Internet of Things (IoT) and Machine to Machine (M2M) communication because of its small code footprint, lower bandwidth requirements and lower power consumption on device.

- ❖ Key Highlights:
- ❖ Publish/Subscribe Architecture
- ❖ Light Weight (Min Header Size: 2 Bytes)
- ❖ TCP/IP Based
- ❖ QoS Support
- ❖ Payload Agnostic
- ❖ Uses SSL/TLS for Security
- ❖ Broker could be Single Point of Failure

2. Advanced Message Queuing Protocol (AMQP):

The Advanced Message Queuing Protocol (AMQP) is an open standard for passing business messages between applications. AMQP has low success rate at low bandwidths, but it increases as bandwidth increases. However, comparing AMQP with REST, AMQP can send a larger amount of messages per second.

- ❖ Key Highlights:
- ❖ Point to Point Message Exchange (Server to Server)
- ❖ Flexible Messaging Patterns
- ❖ TCP/IP Based
- ❖ Smallest Packet Size: 60 Bytes
- ❖ QoS Support
- ❖ Uses SSL/TLS for Security

3. Constrained Application Protocol (CoAP):

The Constrained Application Protocol (CoAP) is a specialized web transfer protocol for use with constrained nodes and constrained networks in the IoT. It runs over UDP (no TCP overhead) to keep the overall implementation lightweight and uses the HTTP commands like GET, POST, PUT, and DELETE to provide resource-oriented interactions.

- ❖ Key Highlights:
- ❖ Request/Response Architecture
- ❖ UDP/IP Based
- ❖ QoS Support
- ❖ Supports Unicast as well as Multicast
- ❖ Uses DTLS for Security
- ❖ Supports Resource Discovery
- ❖ CoAP Node plays Server Role too (NAT Issues)
- ❖ Decentralised (No Single Point of Failure)

4. Extensible Messaging and Presence Protocol (XMPP):

XMPP is the Extensible Messaging and Presence Protocol, a set of open technologies for instant messaging, presence, multi-party chat, voice and video calls, collaboration, lightweight middleware, content syndication, and generalized routing of XML data.

- ❖ Key Highlights:
- ❖ Publish/Subscribe and Request/Response Architecture
- ❖ Widely used for Instant Messaging, Presence, Voice/Video Calls etc.

- ❖ No QoS Support
- ❖ Text Based Communication (XML Payload)
- ❖ Secure Authentication (SASL) and TLS Based Security
- ❖ XML Payload creation and parsing may need extra compute and battery power.

5. Very Simple Control Protocol (VSCP):

VSCP is an open source standard protocol for M2M, IoT and other remote control and measurement applications. It enables simple, low-cost devices to be networked together with high-end computers and/or to work as an autonomous system, whatever the communication media is.

- ❖ Key Highlights:
- ❖ Event Based Architecture
- ❖ Leverages RS-323, CAN and/or TCP/IP
- ❖ Support for Device Discovery, Identification and Configuration
- ❖ Support for Secure Device firmware update
- ❖ A solution from sensor to UI

6. Data Distribution Service (DDS):

The Data Distribution Service (DDS) is a middleware protocol and API standard for data-centric connectivity from the Object Management Group. It integrates the components of a system together, providing low-latency data connectivity, extreme reliability, and a scalable architecture that business and mission-critical Internet of Things (IoT) applications need.

- ❖ Key Highlights:
- ❖ Publish/Subscribe and Request/Response Architecture
- ❖ Relational data modeling
- ❖ QoS Support
- ❖ Multicast support over plain UDP sockets
- ❖ Leverages both TCP/IP and UDP/IP Transport
- ❖ Uses TLS and DTLS for Security
- ❖ Decentralised (No Single Point of Failure)

7. Restful Http (REST):

The Representational State Transfer (REST) is not really a protocol but an architectural style. REST uses the HTTP methods GET, POST, PUT, and DELETE to provide a resource oriented messaging system where all actions can be performed simply by using the synchronous request/response

HTTP commands.

- ❖ Key Highlights:
- ❖ Request/Response Architecture
- ❖ TCP/IP Based
- ❖ No QoS Support
- ❖ Complex Implementation at Client Side
- ❖ Larger Header compared to other IoT Protocols (Higher Bandwidth Requirement)
- ❖ Uses SSL/TLS for Security

8. OMA LightweightM2M (LWM2M):

LightweightM2M is primarily a device management protocol, but it should be designed to be able to extend to meet the requirements of applications. LightweightM2M is not restricted to device management, it should be able transfer service / application data.

CHAPTER 4

DATA ANALYTICS AND SUPPORTING SERVICES

4.1 Introduction

Traditional data management systems are simply unprepared for the demands of what has come to be known as “big data.” As discussed throughout this book, the real value of IoT is not just in connecting things but rather in the data produced by those things, the new services you can enable via those connected things, and the business insights that the data can reveal. However, to be useful, the data needs to be handled organized and controlled. Thus, a new approach to data analytics is needed for the Internet of Things.

In IoT, the creation of massive amounts of data from sensors is common and one of the biggest challenges—not only from a transport perspective but also from a data management standpoint. A great example of the deluge of data that IoT can generate is in the commercial aviation industry and the deployed sensors throughout an aircraft. Modern jet engines are fitted with thousands of sensors that generate 10GB of data per second.

For example, modern jet engines, similar to the one shown in Figure 1, may be equipped with around 5000 sensors. Therefore, a twin engine commercial aircraft with these engines operating on average 8 hours a day will generate over 500 TB of data daily, and this is just the data from the engines! Aircraft today have thousands of other sensors connected to the airframe and other systems. A single wing of a modern jumbo jet is equipped with 10,000 sensors.

The potential for a petabyte (PB) of data per day per commercial airplane is not farfetched—and this is just for one airplane. Across the world, there are approximately 100,000 commercial flights per day. The amount of IoT data coming from the commercial airline business is overwhelming. This example is one of many highlighting the big data problem that IoT exacerbates. Analyzing this amount of data in the most efficient manner possible falls under the umbrella of data analytics. Data analytics must offer actionable insights and knowledge from data, no matter the amount or style, on time, or the full benefits of IoT cannot be realized.

Before diving deeper into data analytics, it is important to define a few key concepts related to data. Not all data is the same; it can be categorized

and thus analyzed in different ways. Depending on how data is categorized, various data analytics tools and processing methods can be applied. Two important categorizations from an IoT perspective are whether the data is structured or unstructured and whether it is in motion or at rest.

4.1.1 Structured VS Unstructured Data

Structured and unstructured data are important classifications as they typically require different toolsets from a data analytics perspective. Figure 2 provides a high-level comparison of structured data and unstructured data.

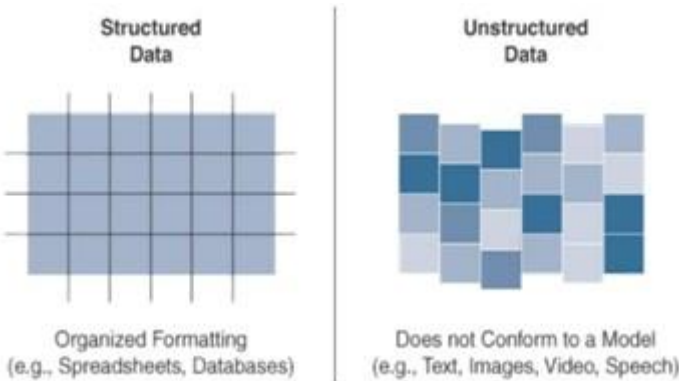


FIG 4.1 Structured Vs Unstructured Data

Structured data follows a model or schema that defines how the data is represented or organized. It fits well with a traditional relational database management system (RDBMS). In many cases you will find structured data in a simple tabular form—for example, a spreadsheet where data occupies a specific cell and can be explicitly defined and referenced. Structured data can be found in most computing systems and includes everything from banking transaction and invoices to computer log files and router configurations. IoT sensor data often uses structured values, such as temperature, pressure, humidity, and so on, which are all sent in a known format. Structured data is easily formatted, stored, queried, and processed; for these reasons, it has been the core type of data used for making business decisions. Because of the highly organizational format of structured data, many data analytics tools are readily available for processing this type of data. From custom scripts to commercial software like Microsoft Excel and Tableau, most people are familiar and comfortable with working with structured data. Unstructured data lacks a logical schema for understanding

and decoding the data through traditional programming. Examples of this data type include text, speech, images, and video. As a general rule, any data that does not fit neatly into a predefined data model is classified as unstructured data. According to some estimates, around 80% of a business's data is unstructured. Because of this fact, data analytics methods applied to unstructured data, such as cognitive computing and machine learning, are deservedly garnering much attention. You can decode speech with machine learning applications, such as natural language processing (NLP). With image/facial recognition applications, you can extract critical information from still images and video.

The handling of unstructured IoT data employing machine learning techniques is covered in more depth later.

Smart objects in IoT networks generate both structured and unstructured data. Structured data is more easily managed and processed due to its well-defined organization. On the other hand, unstructured data can be harder to deal with and typically requires very different analytics tools for processing the data. Being familiar with both of these data classifications is important because knowing which data classification you are working with integrates with the appropriate data analytics solution.

Data in Motion Versus Data at Rest

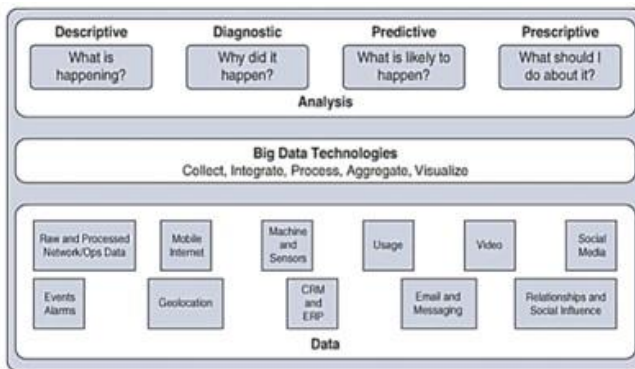


FIG 4.2 Data In Motion Versus Data At Rest

As in most networks, data in IoT networks is either in transit (“data in motion”) or being held or stored (“data at rest”). Examples of data in motion include traditional client/server exchanges, such as web browsing, file transfers, and email. Data saved to a hard drive, storage array, or USB drive is data at rest. From an IoT perspective, the data from smart objects is considered data in motion as it passes through the network en route to its

final destination. This is often processed at the edge, using fog computing. When data is processed at the edge, it may be filtered and deleted or forwarded on for further processing and possible storage at a fog node or in the data center. Data does not come to rest at the edge.

When data arrives at the data center, it is possible to process it in real-time, just like at the edge, while still in motion. Tools with this sort of capability, such as Spark, Storm, and Flink, are relatively nascent compared to the tools for analyzing stored data. Later sections of this chapter provide more information on these real-time streaming analysis tools that are part of the Hadoop ecosystem. Data at rest in IoT networks can be typically found in IoT brokers or some storage array at the data center. Myriad tools, especially tools for structured data in relational databases, are available from a data analytics perspective. The best known of these tools is Hadoop. Hadoop not only helps with data processing but also data storage. It is discussed in more detail later.

Descriptive: Descriptive data analysis tells you what is happening, either now or in the past. For example, a thermometer in a truck engine reports temperature values every second. From a descriptive analysis perspective, you can pull this data at any moment to gain insight into the current operating condition of the truck engine. If the temperature value is too high, there may be a cooling problem or the engine experiencing too much load.

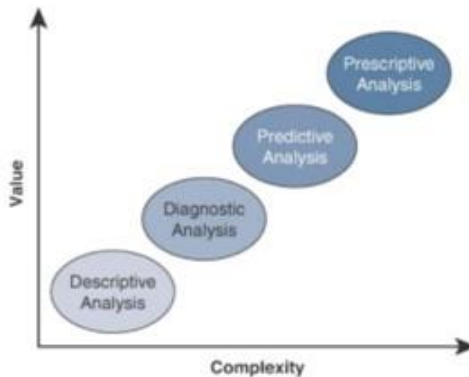


FIG 4.3 Prescriptive

Diagnostic: When you are interested in the “why,” diagnostic data analysis can provide the answer. Continuing with the example of the temperature sensor in the truck engine, you might wonder why the truck

engine failed. Diagnostic analysis might show that the engine's temperature was too high and overheated. Applying diagnostic analysis across the data generated by a wide range of smart objects can provide a clear picture of why a problem or an event occurred.

Predictive: Predictive analysis aims to foretell problems or issues before they occur. For example, with historical values of temperatures for the truck engine, predictive analysis could estimate the remaining life of certain components in the engine. These components could then be proactively replaced before failure occurs. Or perhaps if temperature values of the truck engine start to rise slowly over time, this could indicate the need for an oil change or some other sort of engine cooling maintenance.

Prescriptive: Prescriptive analysis goes beyond predictive and recommends solutions for upcoming problems. A prescriptive analysis of the temperature data from a truck engine might calculate various alternatives to cost-effectively maintain our truck. These calculations could range from the cost necessary for more frequent oil changes and cooling maintenance to installing new cooling equipment or upgrading to a lease on a model with a more powerful engine. Prescriptive analysis looks at a variety of factors and makes the appropriate recommendation.

4.2 IoT Data Analytics Challenges

As IoT has grown and evolved, it has become clear that traditional data analytics solutions were not always adequate. For example, traditional data analytics typically employs a standard RDBMS and corresponding tools. However, the world of IoT is much more demanding. While relational databases are still used for certain data types and applications, they often struggle with the nature of IoT data. IoT data places two specific challenges on a relational database:

Scaling problems: Due to the large number of smart objects in most IoT networks that continually send data, relational databases can grow incredibly quickly. This can result in performance issues that can be costly to resolve, often requiring more hardware and architecture changes.

Volatility of data: The schema must be designed correctly with relational databases. Changing it later can slow or stop the database from operating. Due to the lack of flexibility, revisions to the schema must be kept minimum. IoT data, however, is volatile in the sense that the data model is likely to change and evolve. A dynamic schema is often required to make daily or hourly data model changes.

A different type of database, known as NoSQL, is being used to deal with scaling and data volatility challenges. Structured Query Language (SQL) is the computer language used to communicate with an RDBMS. As the name implies, a NoSQL database is a database that does not use SQL. It is not set up in the traditional tabular form of a relational database. NoSQL databases do not enforce a strict schema, and they support a complex, evolving data model. These databases are also inherently much more scalable.

In addition to the relational database challenges that IoT imposes, with its high volume of smart object data that frequently changes, IoT also brings challenges with the live streaming nature of its data and with managing data at the network level. Streaming data, which is generated as smart objects transmit data, is challenging because it is usually of a very high volume. It is valuable only if it is possible to analyze and respond to it in real-time. Real-time analysis of streaming data allows you to detect patterns or anomalies that could indicate a problem or a situation that needs some kind of immediate response. To have a chance of affecting the outcome of this problem, you naturally must be able to filter and analyze the data while it is occurring, as close to the edge as possible. The market for analyzing streaming data in real-time is growing fast.

Major cloud analytics providers, such as Google, Microsoft, and IBM, have streaming analytics offerings. Various other applications can be used in house. (Edge streaming analytics is discussed in depth later in this chapter.) Another challenge that IoT brings to analytics is network data, referred to as network analytics. With the large numbers of smart objects in IoT networks that communicate and stream data, it can be challenging to ensure that these data flows are effectively managed, monitored, and secure.

4.3 Network analytics tools IN IoT

Network analytics tools such as Flexible NetFlow and IPFIX can detect irregular patterns or other problems in the flow of IoT data through a network. Network analytics, including Flexible NetFlow and IPFIX, is covered more detail.

Data acquiring

Having learnt about devices, devices-network data, messages and packet communication to the Internet, let us understand the functions required for applications, services and business processes at application-

support and application layers. These functions are data acquiring, data storage, data transactions, analytics, results visualisations, IoT applications integration, services, processes, intelligence, knowledge discovery and knowledge management.

First, let us discuss the following terms and their meanings in IoT application layers.

Application refers to application software or a collection of software components. An application enables a user to perform a group of coordinated activities, functions and tasks. Streetlights control and monitoring is an example of an application. Software for tracking and inventory control are other examples of applications. Tracking applications use tags and locations data of the RFIDs.

An application enables a user to withdraw cash using an Automatic Teller Machine (ATM). An umbrella sending warning messages for weather (Example 1.1), a waste container management, health monitoring, traffic lights control, synchronisation and monitoring are other examples of IoT applications. Service denotes a mechanism, which enables the provisioning of access to one or more capabilities. An interface for the service provides the access to capabilities. The access to each capability is consistent with constraints and policies, which a service-description specifies. Examples of service capabilities are automotive maintenance or service capabilities for the Automatic Chocolate Vending Machines (ACVMs) for timely filling of chocolates into the machines.

Service consists of a set of related software components and their functionalities. The set is reused for one or more purposes. Usage of the set is consistent with the controls, constraints, and policies specified in the service description for each service. A service also associates a Service Level Agreement (SLA).

A service consists of a collection of self-contained, distinct and reusable components. It provides logically grouped and encapsulated functionalities. Traffic lights synchronising service, automobile maintenance service, devices location, detection and tracking service, home security-breach detection and management service, waste containers substitution service, and health-alerts service are the examples of IoT services.

Service Oriented Architecture (SOA) is a software architecture model consisting of services, messages, operations, and processes. SOA components are distributed over a network or the Internet in a high-level business entity. New business applications and applications integration architecture in an enterprise can be developed using an SOA.

Message Means a Communicating Entity or Object.

Operation means action or set of actions. For example, actions during a bank transaction. Transaction (trans + action) refers to two inter-related sets of operations or actions or instructions. For example, a transaction may access sales data to select and get the annual sales in a specific year. One operation is access to sales data and other is annual sales in return. Another example of a transaction is a query transaction with a Database Management System (DBMS).

Query is a command for getting select values from a database that transfers the answer to the query after processing. A query example is a command to the ACVMs database to provide sales data of ACVMs on Sundays near city gardens in a specific festival period in a year. Another example is a query to the service center database to provide the list of automobile components needing replacement that have completed expected service-life in a specific vehicle.

Query Processing is a group of structured activities undertaken to get the results from a data store as per the query Key Value Pair (KVP) refers to two linked entities. One is the key, which is a unique identifier for a linked entity. The other is the value, which is either the identified entity or a pointer to the location of that entity. A KVP example is birthday-date pair. KVP is birthday: July 17, 2000. Birthday is the key for a table and date July 17, 2000 is the value. KVP applications create the look-up tables, hash tables and the network or device configuration files.

Hash Table (also called hash map) refers to a data structure that maps the KVPs and implements an associative array (for example, array of KVPs). A hash table may use an index (key) computed using a hash function and key maps to the value. Index is used to get or point to the desired value.

Bigtable maps two arbitrary string values into an associated arbitrary byte array. One is used as row key and the other as column key. Time stamp associates in three-dimensional mapping. Mapping is unlike a relational database but can be considered a sparse, distributed multi-dimensional sorted map. The table can scale up to 100s to 1000s of distributed computing nodes with ease of adding more nodes.

In database theory, business transaction (BT) refers to a (business) process that requests information from or changes the data in a database. One operation in a BT is a command 'connect' that connects a DBMS and database, which also connects with the DBMS. Similarly, BTs are processes using commands 'insert', 'delete', 'append', and 'modify'.

Process means a composition of a group of structured activities or tasks that lead to a particular goal (or that interact to achieve a result). For example, streetlights control the purchase process for an airline ticket. A process specifies activities with relevance rules based on data in the process.

Process Matrix is a multi-element entity. Each element relates a set of data or inputs to an activity (or subset of activities).

Business Process (BP) is an activity or series of activities or a collection of inter-related structured activities, tasks or processes. A BP serves a particular goal or specific result or service or product. The BP is a representation or process matrix or flowchart of a sequence of activities with interleaving decision points; interleaving means putting in between. Decision point means an instance in a series of activities when decisions are taken for further activities.

A web definition states, “a BP is a specific event in a chain of structured business activities that typically change the state of data and/or a product and generate some type of output”. Examples of BPs include finding the annual sales growth and managing the supplies. Another definition² of BP is that “business process is an activity or set of activities that will accomplish a specific organizational goal”. One more definition³ of BP states, “BP is a series of logically related activities or tasks (such as planning, production, or sales) performed together to produce a defined set of results.”

4.4 Business Intelligence

Business Intelligence (BI) is a process which enables a business service to extract new facts and knowledge, and then undertake better decisions. These new facts and knowledge follow from earlier data processing results, aggregation, and analysis.

Data Acquiring

1. Data Generation

Data generates at devices that transfer to the Internet through a gateway later on.

Data generates as follows:

Passive devices data: Data generate at the device or system, following the result of interactions. A passive device does not have its power source. An external source helps such a device to generate and send data. Examples are an RFID or an ATM debit card. The device may or may not have an

associated microcontroller, memory and transceiver. A contactless card is an example of the former and a label or barcode is the example of the latter.

Active devices data: Data generates at the device or system or following the result of interactions. An active device has its power source. Examples are active RFID, streetlight sensor or wireless sensor node. An active device also has an associated microcontroller, memory and transceiver.

Event data: A device can generate data on an event only once. For example, on detection of the traffic or on dark ambient conditions, which signals the event. The event on darkness communicates a need for lighting up a group of streetlights (Example 1.2). A system consisting of security cameras can generate data on an event of security breach or detection of an intrusion. A waste container with associate circuit can generate data in the event of getting it filled up 90% or above. The components and devices generate data of their performance and functioning. For example, a play in steering wheel and reduced air-conditioning is felt on wearing out of a brake lining. The data communicates to the Internet. The communication takes place when the automobile reaches near a Wi-Fi access point.

Device real-time data: An ATM generates data and communicates it to the server instantaneously through the Internet. This initiates and enables Online Transactions Processing (OLTP) in real time.

Event-driven device data: A device data can generate on an event only once. Examples are: (i) a device receives command from Controller or Monitor and acts (s) using an actuator. When the action completes, the device sends an acknowledgement; (ii) when an application seeks the status of a device, then the device communicates the status.

Data Acquisition

Data acquisition means acquiring data from IoT or M2M devices. The data communicates after the interactions with a data acquisition system (application). The application interacts and communicates with many devices for acquiring the needed data. The devices send data on demand or at programmed intervals. Data of devices communicate using the network, transport and security layers.

An application can configure the devices for the data when devices have configuration capability. For example, the system can configure devices to send data at defined periodic intervals. Each device configuration controls the frequency of data generation. For example, system can configure an umbrella device to acquire weather data from the

Internet weather service, once each working day in a week. An ACVM can be configured to communicate machine and other information sales data every hour. The ACVM system can be configured to communicate instantaneously in case of fault or if a specific chocolate flavour needs the Fill service.

Application can configure sending of data after filtering or enriching at the gateway at the data-adaptation layer. The gateway in-between application and the devices can provision for one or more of the following functions— transcoding, data management and device management. Data management may be provisioning of the privacy and security, and data integration, compaction and fusion

Device-management software provisions for device ID or address, activation, configuring (managing device parameters and settings), registering, deregistering, attaching, and detaching

Data validation

Data acquired from the devices does not mean that data are correct, meaningful or consistent. Data consistency means within expected range data or as per pattern or data not corrupted during transmission. Therefore, data needs validation checks. Data validation software do the validation checks on the acquired data. Validation software applies logic, rules and semantic annotations. The applications or services depend on valid data. Then only the analytics, predictions, prescriptions, diagnosis and decisions can be acceptable.

Large data is acquired from a large number of devices, especially, from machines in industrial plants or embedded components data from many automobiles or health devices in ICUs or wireless sensor networks, and so on. Validation software, therefore, consumes significant resources. An appropriate strategy needs to be adopted. For example, the adopted strategy may be filtering out the invalid data at the gateway or device itself or controlling the frequency of acquiring or cyclically scheduling the set of devices in industrial systems. Data enriches, aggregates, fuses or compacts at the adaptation layer.

Data Categorization for Storage

Services, business processes and business intelligence use data. Valid, useful and relevant data can be categorized into three categories for storage—data alone, data and results of processing, only the results of data analytics are stored. Following are three cases for storage:

Data that needs to be repeatedly processed, referenced, or audited in the future, data alone needs to be stored. Data needs processing only once, and the results are used later using the analytics. Both the data and results of processing and analytics are stored. Advantages of this case are quick visualization and reports generation without reprocessing. Also the data is available for reference or auditing in future.

Online, real-time or streaming data need to be processed and the results of this processing and analysis need storage.

Data from many devices and sources is categorized into a fourth category called Big data. Data is stored in databases at a server, in a data warehouse, or on a Cloud as Big data.

4.5 Assembly Software for the Events

A device can generate events. For example, a sensor can generate an event when temperature reaches a preset value or falls below a threshold. A pressure sensor in a boiler generates an event when pressure exceeds a critical value which warrants attention. Each event can be assigned an ID. A logic value sets or resets for an event state. Logic 1 refers to an event generated but not yet acted upon. Logic 0 refers to an event generated and acted upon or not yet generated. A software component in applications can assemble the events (logic value, event ID and device ID) and add the date stamp. Events from IoTs and logic-flows assemble using software.

Data Store

A data store is a data repository of a set of objects which integrate into the store.

Features of data store are:

Objects in a data-store are modeled using Classes which are defined by the database schemas

A data store is a general concept. It includes data repositories such as database, relational database, flat file, spreadsheet, mail server, web server, directory services and VMware

A data store may be distributed over multiple nodes. Apache Cassandra is an example of distributed data store.

A data store may consist of multiple schemas or data in only one scheme. Example of only one scheme data store is a relational database.

Repository in English means a group related to looking for required things, for special information or knowledge. For example, a repository of paintings of artists. A database is a repository of data which can be relied

upon for reporting, analytics, process, knowledge discovery and intelligence. A flat file is another repository.

Data Centre Management

A data centre is a facility which has multiple banks of computers, servers, large memory systems, high speed network and Internet connectivity. The centre provides data security and protection using advanced tools, full data backups, recovery, redundant data communication connections, full system power, and electricity supply backups.

Large industrial units, banks, railways, airlines and units for whom data are the critical components use the services of data centres. Data centres also possess a dust free, heating, ventilation and air conditioning (HVAC), cooling, humidification and dehumidification equipment, pressurisation system with a physically highly secure environment. The data center manager is responsible for all technical and IT issues, operations of computers and servers, data entries, data security, data quality control, network quality control and the management of the services and applications used for data processing.

Server Management

Server management means managing services, setup and maintenance of systems of all types associated with the server. A server needs to serve around the clock. Server management includes managing the following:

- Short reaction times when the system or network is down

- High security standards by routinely performing system maintenance and updation

- Periodic system updates for state-of-the art setups

- Optimised performance

- Monitoring of all critical services, with SMS and email notifications

- Security of systems and protection

- Maintaining confidentiality and privacy of data

- High degree of security and integrity and effective protection of data, files and databases at the organisation Protection of customer data or enterprise internal documents by attackers which includes spam mails, unauthorised use of the access to the server, viruses, malwares and worms

- Strict documentation and audit of all activities.

9. Spatial Storage

Consider goods with RFID tags. When goods move from one place to another, the IDs of goods and locations are needed in tracking or inventory control applications. Spatial storage is a spatial database that is optimised to store and receives queries from the applications. Suppose a digital map is required for parking slots in a city. Spatial data refers to data which represents objects defined in a geometric space. Points, lines and polygons are common geometric objects represented in spatial databases. Spatial database can also represent database for 3D objects, topological coverage, linear networks, triangular irregular networks and other complex structures. Additional functionality in spatial databases enables efficient processing. Internet communication by RFIDs, ATMs, vehicles, ambulances, traffic lights, streetlights, waste containers are examples of where spatial database are used.

Spatial database functions optimally for spatial queries. A spatial database can perform typical SQL queries, such as select statements, and various spatial operations. Spatial database has the following features:

- ❖ Can perform geometry constructors. For example, creating new geometries
- ❖ Can define a shape using the vertices (points or nodes)
- ❖ Can perform observer functions using queries which replies specific spatial information such as location of the centre of a geometric object
- ❖ Can perform spatial measurements which mean computing distance between geometries, lengths of lines, areas of polygons and other parameters
- ❖ Can change the existing features to new ones using spatial functions and predicate spatial relationships between geometries using true or false type queries.

4.6 Computing Using A Cloud Platform For Iot/M2m Applications/Services

A few conventional methods for data collection and storage are as follows:

Saving devices' data at a local server for the device nodes

Communicating and saving the devices' data in the files locally on removable media, such as micro SD cards and computer hard disks

Communicating and saving the data and results of computations in a dedicated data store or coordinating node locally

Communicating and saving data at a local node, which is a part of a distributed DBMS Communicating and saving at a remote node in the distributed DBMS

Communicating on the Internet and saving at a data store in a web or enterprise serverCommunicating on the Internet and saving at data centre for an enterprise

Cloud is a new generation method for data collection, storage and computing. cloud computing paradigm for data collection, storage, computing and services. describes cloud-computing service models in a software architectural concept, 'everything as a service'.Describes IoT specific cloud based services, Xively, Nimbits. describes platforms such as AWS IoT, Cisco IoT, IOx and Fog, IBM IoT Foundation, TCS Connected Universe (TCS CUP).

Different methods of data collection, storage and computing are shown in Figure 4.4 The figure shows (i) Devices or sensor networks data collection at the device web server, (ii) Local files, (iii) Dedicated data store at coordinating node, (iii) Local node in a distributed DBMS, (iv) Internet- connected data centre, (v) Internet-connected server, Internet-connected distributed DBMS nodes, and (vii) Cloud infrastructure and services.

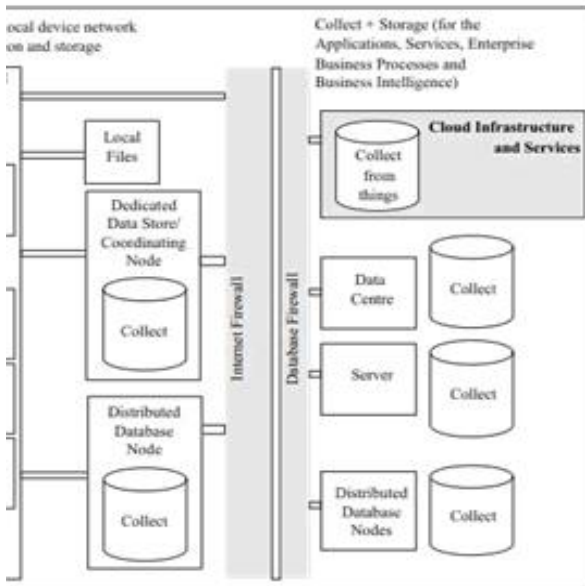


FIG 4.4 Cloud Platform For Iot/M2m

Cloud computing paradigm is a great evolution in Information and Communications Technology (ICT). The new paradigm uses XAAS at the Internet connected clouds for collection, storage and computing.

The key terms and their meanings need to be understood before learning about the cloud computing platform.

Resource refers to one that can be read (used), written (created or changed) or executed (processed). A path specification is also a resource. The resource is atomic (not-further divisible) information, which is usable during computations. A resource may have multiple instances or just a single instance. The data point, pointer, data, object, data store or method can also be a resource.

Devices or sensors network data collection at a device local-server, local files, dedicated data store, at a coordinating node, a local node of a distributed DBMS, Internet-connected server of data centre, server or distributed database nodes or a cloud infrastructure

System resource refers to an operating system (OS), memory, network, server, software or application. Environment refers to an environment for programming, program execution or both. For example, cloud9 online provides an open programming environment for BeagleBone board for developing IoT devices; Windows environment for programming and execution of applications; Google App Engine environment for creating and executing web applications in Python or Java. Platform denotes the basic hardware, operating system and network, and is used for software applications or services over which programs can be run or developed.

A platform may provide a browser and APIs which can be used as a base on which other applications can be run or developed.

Edge computing is a type of computing that pushes the frontier of computing applications, data and services away from centralised nodes to IoT data generating nodes, that means at logical extremes of the network. Events push 2 IoT device nodes, triggers, alerts, messages and data is collected for enrichment, storage and computations from the remote centralised database nodes. Pushing the computations from centralised nodes enables the usage of resources at device nodes, which could be a requirement in case of low power lossy networks. The processing can also be classified as edge computing at local cloud, grid or mesh computing. The nodes may be mobile or of a wireless sensor network or cooperative distributed in peer-to-peer and ad-hoc networks.

Distributed computing refers to computing and using resources distributed at multiple computing environments over the Internet. The

resources are logically-related, which means communicating among themselves using message passing and transparency concepts. They cooperate, movable without affecting the computations, and can be considered one (location independent) computing system.

Service is a software which provides the capabilities and logically grouped and encapsulated functionalities. An application calls a service for utilising the capabilities. A service has a description and discovery methods, such as advertisement for direct use or through a service broker. The service binds to Service Level Agreement (SLA) between service (provider end point) and application (end point). One service can also use another service.

According to the W3C definition, Web Service is an application identified by a URI, described and discovered using the XML based Web-Service Description Language (WSDL). A web service interacts with other services and applications using XML messages and exchanges the objects using Internet protocols.

Service-oriented architecture consists of components implemented as independent services that can be dynamically bonded and orchestrated and possess loosely coupled configurations. At the same time, the communication between them uses messages. Orchestrating means a process which predefines an order of calling the services (in sequences and parallel) and the data and message exchanges.

Web computing refers to computing using resources at computing environment of web server(s) or web services over the Internet.

Grid computing refers to computing using the pooled interconnected grid of computing resources and environments in place of web servers.

Utility computing refers to computing using focus on service levels with optimum amount of resources allotted when required and takes the help of pooled resources and environments for hosting applications. The applications utilise the services.

Cloud computing refers to computing using a collection of services available over the Internet that deliver computational functionality on a service provider's infrastructure for connected systems and enables distributed grid and utility computing.

Key Performance Indicator (KPI) refers to values consisting of one or more raw monitored values, including minimum, average, and maximum values specifying the scale. A service is expected to be fast, reliable and secure. The KPIs monitor the fulfillment of these objectives. For example, a set of values can relate to Quality of Service (quality of service)

characteristics, such as bandwidth availability, data backup capability, peak and average workload handling capacity, ability to handle defined volume of demand at different times of the day, and the ability to deliver defined total volume of service. A cloud service should be able to fulfill the defined minimum, average and maximum KPI values agreed in the SLA

Localisation means cloud computing content usage is monitored by determining localisation of the quality of service level and KPIs.

Seamless cloud computing means during computing the content usages and computations continue without any break when the service usage moves to a location with similar quality of service level and KPIs. For example, continue using same cloud platform when developer of software shifts.

Elasticity denotes that an application can deploy local and remote applications or services and release them after the application usage. The user incurs the costs as per the usages and KPIs.

Measurability (of a resource or service) can be measured for controlling or monitoring and enables report of the delivery of resource or service.

Homogeneity of different computing nodes in a cluster or clusters refers to integration with the kernel providing the automatic migration of the processes from one to other homogeneous nodes. System software on each computing node should ensure same storage representation and same results of processing

Resilient computing refers to offering and maintaining the accepted quality of service and KPIs in the presence of the identified challenges, defined and appropriate resilience metrics, and protecting the service.⁴ The challenges may be small to big, such as misconfiguration of a computing node in a network to natural disasters. An English Cambridge dictionary defines resilience as the power or ability to return to the original form, position, etc., after being bent, compressed or stretched.

Scalability in cloud services refers to how an application can deploy smaller local resources and remotely distributed servers and resources, increasing or decreasing the usage while incurring the cost as per the usage on increasing scales.

Maintainability in cloud services refers to the storage, applications, computing infrastructure, services, data centres and servers maintenances which are responsibilities of the remotely connected cloud services with no costs to the user.

XAAS is a software architectural concept that enables deployment and development of applications, and offers services using web and SOA. A computing paradigm is to integrate complex applications and services (Section 4.3.5) and use XAAS concept for deploying a cloud platform

Multitenant cloud model refers to accessibility to a cloud platform and computing environment by multiple users who pay as per the agreed quality of service and KPIs, defined at separate SLAs with each user. The users do resource pooling but each user pays separately. The following subsections describe the cloud computing paradigm and deployment models.

4.7 Cloud Computing Paradigm

Cloud computing means a collection of services available over the Internet. Cloud delivers the computational functionality. Cloud computing deploys infrastructure of a cloud-service provider. The infrastructure deploys on a utility, grid computing, or webservices environment that includes network, system, or grid of computers or servers or data centres. We—electricity users—do not need to know about the source and underlying infrastructure for electricity supply service.

Similarly, a computing service or application user need not know how the infrastructure deploys or the details of the computing environment. Just as the user does not need to know Intel processor inside a computer, similarly, the user uses the data, computing and intelligence in the cloud, as part of the services. Similarly, the services are used as a utility at the cloud.

4.8 Cloud Platform Services

Cloud platform offers the following:

- ❖ infrastructure for large data storage of devices, RFIDs, industrial plant machines, automobiles and device networks
- Computing capabilities, such as analytics, IDE (Integrated Development Environment)
- Collaborative computing and data store sharing

4.8.1 Cloud Platform Usages

Cloud platform usages connect devices, data, APIs, applications and services, persons, enterprises, businesses and XAAS.

The following describes a simple conceptual framework of the Internet Cloud:

Internet Cloud + Clients = User applications and services with ‘no boundaries and no walls’

An application or service executes on a platform which includes the operating system (OS), hardware and network. Multiple applications may initially be designed to run on diversified platforms (OSs, hardware and networks). Applications and services need to integrate them on a common platform and running environment. Cloud storage and computing environment offers a virtualized environment, which refers to a running environment made to appear as one to all applications and services. However, physically two or more running environments and platforms may be present.

Virtualization

A characteristic of virtualized environment is that it enables applications and services to execute in an independent execution environment (heterogeneous computing environment). Each one of them stores and executes in isolation on the same platform. It may execute or access a set of data centers or servers or distributed services and computing systems. The applications or services which are hosted remotely and are accessible using the Internet can easily be deployed at a user application or service in a virtualized environment, provided the Internet or other communications are present.

Applications need not be aware of the platform, just Internet connectivity to the platform, called cloud platform, is required. The storage is called cloud storage. The computing is called cloud computing. The services are called cloud services in line with the web services which host on web servers.

Virtualization of storage means user application or service accesses physical storage using abstract database interface or file system or logical drive or disk drive. Storage may be accessible using multiple interfaces or servers. For example, Apple iCloud offers storage to a user or user group that enables sharing albums, music, videos, data store, editing files, and collaboration among the user group members.

Network Function Virtualisation (NFV) means a user application or service accesses the resources appearing as just one network. However, the network access to the resources may be through multiple resources and networks. Virtualisation of server means user application accesses not only one server but accesses multiple servers.

Virtualised desktop means the user application can change and deploy multiple desktops, though the access by the user is through their computer platform (OS) that in fact may be through multiple OSs and platforms or remote computers

4.8.2 Cloud Computing Features and Advantages

Essential features of cloud storage and computing are:

On demand self-service to users for the provision of storage, computing servers, software delivery and server time

Resource pooling in multi-tenant model

Broad network accessibility in virtualised environment to heterogeneous users, clients, systems and devices Elasticity

Massive scale availability Scalability Maintainability Homogeneity Virtualisation Interconnectivity platform with virtualised environment for enterprises and provisioning of in-between Service Level Agreements (SLAs) Resilient computing Advanced security Low cost

4.8.3 Cloud Computing Concerns

Concerns in usage of cloud computing are:

Requirement of a constant high-speed Internet connection

Limitations of the services available

Possible data loss

Non delivery as per defined SLA specified performance

Different APIs and protocols used at different clouds

Security in multi-tenant environment needs high trust and low risks

Loss of users' control

4.9 Cloud Deployment Models

Following are the four cloud deployment models:

Public cloud: This model is provisioned by educational institutions, industries, government institutions, businesses, or enterprises and is open for public use.

Private cloud: This model is exclusive for institutions, industries, businesses or enterprises and is meant for private use in the organisation by the employees and associated users only.

Community cloud: This model is exclusive for use by a community formed by institutions, industries, businesses or enterprises, and for use within the community organisation, employees and associated users. The community specifies security and compliance considerations.

Hybrid cloud: A set of two or more distinct clouds (public, private or community) with distinct data stores and applications that bind to deploy the proprietary or standard technology.

Cloud platform architecture is a virtualised network architecture consisting of a cluster of connected servers over the data centres and Service Level Agreements (SLAs) between them. A cloud platform

controls and manages resources, and dynamically provisions the networks, servers and storage. Cloud platform applications and network services are utility, grid and distributed services. Examples of cloud platforms are Amazon EC2, Microsoft Azure, Google App Engine, Xively, Nimbits, AWS IoT, CISCO IoT, IOx and Fog, IBM IoT Foundation, TCS Connected Universe Platform.

Everything as a service and cloud service models Cloud connects the devices, data, applications, services, persons and business. Cloud services can be considered a distribution service for linking the resources (computing functions, data store, processing functions, networks, servers and applications) and coordinating between the resources.

Cloud computing can be considered by a simple equation: Cloud Computing = SaaS + Paas + IaaS + DaaS

SaaS means Software as a Service. The software is made available to an application or service on demand. SaaS is a service model where the applications or services deploy and host at the cloud, and are made available through the Internet on demand by the service user. The software control, maintenance, updation to new version and infrastructure, platform and resource requirements are the responsibilities of the cloud service provider.

PaaS means Platform as a Service. The platform is made available to a developer of an application on demand. PaaS is a service model where the applications and services develop and execute using the platform (for computing, data store and distribution services) which is made available through the Internet on demand for the developer of the applications. The platform, network, resources, maintenance, updation and security as per the developers' requirements are the responsibilities of the cloud service provider.

IaaS means Infrastructure as a Service. The infrastructure (data stores, servers, data centres and network) is made available to a user or developer of application on demand. Developer installs the OS image, data store and application and controls them at the infrastructure. IaaS is a service model where the applications develop or use the infrastructure made available through the Internet on rent demand (pay as per use in multi-tenancy model) by a developer or user. IaaS computing systems, network and security are the responsibilities of the cloud service provider.

DaaS means Data as a Service. Data at a data centre is made available to a user or developer of application on demand. DaaS is a service model where the data store or data warehouse is made available through the Internet on demand on rent (pay as per use in multi tenancy model) to an enterprise. The data centre management, 24×7 power, control, network, maintenance, scale up, data replicating and mirror nodes and systems, and physical security are the responsibilities of the data centre service provider.

CHAPTER 5

IOT PHYSICAL DEVICES AND ENDPOINTS

5.1 Introduction

Device for the Internet of Things

Any object with a unique identifier and the ability to send/receive data (including user data) over a network qualifies as a "Thing" in the Internet of Things (IoT) (e.g., smart phone, smartTV, computer, refrigerator, car, etc.).

Internet-connected IoT devices transmit information about themselves or their surroundings (e.g., information sensed by connected sensors) over a network (to other devices or servers/storage) or enable remote actuation of physical entities/environments.

Examples of IoT Devices

A home automation device that allows you to remotely track and manage the status of your appliances. • An industrial computer that sends data to a server about its activity and health monitoring. • A vehicle that transmits data about its position to a cloud-based service. • A wireless-enabled wearable device that collects and sends data about a person, such as the number of steps taken, to a cloud-based service.

5.2 IoT Building Blocks

Sensors, processors, gateways, and applications are the four fundamental building blocks of the IoT framework. To form a useful IoT structure, each node must have its characteristics.

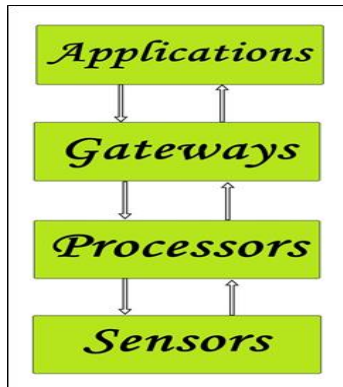


Figure 5.1: Simplified Block Diagram of the Basic Building Blocks of the IoT

Sensors:

These are the IoT devices' front ends. These are the system's so-called "Stuff." Their primary function is to gather data from the environment (sensors) or transmit data to the environment (actuators).

- ❖ These must be individually recognisable devices with their IP address to be easily identified over a vast network.
- ❖ They must be interested in nature, which means they must be able to gather data in real time. This can either function on their own (autonomous) or be programmed by the user to meet their specific requirements (user-controlled).
- ❖ Gas, water quality, moisture, and other sensors are examples of sensors.
- ❖ Processors: Processors are the IoT system's brain. Their primary purpose is to process the data collected by the sensors in order to derive useful information from the massive data of raw data collected. In a nutshell, it provides intelligence to the data.
- ❖ Processors are often real-time devices that can be easily operated by software. These are also in charge of data security, including encryption and decryption.
- ❖ Embedded hardware devices, such as microcontrollers, process data because they have processors connected to them. Gateways:
- ❖ Gateways are in charge of routing processed data and directing it to the appropriate locations for proper (data) use.
- ❖ In other words, a gateway facilitates data exchange between two points. It allows data to be accessed over a network. Any IoT machine that wants to communicate needs to be connected to the internet.
- ❖ Network gateways include things like LANs, WANs, and PANs.

Applications:

- ❖ An IoT system's applications are another end. Applications are needed for the correct use of all collected data.
- ❖ These cloud-based systems are in charge of giving the data obtained a meaningful value. Users manage applications, which serve as a delivery point for specific services.
- ❖ Home automation software, security systems, industrial control hubs,
- ❖ and other applications are examples.
- ❖ In Figure 2, the extreme right block forms the application end of the IoT system.

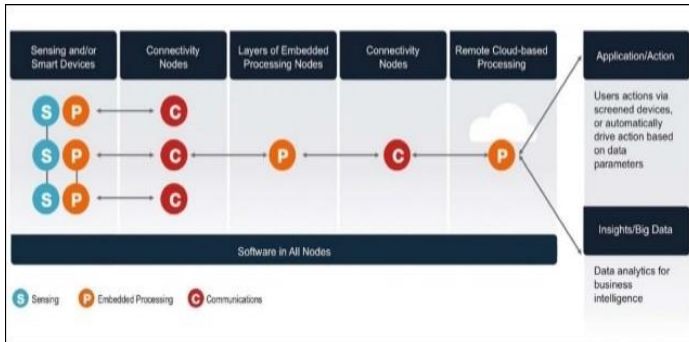


Figure 5.2: Basic Building Blocks of IoT

In a nutshell, the information gathered by the sensing node (end node) is processed first, then transmitted via connectivity to the embedded processing nodes, which can be any embedded hardware devices, and processed there. It then passes through the networking nodes again, arriving at remote cloud-based processing, which can be any software, before being sent to the application node for proper application of the data collected and big data analysis.

5.3 IoT Works

It's easy to understand how the Internet of Things functions.

In other words, sensors, RFID tags, and all other uniquely identifiable objects or "things" acquire real-time information (data) by acquiring basic resources (names, addresses, and so on) and related attributes of objects through automatic identification and perception technologies such as RFID, wireless sensor, and satellite positioning.

Second, it incorporates object-based information into the information network and realises intelligent indexing and aggregation of information related to masses of objects by relying on fundamental resource resources, thanks to various communications technologies (similar to the resolution, addressing and discovery of the internet).

Finally, it analyses and processes information related to masses of objects using intelligent computing technologies like cloud computing, fuzzy recognition, data mining, and semantic analysis to achieve intelligent decision and control in the physical world.

Let's take a look at the diagram below.

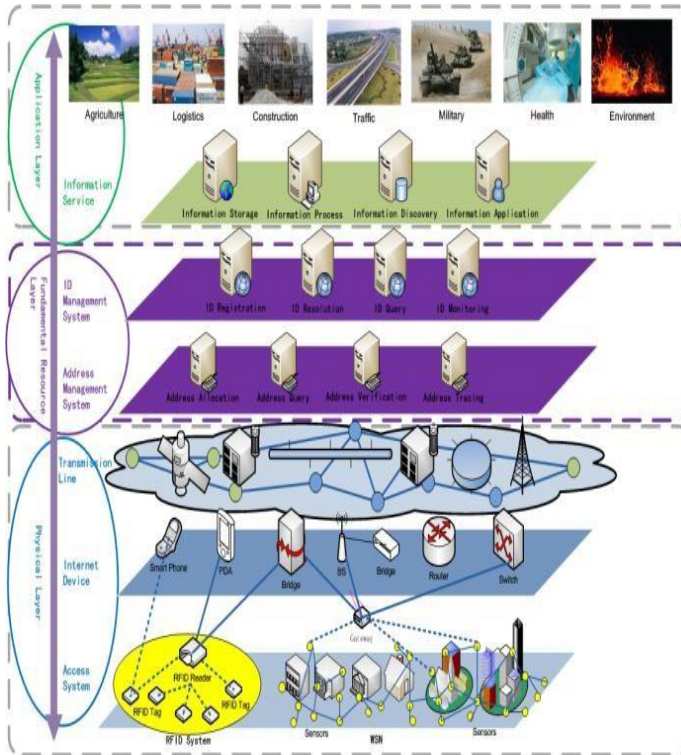


Figure 5.3: Layers of the IoT

All data collected by the access system (uniquely recognisable "things") is collected in the Physical layer and sent to internet devices (like smartphones). The data is then sent to the management layer through transmission lines (fiber-optic cable). It is separated from the raw data and handled separately (stream analytics and data analytics). The application layer receives all of the managed data and uses it appropriately.

5.4 IoT Architecture Layers

There are four major layers.

The Sensors and Connectivity network, which collects data, is at the bottom of the IoT architecture. The Gateway and Network Layer are the next two layers. Over that, there's the Management Service layer. Finally, there's the application layer, where the data is processed to meet the needs of different applications.

Let's look at the characteristics of each of these architectural layers individually.

Layer of Sensors, Connectivity, and Networking

- ❖ RFID tags and sensors are included in this layer (an essential part of an IoT system and responsible for collecting raw data). These are the “things” that make up an IoT scheme.
- ❖ Wireless Sensor Networks are sensors, RFID tags, and other wireless devices (WSN).
- ❖ Since sensors are active, they must capture and process data in real time.
- ❖ This layer also includes network access (such as WAN, PAN, and others) for transmitting raw data to the next layer, the Gateway and Network Layer.
- ❖ The computers that make up a WSN have limited storage space, limited communication bandwidth, and slow processing speeds.

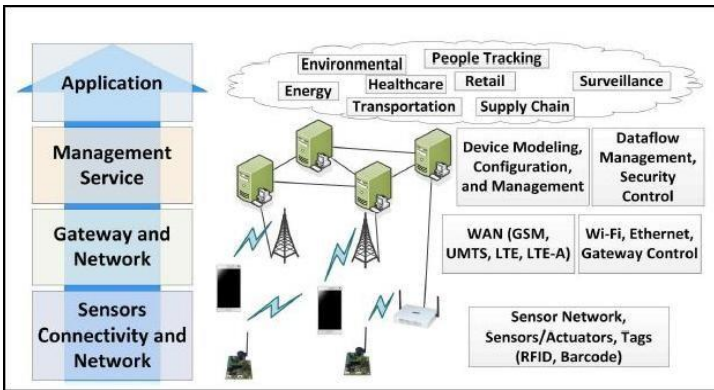


Figure 5.4: IoT Architecture Layers

- ❖ We have various sensors for various applications, such as a temperature sensor for collecting temperature data, a water quality sensor for analysing water quality, and a moisture sensor for measuring the moisture content of the atmosphere or soil.

The tags, RFID tags or barcode readers, are at the bottom of this layer, followed by sensors/actuators and communication networks, as shown in the diagram below.

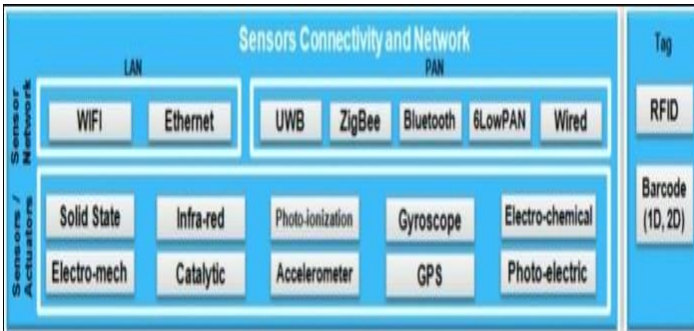


Figure 5.5: Sensor, Connectivity And Network layer gateway and network layer

- ❖ Gateways are in charge of routing data from the Sensor, Connectivity, and Network layers and passing it on to the Management Service Layer's next layer.
- ❖ This layer necessitates a huge storage space to store the massive amounts of data generated by sensors, RFID tags, and other devices. This layer must also provide consistent, reliable performance across public, private, and hybrid networks.
- ❖ Different IoT devices use various types of network protocols. Both of these protocols must be consolidated into a single layer. This layer is in charge of integrating different network protocols.

The gateway, which comprises the embedded OS, Signal Processors and Modulators, Micro-Controllers, and other components, is shown at the bottom of the diagram. The Gateway Networks, including LAN (Local Area Network), WAN (Wide Area Network), and others, are located above the gateway.

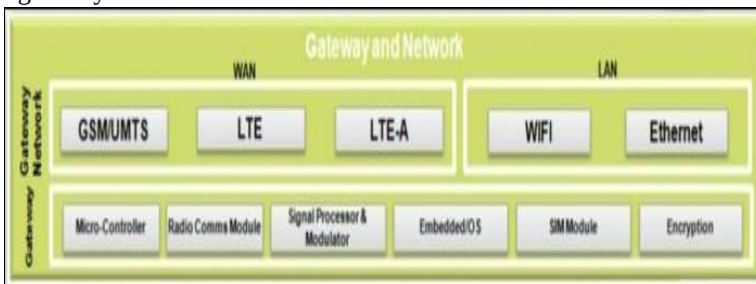


Figure 5.6: Gateway and Network Layer

Management Service Layer

This layer is used to manage Internet of Things (IoT) services. The management service layer handles security analysis of IoT devices, Information Analysis (Stream Analytics, Data Analytics), and device management.

- ❖ Data management is needed to extract the appropriate information from the massive amounts of raw data obtained by sensor devices to produce a useful result. This layer is where this operation is carried out.
- ❖ In addition, a specific circumstance necessitates an urgent response. By abstracting data, extracting knowledge, and managing the data flow, this layer aids in this endeavour.
- ❖ This layer is also in charge of data mining, text mining, and service analytics, among other things.

The management service layer has Operational Support Service (OSS), which includes Device Modeling, Device Configuration and Management, and more, as shown in the diagram below. The Billing Support System (BSS) also assists in billing and reporting.

We can also see from the diagram that there are IoT/M2M Application Services, which include Analytics Platform; Data – which is the most important part; Security, which includes Access Controls, Encryption, Identity Access Management, and so on; and finally, BRM and BPM (BPM).



Figure 5.7: Management Service Layer Application Layer

The application layer is the topmost layer of IoT architecture. It is in charge of making efficient use of the data collected.

- ❖ Home automation, e-health, e-government, and other IoT applications are only a few examples.

- ❖ As seen in the diagram below, there are two types of IoT applications: horizontal market applications such as fleet management and supply chain, and sector-specific applications such as oil, healthcare, and transportation.



**Figure 5.8: Application Layer
Smart Environment Application Domains**

	Smart Home	Smart Office	Smart Retail	Smart City	Smart Agriculture	Smart Energy & Fuel	Smart Transportation	Smart Military
Network Size	Small	Small	Small	Medium	Medium /Large	Large	Large	Large
Network Connectivity	WPAN, WLAN, 3G, 4G, Internet	WPAN, WLAN, 3G, 4G, Internet	RFID, NFC, WPAN, WLAN, 3G, 4G, Internet	RFID, NFC, WLAN, 3G, 4G, Internet	WLAN, Satellite Comm., Internet	WLAN, 3G, 4G, Microwave links, Satellite Comm.,	WLAN, 3G, 4G, Satellite Comm.	RFID, NFC, WPAN, WLAN, 3G, 4G, Satellite Comm.
Bandwidth Requirement	Small	Small	Small	Large	Medium	Medium	Medium~Large	Medium~Large

Figure 5.9: Smart Environment Application Domains

- ❖ WLAN stands for Wireless Local Area Network, encompassing Wi-Fi, WAVE, IEEE 802.11 a/b/g/p/n/ac/ad, and other wireless technologies.
- ❖ Bluetooth, ZigBee, 6LoWPAN, IEEE 802.15.4, UWB, and other wireless
- ❖ personal area networks are examples of WPAN.

Service Domain	Services
Smart Home	Entertainment, Internet Access
Smart Office	Secure File Exchange, Internet Access, VPN, B2B
Smart Retail	Customer Privacy, Business Transactions, Business Security, B2B, Sales & Logistics Management
Smart City	City Management, Resource Management, Police Network, Fire Department Network Transportation Management, Disaster Management
Smart Agriculture	Area Monitoring, Condition Sensing, Fire Alarm, Trespassing
Smart Energy & Fuel	Pipeline Monitoring, Tank Monitoring, Power Line Monitoring, Trespassing & Damage Management
Smart Transportation	Road Condition Monitoring, Traffic Status Monitoring, Traffic Light Control, Navigation Support, Smart Car Support, Traffic Information Support, ITS (Intelligent Transportation System)
Smart Military	Command & Control, Communications, Sensor Network, Situational Awareness, Security Information, Military Networking

Figure 5.10: Smart Environment Application Domains: Service Domain And Their Services Classified.

5.5 Exemplary Device: Raspberry PI

The Raspberry Pi is a small, low-cost device about the size of a credit card. Raspberry Pi runs a variety of Linux distributions and can perform nearly all tasks that a typical desktop computer can. The general purpose I/O pins on the Raspberry Pi can also be used to attach sensors and actuators. Python is supported "out of the box" on the Raspberry Pi since it runs the Linux operating system. The Raspberry Pi is a small, low-cost device about the size of a credit card. Raspberry Pi runs a variety of Linux distributions and can perform nearly all tasks that a typical desktop computer can. The general purpose I/O pins on the Raspberry Pi can also be used to attach sensors and actuators. Python is supported "out of the box" on the Raspberry Pi since it runs the Linux operating system.

5.5.1 Fundamentals of the Raspberry PI

The Raspberry Pi is a platform, not a single device. It's a group of single-board computers (SBC). Each SBC has its own set of characteristics and capabilities.

The Raspberry Pi model 1 A had a single-core ARM processor, 256MB RAM,

USB port, composite video out, HDMI output, and no networking capabilities when it was first released. The most recent version of the Raspberry Pi 3 model B has a quad-core processor, 1GB RAM, four USB

ports, ethernet, built-in Wi-Fi, and a 40-pin header for controlling external circuits.

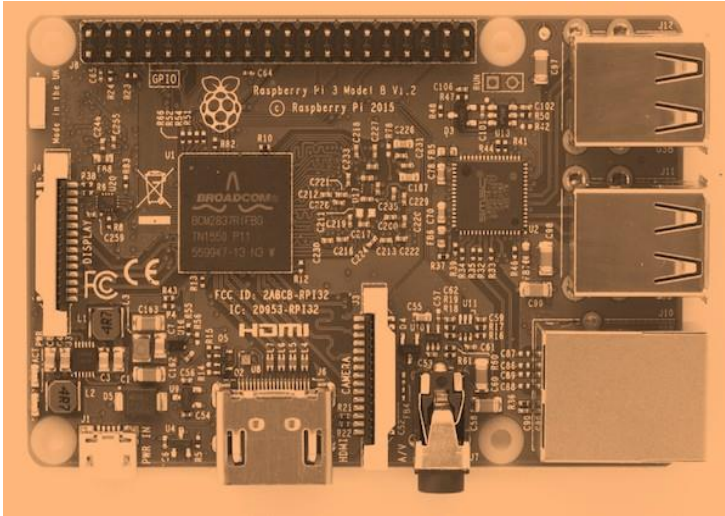


FIG 5.12 Fundamentals of the Raspberry Pi

With so many different Pi computers to choose from, deciding which one is best for a project can be difficult. If you're new to the Raspberry Pi, the most feature-packed version is best because it will allow you to do the most projects. The Raspberry Pi 3 Model B features networking, a powerful CPU, and GPIO ports. The Raspberry Pi 1 range is ideal for projects that do not need networking but need a monitor and GPIO. In contrast, the Raspberry Pi Zero is ideal for projects that only need a minimal system.

Linux on Raspberry Pi

1. Raspbian: Raspbian Linux is a RaspberryPi-optimized Debian Wheezy port.
2. Arch: Arch is an AMD-optimized version of Arch Linux.
3. Pidora: Pidora Linux is a RaspberryPi-optimized version of Fedora Linux.
4. RaspBMC: RaspBMC is a RaspberryPi-based XBMC media centre distribution.
5. OpenELEC: OpenELEC is an XBMC media centre distribution that is fast and easy to use.
6. RISC OS: RISC OS is a small, fast operating system.

Once you've decided on a Raspberry Pi, you'll need to learn a few things about it, beginning with understanding the various parts on the PCB and what - one does.

GPIO is the Raspberry Pi's most critical function, and it's the Arduino's

The Raspberry Pi will connect to a serial display similar to those found

The CSI camera port is a connector that connects a Raspberry Pi

This slot stores the Raspberry Pi operating system on a microSD

HDMI / USB / Network

These ports link the Pi to an HDMI display, USB devices including mice and keyboards, and an ethernet connection for internet access. However, since the Raspberry Pi 3 has built-in Wi-Fi, the ethernet port is seldom used.

Micro USB Power

Power can be supplied to the Raspberry Pi via a micro USB cable connected to the micro USB connector (recommended) or directly feeding 5V into the

5V GPIO pin. Now that you're familiar with the components that make up the Raspberry Pi, you might discover that when it's turned on, it does nothing. This is since we must first instal an operating system! I'll show you how to put Raspbian, the Raspberry Pi's most common operating system, on a microSD card in the following post.

5.5.2 Raspberry PI Interfaces

The Raspberry Pi has Serial, SPI, and I2C interfaces for data transfer. Receive (Rx) and transmit (Tx) pins on the Serial interface on the Raspberry

Pi are used to communicate with serial peripherals.

Serial Peripheral Interface (SPI) is a synchronous serial data protocol that allows one or more peripheral devices to communicate with one another. In an SPI link, the Raspberry Pi has five pins for the SPI interface:

MISO (Master in slave out) – Master line for data transmission to peripherals.

Slave line for sending data to the owner, MOSI (Master out slave in).

SCK (Serial Clock) – Clock generated by the master for data transmission synchronisation.

To allow or disable devices, use CE0 (Chip Enable 0).

CE0 (Chip Enable 1) – This command enables or disables devices. I2C (Inter-Industry Communication):

The Raspberry Pi's I2C interface pins allow you to attach hardware modules. The I2C interface allows for synchronous data transfer using only two pins: SDA (data line) and SCL (control line) (Clock Line).

Python is an excellent language for programming the Raspberry Pi because of its ease of use and accessibility to hardware, such as GPIO. This tutorial will look at running Python programmes on the Raspberry Pi.

What You Would Need

- ❖ Pi (Raspberry Pi)
- ❖ SD card with Raspbian OS
- ❖ Display with HDMI input
- ❖ Mouse and keyboard

A Python programme that requires external libraries could not operate on Raspbian using any of the methods mentioned below. PIP can be used to instal additional libraries. However, it only does so for programmes that run in the terminal window. This could be because the Python IDE programmes have their local copy of Python. As a result, it is recommended that you run Python programmes that require external libraries installed using PIP from the terminal window.

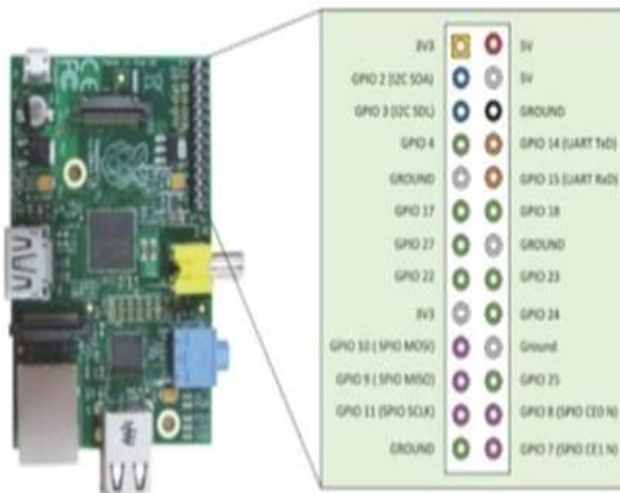


FIG 5.15 Raspberry Pi Example: Interfacing Led and Switch With Raspberry PI from time import sleep import RPi.GPIO as GPIO GPIO.setmode(GPIO.BCM) #Switch Pin GPIO.setup(25,GPIO.IN)

```
#LEDPin GPIO.setup(18,GPIO.OUT) state=false
```

```
deftoggleLED(pin):
```

```
state = not state GPIO.output(pin,state) whileTrue:
```

```
try:
```

```
if (GPIO.input(25) ==True):
```

```
toggleLED(pin) sleep(.01) exceptKeyboardInterrupt: exit()
```

Other Devices pcDuino BeagleBoneBlack Cubieboard pcDuino

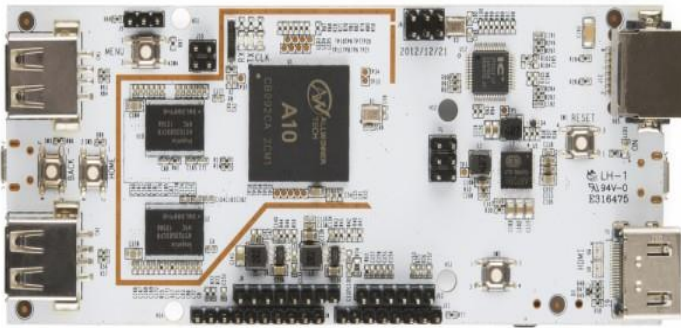


FIG 5.16 Pcdduino

The pcDuino's elegance lies in its exceptionally well-exposed hardware peripherals. On the other hand, using these peripherals is more difficult than using them on an Arduino-compatible circuit.

- ❖ pcDuino - some familiarity with the pcDuino basics is needed before proceeding. Before continuing, please check our Getting Started with pcDuino tutorial.
- ❖ Linux - The Linux operating system is the most important thing you should be familiar with. Remember that pcDuino isn't an Arduino; it's a modern microcomputer with a completely functional, albeit compact, operating system.
- ❖ SPI is a synchronous (clocked) serial peripheral interface that allows chips on a board to communicate. A minimum of four wires are required (clock, master-out-slave-in data, master-in-slave-out data, and slave chip select), with one additional chip select line required for each additional chip added to the bus.
- ❖ I2C - also known as IIC (inter-integrated circuit), SMBus, or TWI (two-wire interface), I2C communicates with different devices using just two wires (bidirectional data and clock lines).
- ❖ Serial Communication - a data interface of at least two wires that is asynchronous (no transmitted clock) (data transmit and data receive; sometimes, additional signals are added to indicate when a device is ready to send or receive).
- ❖ Pulse Width Modulation (PWM) is a digital-to-analog conversion technique that employs a fixed frequency square wave with varying duty cycle that can be easily converted to an analogue signal ranging

from 0V to the maximum amplitude the digital IC driving the signal.

- ❖ Analog-to-Digital Conversion measures an analogue voltage and converts it to a digital value.

BeagleBoneBlack is a character in the game BeagleBoneBlack

The BeagleBone Black is the BeagleBoard family's newest member. It's a low-cost, high-expansion BeagleBoard powered by a Texas Instruments Sitara XAM3359AZCZ100 Cortex A8 ARM processor. It's similar to the Beaglebone, but it has certain features that the Beaglebone doesn't. The distinctions between the BeagleBone and the BeagleBone Black are summarised below.

	BeagleBone Black \$55	BeagleBone \$89
Processor	AM3358BZCZ100, 1GHZ	AM3359ZCZ72, 720MHz
Video Out	HDMI	None
DRAM	512MB DDR3L 800MHZ	256MB DDR2 400MHz
Flash	4GB eMMC, uSD	uSD
Onboard JTAG	Optional	Yes, over USB
Serial	Header	Via USB
PWR Exp Header	No	Yes
Power	210-460 mA@5V	300-500 mA@5V

BeagleBone Black Features

The following table lists the key features of the BeagleBone Black.

	Feature	
Processor	Sitara AM3358BZCC100	
Graphics Engine	1GHz, 2000 MIPS	
SDRAM Memory	SGX530 3D, 20M Polygons/S	
Onboard Flash	512MB DDR3L 800MHZ	
PMIC	4GB, 8bit Embedded MMC	
Debug Support	TPS65217C PMIC regulator and one additional LDO.	
Power Source	Optional Onboard 20-pin CTI JTAG, Serial Header	
PCB	miniUSB USB or DC Jack	5VDC External Via Expansion Header
Indicators	3.4" x 2.1"	6 layers
HS USB 2.0 Client Port	1-Power, 2-Ethernet, 4-User Controllable LEDs	
HS USB 2.0 Host Port	Access to USB0, Client mode via miniUSB	
Serial Port	Access to USB1, Type A Socket, 500mA LS/FS/HS	
Ethernet	UART0 access via 6 pin 3.3V TTL Header. Header is populated	
SD/MMC Connector	10/100, RJ45	
User Input	microSD, 3.3V	
Video Out	Reset Button	
Audio	Boot Button	
Expansion Connectors	Power Button	
Weight	16b HDMI, 1280x1024 (MAX)	
Power	1024x768, 1280x720, 1440x900, 1920x1080@24Hz	
	w/EDID Support	
	Via HDMI Interface, Stereo	
	Power 5V, 3.3V, VDD_ADC(1.8V)	
	3.3V I/O on all signals	
	McASP0, SPI1, I2C, GPIO(69 max), LCD, GPMC, MMC1, MMC2, 7	
	AIN(1.8V MAX), 4 Timers, 4 Serial Ports, CAN0,	
	EHRPWM(0.2), XDMA Interrupt, Power button, Expansion Board ID	
	(Up to 4 can be stacked)	
	1.4 oz (39.68 grams)	
	Refer to Section 6.1.7	

In the box is (1)BeagleBone Black board, (1)USB cable, and (1)card that should be read.

BeagleBone Black Picture

Here is a picture of the Rev A5A board.

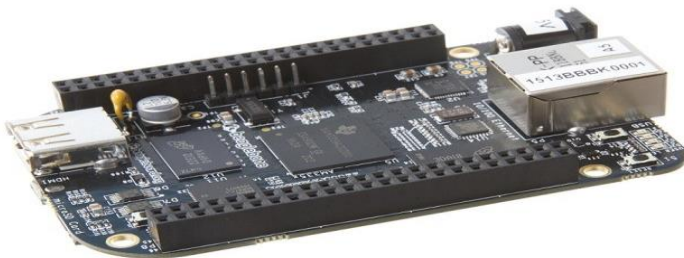


FIG 5.17 Beaglebone Black Picture

BeagleBone Black Key Component Locations

Here are the locations of the key components on the Rev A5A.

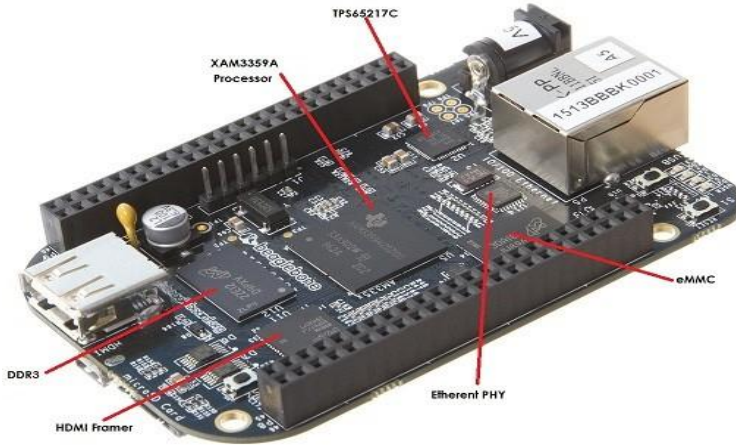


FIG 5.18 Beaglebone Black Key Component Locations BeagleBone Black Connector and Switch Locations

Below is the connectors and switches on the Rev A5A board. The Power Button and Battery Connections are new additions to the Rev A5A.

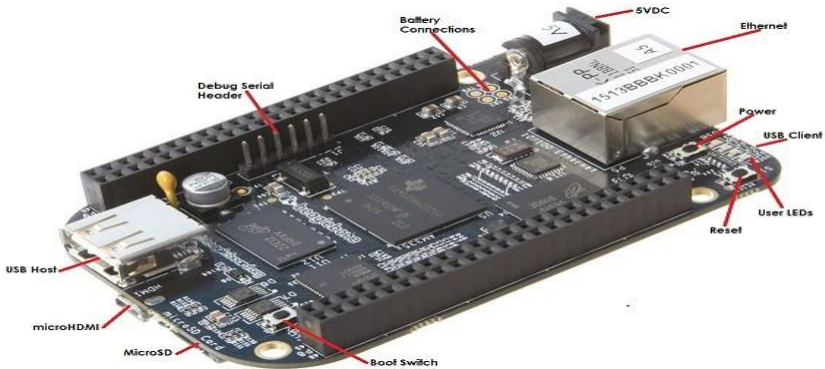


Fig 5.19 Beaglebone Black Connector and Switch Locations Cubieboard

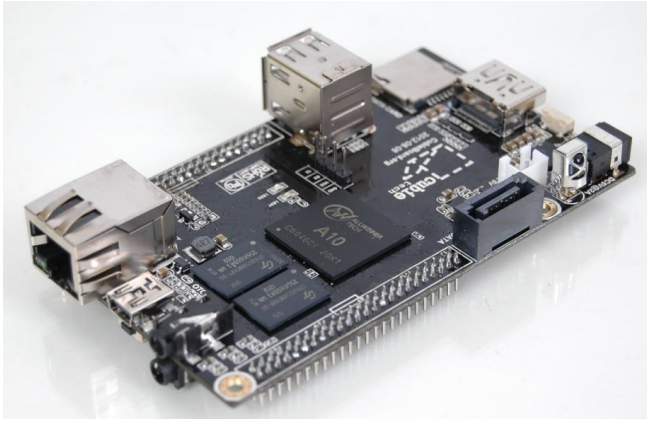


FIG 5.20 Cubieboard

Cubieboard is a Chinese single-board machine manufactured in Zhuhai, Guangdong. In September 2012, the first short run of prototype boards was sold internationally, and in October 2012, the production version was released. [1] It can run Android 4 ICS, Ubuntu 12.04 desktop, and other operating systems. [2] Remixed Fedora 19 for ARM [3] Armbian, Arch Linux ARM, desktop [4] a Cubian distribution based on Debian, [5] FreeBSD,[6] or OpenBSD are two options. [nine]

It is powered by the AllWinner A10 SoC, commonly used in low-cost tablets, phones, and media PCs. Developers of the lima driver, an open-source driver for the ARM Mali GPU, use this SoC. [eight] It ran ioquake 3 at 47 frames per second in 1024x600 at the 2013 FOSDEM demo. [nine] Using the Ubuntu Linux distribution, the Cubieboard team was able to run an Apache Hadoop computer cluster.

CHAPTER 6

IOT PHYSICAL SERVERS AND CLOUD OF FERINGS

6.1 Introduction To Cloud Computing

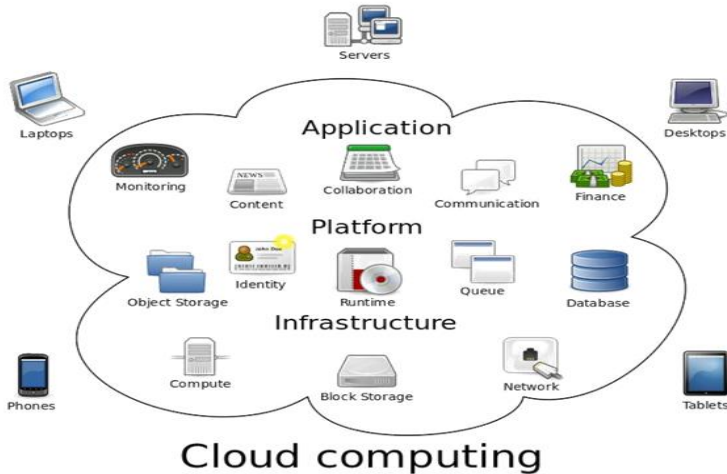


FIG 6.1 Cloud Computing Applications

The Internet of Things (IoT) involves the internet-connected devices we use to perform the processes and services that support our way of life. Another component set to help IoT succeed is cloud computing, which acts as a front end. Cloud computing is an increasingly popular service that offers several advantages to IOT. It is based on the concept of allowing users to perform normal computing tasks using services delivered entirely over the internet. A worker may need to finish a major project that must be submitted to a manager. However, perhaps they encounter problems with memory or space constraints on their computing device. Memory and space constraints can be minimized if an application is instead hosted on the internet. The worker can use a cloud computing service to finish their work because the data is managed remotely by a server.

Another example is that you have a problem with your mobile device and need to reformat it or reinstall the operating system. You can use Google Photos to upload your photos to internet-based storage. After the reformat or reinstall, you can either move the photos back to your device or view the photos on your device from the internet when you want.

Concept

In truth, cloud computing and IoT are tightly coupled. The growth of IoT and the rapid development of associated technologies create a widespread connection of —things. This has led to the production of large amounts of data, which needs to be stored, processed and accessed. Cloud computing as a paradigm for big data storage and analytics. While IoT is exciting, the real innovation will come from combining it with cloud computing. The combination of cloud computing and IoT will enable new monitoring services and powerful processing of sensory data streams. For example, sensory data can be uploaded and stored with cloud computing, later used intelligently for smart monitoring and actuation with other smart devices. Ultimately, the goal is to be able to transform data to insight and drive productive, cost-effective action from those insights. The cloud effectively serves as the brain to improved decision-making and optimized internet-based interactions. However, when IoT meets cloud, new challenges arise. There is an urgent need for novel network architectures that seamlessly integrate them. During integration, the critical concerns are quality of service (quality of service) and quality of experience (QoE) and data security, privacy, and reliability. The virtual infrastructure for practical mobile computing and interfacing includes integrating applications, storage devices, monitoring devices, visualization platforms, analytics tools and client delivery. Cloud computing offers a practical utility-based model that will enable businesses and users to access applications on demand anytime and from anywhere.

6.2 Characteristics

First, the cloud computing of IoT is an on-demand self service, meaning it's there when you need it. Cloud computing is a web-based service that can be accessed without any special assistance or permission from other people; however, you need at minimum some sort of internet access.

Second, the cloud computing of IoT involves broad network access, meaning it offers several connectivity options. Cloud computing resources can be accessed through many internet-connected devices such as tablets, mobile devices and laptops. This level of convenience means users can access those resources in a wide variety of manners, even from older devices. Again, though, this emphasizes the need for network access points. Third, cloud computing allows for resource pooling, meaning information can be shared with those who know where and how (have permission) to

access the resource, anytime and anywhere. This lends to broader collaboration or closer connections with other users. From an IoT perspective, just as we can easily assign an IP address to every "thing" on the planet, we can share the "address" of the cloud-based protected and stored information with others and pool resources.

Fourth, cloud computing features rapid elasticity, meaning users can readily scale the service to their needs. You can easily and quickly edit your software setup, add or remove users, increase storage space, etc. This characteristic will further empower IoT by providing elastic computing power, storage and networking.

Finally, the cloud computing of IoT is a measured service, meaning you get what you pay for. Providers can easily measure usage statistics such as storage, processing, bandwidth and active user accounts inside your cloud instance. This pay per use (PPU) model means your costs scale with your usage. In IoT terms, it's comparable to the ever-growing network of physical objects that feature an IP address for internet connectivity, and the communication that occurs between these objects and other internet-enabled devices and systems; just like your cloud service, the service rates for that IoT infrastructure may also scale with use.

6.3 Service and Deployment

Service models

Service delivery in cloud computing comprises three different service models: software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS).

Software as a service (SaaS) provides applications to the cloud's end user mainly accessed via a web portal or service-oriented architecture-based web service technology. These services can be seen as ASP (application service provider) on the application layer. Usually, a specific company that uses the service would run, maintain and give support to be reliably used over a long time.

Platform as a service (PaaS) consists of the actual environment for developing and provisioning cloud applications. This layer's main users are developers who want to develop and run a cloud application for a particular purpose. A proprietary language was supported and provided by the platform (a set of important basic services) to ease communication, monitoring, billing and other aspects such as startup and ensure an application's scalability and flexibility. Limitations regarding the programming languages supported, the programming model, the ability to access resources, and the long-term persistence are possible disadvantages.

Infrastructure as a service (IaaS) provides the necessary hardware and software upon which a customer can build a customized computing environment. Computing resources, data storage resources and the communications channel are linked together with these essential IT resources to ensure the stability of applications used on the cloud. Those stack models can be referred to as the medium for IoT, being used and conveyed by the users in different methods for the greatest chance of interoperability. This includes connecting cars, wearables, TVs, smartphones, fitness equipment, robots, ATMs, vending machines, the vertical applications, security and professional services, and analytics platforms that come with them.

6.3.1 Deployment Models

Deployment in cloud computing comprises four deployment models: private cloud, public cloud, community cloud and hybrid cloud.

A private cloud has infrastructure provisioned for exclusive use by a single organization comprising multiple consumers such as business units. It may be owned, managed and operated by the organization, a third party or some combination of them. It may exist on or off premises.

A public cloud is created for open use by the general public. Public cloud sells services to anyone on the internet. (Amazon Web Services is an example of a large public cloud provider.) This model is suitable for business requirements that require management of load spikes and the applications used by the business, activities that would otherwise require greater infrastructure investment. The public cloud also helps reduce capital expenditure and bring down operational IT costs.

A community cloud is managed and used by a group or organizations with shared interests, such as specific security requirements or a common mission.

Finally, a hybrid cloud combines two or more distinct private, community or public cloud infrastructures. They remain unique entities but are bound together by standardized or proprietary technology enabling data and application portability. Normally, information that's not critical is outsourced to the public cloud. At the same time, business-critical services and data are kept within the organization's control.

6.4 Cloud Storage API

A cloud storage API is an application program interface that connects a locally-based application to a cloud-based storage system. A user can send

data to it and access and work with data stored. To the application, the cloud storage system is just another target device, like tape or disk-based storage. An application program interface (API) is code that allows two software programs to communicate with each other. The API defines the correct way for a developer to write a program that requests services from an operating system (OS) or other application. APIs are implemented by function calls composed of verbs and nouns. The required syntax is described in the documentation of the application being called.

6.4.1 APIS work

APIs are made up of two related elements. The first is a specification that describes how information is exchanged between programs, done in the form of a request for processing and a return of the necessary data. The second is a software interface written to that specification and published for use. The software that wants to access the features and capabilities of the API is said to call it, and the software that creates the API is said to publish it.

APIS are Important for Business

The web, software designed exchange information via the internet and cloud computing have all combined to increase the interest in APIs in general and services in particular. Software that was once custom-developed for a specific purpose is now often written referencing APIs that provide broadly useful features, reducing development time and cost and mitigating the risk of errors. APIs have steadily improved software quality over the last decade. The growing number of web services exposed through APIs by cloud providers also encourages the creation of cloud-specific applications, internet of things (IoT) efforts, and apps to support mobile devices and users.

6.4.2 Three Basic Types of APIS

APIs take three basic forms: local, web-like and program-like.

1. Local APIs is the original form, from which the name came. They offer OS or middleware services to application programs. Microsoft's .NET APIs, the TAPI (Telephony API) for voice applications, and database access APIs are the local API form examples.

Web APIs are designed to represent widely used resources like HTML pages and are accessed using a simple HTTP protocol. Any web URL activates a web API. Web APIs are often called REST (representational

state transfer) or RESTful because the publisher of REST interfaces doesn't save any data internally between requests. As such, requests from many users can be intermingled as they would be on the internet.

Program APIs are based on remote procedure call (RPC) technology that makes a remote program component appear local to the rest of the software. Service-oriented architecture (SOA) APIs, such as Microsoft's WS- series, are program APIs.

6.5 IoT / Cloud Convergence

Internet-of-Things can benefit from cloud computing infrastructures' scalability, performance, and pay-as-you-go nature. Indeed, as IoT applications produce large volumes of data and comprise multiple computational components (e.g., data processing and analytics algorithms), their integration with cloud computing infrastructures could provide them with opportunities for cost-effective on-demand scaling. As prominent examples consider the following settings:

A Small Medium Enterprise (SME) developing an energy management IoT product, targeting smart homes and smart buildings. By streaming the product's data (e.g., sensors and WSN data) into the cloud it can accommodate its growing needs in a scalable and cost-effective fashion.

As the SMEs acquires more customers and performs more deployments of its product, it is able A smart city can benefit from the cloud-based deployment of its IoT systems and applications. A city is likely to deploy many IoT applications, such as smart energy management, smart water management, smart transport management, urban mobility of the citizens, and more. These applications comprise multiple sensors and devices, along with computational components. Furthermore, they are likely to produce very large data volumes. Cloud integration enables the city to host these data and applications cost-effectively. Furthermore, the elasticity of the cloud can directly support expansions to these applications and the rapid deployment of new ones without major concerns about the provisioning of the required cloud computing resources.

A cloud computing provider offering pubic cloud services can extend them to the IoT area, by enabling third parties to access its infrastructure to integrate IoT data and/or computational components operating over IoT devices. The provider can offer IoT data access and services in a pay-as-you fashion, by enabling third parties to access resources of its infrastructure and accordingly to charge them in a utility-based fashion.

These motivating examples illustrate the merit and need for converging IoT and cloud computing infrastructure. Despite these merits, this convergence has always been challenging mainly due to the conflicting properties of IoT and cloud infrastructures, in particular, IoT devices tend to be location- specific, resource-constrained, expensive (in terms of development/ deployment cost), and generally inflexible (in terms of resource access and availability). On the other hand, cloud computing resources are typically location-independent and inexpensive, while at the same time providing rapid and flexibly elasticity. To alleviate these incompatibilities, sensors and devices are virtualized before integrating their data and services in the cloud, to enable their distribution across cloud resources. Furthermore, service and sensor discovery functionalities are implemented on the cloud to discover services and sensors in different locations.

Based on these principles, IoT/cloud convergence efforts have started over a decade, i.e. since the early days of IoT and cloud computing. Early efforts in the research community (i.e. during 2005-2009) have focused on streaming sensor and WSN data in a cloud infrastructure. Since 2007 we have also witnessed the emergence of public IoT clouds, including commercial efforts. One of the earliest efforts has been the famous Pachube.com infrastructure (used extensively for radiation detection and production of radiation maps during earthquakes in Japan). Pachube.com has evolved (following several evolutions and acquisitions of this infrastructure) to Xively.com, which is nowadays one of the most prominent public IoT clouds.

Nevertheless, there are tens of other public IoT clouds, such as ThingsWorx, ThingsSpeak, Sensor-Cloud, Realtime.io, etc. The list is certainly non-exhaustive. These public IoT clouds offer commercial pay-as-you-go access to end-users wishing to deploy IoT applications on the cloud. Most of them come with developer-friendly tools, which enable the development of cloud applications, thus acting like a PaaS for IoT in the cloud. Similarly to cloud computing infrastructures, IoT/cloud infrastructures and related services can be classified into the following models:

Infrastructure-as-a-Service (IaaS) IoT/Clouds: These services provide access to sensors and actuators in the cloud. The associated business model involves the IoT/Cloud provides to act either as a data or sensor provider. IaaS services for IoT provide access control to resources as a prerequisite for offering related pay-as-you-go services.

Platform-as-a-Service (PaaS) IoT/Clouds: This is the most widespread model for IoT/cloud services. It is the model provided by all public IoT/cloud infrastructures outlined above. Most public IoT clouds come with a range of tools and related environments for application development and deployment in a cloud environment. The main characteristic of PaaS IoT services is that they provide access to data, not to hardware. This is a clear differentiator comparing to IaaS.

Software-as-a-Service (SaaS) IoT/Clouds: SaaS IoT services enable their users to access complete IoT-based software applications through the cloud, on-demand, and in a pay-as-you-go fashion. As soon as sensors and IoT devices are not visible, SaaS IoT applications resemble conventional cloud-based SaaS applications. However, there are cases where the IoT dimension is strong and evident, such as applications involving the selection of sensors and the combination of data from the selected sensors in integrated applications. Several of these applications are commonly called Sensing-as-a-Service. They provide on-demand access to the services of multiple sensors. SaaS IoT applications are typically built over a PaaS infrastructure and enable utility-based business models involving IoT software and services.

These definitions and examples provide an overview of IoT and cloud convergence and why it is important and useful. More and more IoT applications are integrated with the cloud to benefit from its performance, business agility, and pay-as-you-go characteristics. In the following chapters of the tutorial, we will present how to maximize the benefits of the

cloud for IoT, through ensuring semantic interoperability of IoT data and services in the cloud, thus enabling advanced data analytics applications, but also the integration of a wide range of vertical (silo) IoT applications that are nowadays available in areas such as smart energy, smart transport, and smart cities. We will also illustrate the benefits of IoT/cloud integration for specific areas and segments of IoT, such as IoT-based wearable computing.

6.6 WAMP for IoT

Web Application Messaging Protocol (WAMP) is a sub-protocol of WebSocket which provides publish-subscribe and remote procedure call (RPC) messaging patterns.

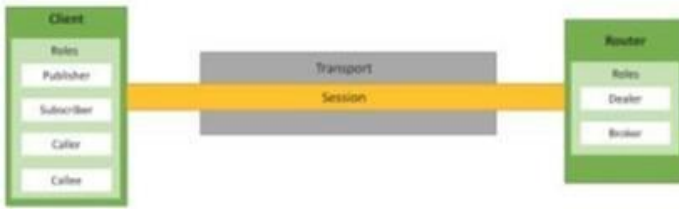


FIG 6.2 Web Application Messaging Protocol

WAMP

Transport: Transport is a channel that connects two peers.

Session: The session is a conversation between two peers that run over a transport.

Client: Clients are peers that can have one or more roles. In the publish- subscribe model client can have the following roles:

Publisher: Publisher publishes events (including payload) to the topic maintained by the broker.

Subscriber: Subscriber subscribes to the topics and receives the events including the payload.

In the RPC model client can have the following roles: –

Caller: Caller issues calls to the remote procedures along with call arguments. Callee executes the procedures to which the calls are issued by the caller and returns the results to the caller. • **Router:** Routers are peers that perform generic call and event routing. In the publish-subscribe model, the Router has the role of a **Broker**: – **Broker:** Broker acts as a router and routes messages published to a topic to all subscribers subscribed to the topic.

In RPC model Router has the role of a **Broker**: – **Dealer:** Dealer acts as a router and routes RPC calls from the Caller to the

Callee and routes results from Callee to Caller.

Application Code: Application code runs on the Clients (Publisher, Subscriber, Callee, or Caller).

6.7 Amazon Ec2 – Python Example

Boto is a Python package that provides Amazon Web Services (AWS) interfaces. In this example, a connection to the EC2 service is first established by calling `boto.ec2.connect_to_region`. The EC2 region, AWS access key, and AWS secret key are passed to this function. After

connecting to EC2, a new instance is launched using the `conn.run_instances` function. The AMI-ID, instance type, EC2 key handle, and security group are passed to this function.

Amazon AutoScaling – Python Example

AutoScaling Service: A connection to AutoScaling service is first established by calling `boto.ec2.autoscale.connect_to_region` function.

Launch Configuration: After connecting to the AutoScaling service, a new launch configuration is created by calling `conn.create_launch_configuration`. Launch configuration contains instructions on launching new instances including the AMI-ID, instance type, security groups, etc.

AutoScaling Group: It is associated with a new AutoScaling group after creating a launch configuration. AutoScaling group is created by calling `conn.create_auto_scaling_group`. The AutoScaling group settings include the maximum and a minimum number of instances, the launch configuration, availability zones, optional load balancer to use with the group, etc.

Amazon AutoScaling – Python Example

#Creating auto-scaling policies

```
scale_up_policy = ScalingPolicy(name='scale_up',
                                adjustment_type='ChangeInCapacity', as_name='My-Group',
                                scaling_adjustment=1, cooldown=180)
scale_down_policy = ScalingPolicy(name='scale_down',
                                   adjustment_type='ChangeInCapacity',
                                   as_name='My-Group', scaling_adjustment=-1, cooldown=180)
conn.create_scaling_policy(scale_up_policy)
conn.create_scaling_policy(scale_down_policy)
```

AutoScaling Policies:

AutoScaling Policies:

After creating an AutoScaling group, the scaling up and scaling down policies are defined.

This example defines a scale-up policy with adjustment type `ChangeInCapacity` and `scaling_adjustment = 1`.

Similarly, a scale-down policy with adjustment type `ChangeInCapacity` and `scaling_adjustment = -1` is defined.

CloudWatch Alarms

#Connecting to CloudWatch

```
cloudwatch = boto.ec2.cloudwatch.connect_to_region(REGION,
                                                    aws_access_key_id=ACCESS_KEY,
```

```
aws_secret_access_key=SECRET_KEY)
alarm_dimensions = {"AutoScalingGroupName": 'My-Group'}
#Creating a scale-up alarm scale_up_alarm = MetricAlarm(
name='scale_up_on_cpu', namespace='AWS/EC2',
metric='CPUUtilization', statistic='Average', comparison='>',
threshold='70',
period='60', evaluation_periods=2,
alarm_actions=[scale_up_policy.policy_arn],
dimensions=alarm_dimensions) cloudwatch.create_alarm(scale_up_alarm)
#Creating a scale-down alarm scale_down_alarm =MetricAlarm(
name='scale_down_on_cpu',namespace='AWS/EC2',
metric='CPUUtilization', statistic='Average', comparison='<',threshold='40',
period='60', evaluation_periods=2,
alarm_actions=[scale_down_policy.policy_arn],
dimensions=alarm_dimensions)
cloudwatch.create_alarm(scale_down_alarm)
```

With the scaling policies defined, the next step is to create Amazon CloudWatch alarms that trigger these policies.

The scale-up alarm is defined using the CPUUtilization metric with the Average statistic and threshold greater than 70% for 60 sec. The scale-up policy created previously is associated with this alarm. This alarm is triggered when the average CPU utilization of the instances in the group becomes greater than 70% for more than 60seconds.

The scaled-down alarm is defined similarly with a threshold of less than50%.

6.8 Python for Mapreduce

#Inverted Index Mapper in Python

The example shows inverted index mapper program. The map function reads the data from the standard input (stdin) and splits the tab-limited data into document-ID and contents of the document. The map function emits key-value pairs where key is each word in the document and value is the document-ID.

Python for MapReduce

#Inverted Index Reducer in Python

```
#!/usr/bin/env python import sys current_word = None current_docids = []
word = None
for line in sys.stdin: # remove leading and trailing whitespace line =
line.strip() # parse the input we got from mapper.py word, doc_id =
```

```
line.split(" ") if current_word
word: current_docids.append(doc_id) else: if current_word: print '%s%s'
%
(current_word, current_docids) current_docids = []
current_docids.append(doc_id) current_word = word
```

The example shows an inverted index reducer program. The key-value pairs emitted by the map phase are shuffled to the reducers and grouped by the key. The reducer reads the key-value pairs grouped by the same key from the standard input (stdin) and creates a list of document-IDs in which the word occurs. The reducer's output contains key-value pairs where the key is a unique word and value is the list of document-IDs in which the word occurs.

Python Packages of Interest

JSON: JavaScript Object Notation (JSON) is easy to read and write data- interchange format. JSON is used as an alternative to XML and it is easy for machines to parse and generate. JSON is built on two structures - a collection of name-value pairs (e.g. a Python dictionary) and ordered lists of values (e.g., a Python list).

XML: XML (Extensible Markup Language) is a data format for structured document interchange. The Python minidom library provides minimal Document Object Model interface implementation. It has an API similar to that in other languages.

HTTPLib & URLLib: HTTPLib2 and URLLib2 are Python libraries used in network/internet programming

SMTPLib: Simple Mail Transfer Protocol (SMTP) is a protocol that handles sending email and routing e-mail between mail servers. The Python smtplib module provides an SMTP client session object that can be used to send emails.

NumPy: NumPy is a package for scientific computing in Python. NumPy provides support for large multi-dimensional arrays and matrices

Scikit-learn: Scikit-learn is an open-source machine learning library for Python that implements various machine learning algorithms for classification, clustering, regression, and dimension reduction problems.

6.9 Python Web Application Framework - Django

Django is an open-source web application framework for developing web applications in Python. A web application framework in general is a collection of solutions, packages, and best practices that allows the development of web applications and dynamic websites. Django is based

on the Model-Template-View architecture and provides a separation of the data model from the business rules and the user interface. Django provides a unified API to a database backend. Thus web applications built with Django can work with different databases without any code changes. With this flexibility in web application design combined with the powerful capabilities of the Python language and the Python ecosystem, Django is best suited for cloud applications. Django consists of an object-relational mapper, a web templating system, and a regular-expression-based URL dispatcher.

Django Architecture

Django is Model-Template-View (MTV) framework.

Model: The model acts as a definition of some stored data and handles the interactions with the database. The data can be stored in a relational database, non-relational database, an XML file, etc. A Django model is a Python class that outlines the variables and methods for a particular data type.

Template: In a typical Django web application, the template is simply an HTML page with a few extra placeholders. Django's template language can be used to create various forms of text files (XML, email, CSS, Javascript, CSV, etc.)

View: The view ties the model to the template. The view is where you write the code that generates the web pages. The view determines what data is to be displayed, retrieves the data from the database, and passes the data to the template.

CHAPTER 7

INDUSTRIAL INTERNET OF THINGS (IIOT)

7.1 Introduction

The Industrial Internet of Things (IIoT), sometimes known as Industrial IoT, is a critical component of Industry 4.0. The Industrial Internet of Things (IIoT) combines the capability of smart devices with real-time analysis to make better use of the data that industrial equipment has been producing for years. Smart machines are the main driver of IIoT for two reasons. The first is that smart machines, unlike humans, can acquire and analyse data in real time. The second benefit is that smart machines show their results in a clear and timely way, which makes it easier for businesses to make faster and more accurate business decisions.



FIG 7.1 IIOT

Manufacturing, logistics, oil and gas, transportation, mining, aviation, energy, and other industries all utilise IIoT. Its major goal is to improve operations, especially through process automation and maintenance. IIoT capabilities improve asset performance and allow for improved maintenance management. There are a lot of ways it helps industries become more digital. In the long run, it moves the industry toward a demand-driven service model that deepens customer relationships and creates new ways to make money.

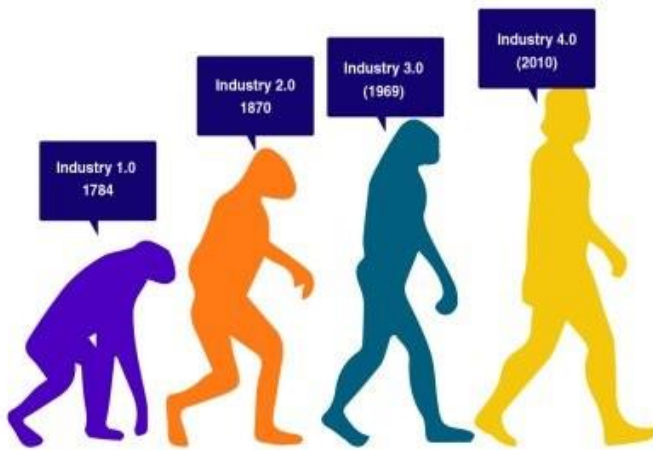


FIG 7.2 Evolution of IIoT

7.2 Evolution of IoT

1. Industry 1.0 (1784): The invention of steam engines ushered in the first industrial industry. The manufacturing, on the other hand, was completely labor-intensive and exhausting.
2. In Industry 2.0 (1870), assembly line production is introduced. For the workers, this invention was a huge relief since their labor was reduced to the bare minimum. Conveyor belts were invented by Henry Ford, the man who invented both mass production and the assembly line. They were used in a vehicle manufacturing plant to speed up production.
3. Industry 3.0 (1969): This was characterized by advances in electronic technology and industrial robots. Industrial robots are being used to simplify, automate, and increase production. Miniaturization of circuit boards with programmable logic controllers. The operations, on the other hand, remained separate from the rest of the company.
4. Industry 4.0 (Industry 4.0): Industry 4.0 realized the vision of a connected organization by connecting industrial assets via the internet. The smart devices connect with one another and provide useful information. Asset optimization, production integration, smart monitoring, remote diagnosis, intelligent decision making, and, most significantly, predictive maintenance are all advantages of IIoT.

7.3 Advantages of Industrial IoT

The Industrial Internet of Things has a lot of benefits, some of which are shown below:

1. Predictive as well as proactive maintenance
2. Real-Time Monitoring
3. Resource and Asset Allocation
4. Diagnosis from a distance

7.4 Drivers of IoT

- ❖ A primary driver for the Industrial Internet is smart sensor technology, robotics and automation, augmented and virtual reality, big data analytics, cloud integration, software applications, mobile, low-power hardware devices, and IPv6-3.4X 10³⁸ IP address scalability, among other things.
- ❖ Customer Behavior: The advantage that IIoT gives enterprises over their competitors enables them to provide value to their customers, resulting in higher customer satisfaction and retention.
- ❖ Macroeconomic Drivers: Government policies such as Industry 4.0, Smart Factories, Make in India, Make in China 2025, and Smart Cities, Japan's Industrial Value Chain Initiative Fom, Support for Green initiatives, Rising Energy and Crude Oil Prices, Favorable FDI Policies, and Policies by Regulatory Bodies, among others, all work in favour of the IIoT evolution.

Industrial Internet of Things: The Magic wand for the Economy

The never-ending development of new technologies has led to a long list of "Next Big Things." Not every emerging technology, however, has the potential to disrupt the business and social environment. The McKinsey Global Institute has highlighted a number of key technologies that have the potential to revolutionise the economy. IIoT is one of the items on the list. According to McKinsey, the Internet of Things will have a potential economic impact of \$6.2 trillion by 2025. In addition, it has the potential to boost productivity and save money in a \$36 trillion market that includes manufacturing, health care, and mining, as well as other things.

What are the security concerns and challenges associated with implementing the IIoT?

Adoption of the IIoT has the potential to transform how industries run. However, it's hard to speed up digital transformation and keep people safe in the face of more and more connectivity.

Industries and enterprises that deal with operational technologies should be well-versed in issues like worker safety and product quality. However, as OT becomes increasingly linked with the internet, organisations are witnessing the introduction of more intelligent and automated equipment at work, posing a bevy of new challenges that will require a thorough grasp of the IIoT's inner workings.

Three aspects of IIoT deployment must be prioritised: availability, scalability, and security. Because industrial operations may have been founded or in business for a long time, availability and scalability may be second nature to them. When it comes to integrating the IIoT into their operations, however, many people run into problems with security. For one reason, many businesses continue to rely on outdated systems and processes. Many of them have been around for a long time and haven't changed, which makes it more difficult to use new technologies.

Furthermore, the widespread use of smart devices has resulted in security flaws and concerns about security responsibility. IIoT adopters are responsible for securing the configuration and usage of their connected devices by default. Device manufacturers, on the other hand, have a responsibility to safeguard their consumers when their products are released. Users should have their safety taken care of by the manufacturers, who should be able to offer preventative measures or remedies if there are security flaws.

As more severe security events emerge over time, the necessity of cybersecurity becomes increasingly apparent. When hackers get access to linked systems, they risk not only exposing the business to a huge data breach, but also shutting down operations. In order to keep physical and digital parts safe, businesses and industries that use the IIoT must plan and work like technology companies at least some of the time.

Adopters are also faced with the difficulty of correctly integrating industrial operations with IT, which necessitates the security of both the connection and the information. Users' data should be treated in accordance with existing privacy legislation, such as the General Data Protection Regulation of the European Union (EU) (GDPR). While acquired data is useful for creating insights for devices and infrastructures, personal information must be separated from log data in general. Information such as personally identifiable information (PII) should be kept in a secure database. Businesses may be at risk if they store unencrypted data and do other things in the cloud.

Technology fragmentation is one of the primary problems surrounding the IoT. As a result, the IIoT is not immune to the cohabitation of many standards, protocols, and architectures. The use of different standards and protocols in IIoT systems, such as Message Queuing Telemetry Transport (MQTT) and Constrained Application Protocol (CoAP), for example, may make it difficult for IIoT systems to work together.

7.5 Risks Associated With Iot Systems

Many of the IIoT's security problems originate from a lack of fundamental security procedures. The rise of risks is aided by security flaws such as open ports, insufficient authentication processes, and old applications. When you combine these factors with the network's being directly linked to the internet, you've got yourself a recipe for disaster.

Businesses may have gotten used to the potential financial systems of IT impact on businesses due to cybercrime or malware infestation. The integration of IT and OT, on the other hand, adds a new major risk factor: real-world threats that might strike even civilians.

Unsecure IIoT systems may result in operational interruption and financial loss, among other things. There are additional security risks in increasingly linked settings, such as:

- ❖ Software flaws that could be exploited to attack systems
- ❖ Publicly searchable Internet-connected devices and systems
- ❖ Malicious behavior, such as data breaches, targeted attacks, and hacking
- ❖ System manipulation can disrupt operations (e.g., product recalls) or sabotage processes (e.g., production line shutdown).
- ❖ A system failure may cause damage to devices and physical facilities, as well as hurt operators and people in the area.
- ❖ IOT systems used for extortion, as a result of the IT environment's penetration.

The December 2015 hack on a power grid in Ukraine is a well-known example of an OT system being infiltrated by the IT environment. The attacker was able to get into the IT system, which caused important systems to go down and thousands of families to lose electricity.

TABLE 1. *Summary of the results of the 1990-1991 survey of the distribution of the 10 most common species of fish in the lower reaches of the River Severn*

• • • • • (300) • • • • •

respond quickly enough to mitigate the consequences of attacks should be a key priority for industries and businesses looking to succeed in the age of IT/OT convergence.

Industries and enterprises would be able to conduct their operations more securely if a whole stack of protection were purposefully incorporated into the various tiers of IIoT installations. The device, the network, and the cloud are all security levels.

IIoT devices and applications from various manufacturers and service providers generally make up the device layer. Adopters of the Industrial Internet of Things should be able to understand how their manufacturers and service providers transmit and retain data. In the case of a security breach, manufacturers and service providers should be able to actively alert businesses about what has to be done.

The gateway, which collects data from devices, is located in the network area. In order to monitor and identify potential attacks, organizations need to have next-generation intrusion prevention systems (IPSs) in place. A control centre generally gives orders to various devices via the gateway. The control center is the most important place where security should be tightened up to protect against malware infections or hacker control.

Finally, providers should have security solutions that operate server-based protection to mitigate the risk of hackers exploiting servers and storing data in the cloud. This reaffirms the concern that organisations might face data protection repercussions.

So, securing IIoT systems requires linked threat defense and full security, from the gateway to the end point, that can do things like:

- ❖ Ongoing detection and monitoring of malware infections
- ❖ A better understanding of threats, as well as the early detection of anomalies
- ❖ Threats and attacks against IT and OT are prevented before they happen.
- ❖ Encrypted data transmission
- ❖ A next-generation intrusion prevention system (IPS) to guard against attacks that take advantage of security flaws.
- ❖ Server and application protection in the data center and the cloud.

7.7 Applications of IoT

The Industrial Internet of Things (IIoT) is a game-changer for any manufacturing industry that creates physical products or handles product

transportation. IIoT may improve operational efficiency, paving the way for new business models to emerge. It has a wide range of applications in a variety of industries.

Production

At the moment, the most IIoT technology is used in the production sector. Smart machines that are connected to the internet of things (IIoT) may self-monitor and predict potential production issues. As a result, there is less downtime and more efficiency.

Supply Chain Management

While maintaining production levels is critical, ensuring seamless delivery across the supply chain is equally crucial. Orders may be automatically replenished as required using IIoT. This cuts down on waste, keeps stock levels in check, and guarantees that the proper quantity of raw materials is always on hand. With the automation of supply chains and ordering, employees may be able to work on more complex tasks.

Building Management

Most concerns with building management can be solved using IIoT technology. Sensor-driven climate control eliminates any uncertainty in maintaining a building's interior environment by taking into account all relevant factors such as the number of people, ventilation spots, equipment, and so on. The IIoT makes buildings more secure by having smart devices that look for threats at every entrance point. For a long time, healthcare has been embracing smart devices. Healthcare experts may keep an eye on patients from afar and be notified if their condition changes. This improves the precision and personalization of healthcare. Artificial intelligence might allow physicians to make more accurate and effective diagnoses, allowing them to treat patients more successfully.

Retail

In retail, IIoT technology allows for swift marketing choices tailored to each location. Companies may update storefronts based on consumer preferences in certain regions, and they can better target audiences with targeted marketing. These data-driven insights make a shop different from its competitors.

Sensors are not a new technology; companies have been using them for years to track cargo and monitor machinery. This is because the IIoT makes

it easier for more people to use sensors, complete wiring networks, and big-data analytics at the same time.

According to IDC, manufacturers spend \$197 billion on IIoT each year. It has received a total investment of \$71 billion from transportation companies alone. Companies are allocating funds for IIoT, but how they spend these funds differs from company to company, depending on their goals.

7.8 Work Flow of IIoT

The Internet of Things (IoT) is a network of smart devices connected to databases through networks. Data is monitored, collected, exchanged, and analyzed using these devices. A typical IIoT system includes the following components:

- ❖ Intelligent information gathering, equipment, and communication devices
- ❖ Internet networks, whether public or private, that serve as a data transmission system.
- ❖ Analytical applications that convert raw data into data insights to improve process optimization.
- ❖ Tools that help decision-makers and employees better understand data in order to improve business results.

Dataflow is critical for IIoT applications to function properly. To facilitate dataflows, companies employ a databus to disseminate and handle real-time data. This technology allows applications and devices to work together as a single entity. A databus maintains data in motion while a database keeps historical data at rest.

The Internet of Things (IIoT) improves business efficiency by streamlining and automating processes. It promotes revenue production by improving operating efficiency and lowering operational expenses. Product quality is improved by higher automation levels, which, when combined with efficient operations, aid in predictive maintenance.

The prospects of generating new income streams are substantially higher with IIoT. Data insights may demonstrate how a well-run company can improve productivity by resolving previously unknown inefficiencies. Data about performance or use leads to the development of innovative products or services. Manufacturers may, for example, work out asset-sharing schemes with other manufacturers. This resource sharing saves money while optimising space and manufacturing capacities. IIoT devices may also be used to monitor and manage an HVAC system.

The IIoT aims to increase production and quality without putting a strain on resources. This is really beneficial to business growth.

7.9 Security Considerations and Challenges When Adopting IoT

Although IIoT adoption has the potential to change industries, it also has the potential to introduce new security concerns. Workers' safety and product quality are important to companies that work with operational technologies. Even though more and more operations, automation, the internet, and smart equipment are being used together, there are new problems in terms of availability and security.

Because availability and scalability are crucial to operations, most industries are well-versed in controlling them, and they can readily be incorporated into an IIoT system. Most organisations fail when it comes to security. Many businesses still use old systems and processes, and new technologies may make it more difficult for them to work together and keep their data safe.

The proliferation of smart devices, especially those used for work, has resulted in a slew of security flaws. Any linked devices must be installed and configured in a secure manner by organisations. However, gadget manufacturers must demonstrate that their devices are safe, which is not always the case.

Cyber-security concerns are on the rise. Successful hackers have the ability to break into linked systems and shut off operations. In order to deal with these security issues, manufacturing companies need to treat the IIoT like any other IT company. They should focus on the security of both physical and digital parts.

Another issue with IIoT adoption is securing the connection between industrial operations and IT. Global privacy standards must be adhered to when it comes to user data. Data collection is essential for a company to provide essential insights. Personal information, on the other hand, must be kept separate and saved in encrypted databases. Storing personal data with business data might expose you to major security concerns.

IIoT is linked to a slew of other security problems. Exposed ports, a lack of adequate authentication methods, or even the usage of old applications might all be contributing factors. All of these little problems, in addition to having an internet network, might pose a risk to companies. IIoT systems that are not secure might result in operational interruption and financial losses.

The higher the security threats, the more networked an environment is:

- ❖ Software flaws are simple prey for hackers to attack.
Internet-connected devices and systems are searchable by the general public.
- ❖ The number of hacking attempts is increasing, resulting in targeted attacks and data loss.
- ❖ System manipulation or sabotage efforts cause operations to be interrupted.
- ❖ System failure causes equipment damage or, even worse, bodily harm to personnel.
- ❖ Extortion attempts as a result of hacked operational technologies
- ❖ Increased penalties for making private information public in violation of the law.

As a result, IoT and IIoT have a lot of different types of technology, which could lead to systems that don't work if processes aren't properly or efficiently managed.

7.10 Companies Can Implement IoT Intelligently

Even with all of these concerns, IIoT is still the way forward for organizations to remain relevant and competitive. Companies can take full advantage of its benefits by using methodical research, implementation, and maintenance.

Without a doubt, one key area of focus is cyber-security. Connecting operational technology to the internet makes any business vulnerable. A lack of security can undermine all IIoT's potential, leaving a business open to tremendous risk. Security design is needed in key areas:

A Centralized Security Operations Centre

Despite these worries, IIoT is still the way to go for organisations to stay relevant and competitive. Methodical research, implementation, and maintenance may help companies maximize the benefits of the system. Without a doubt, cyber-security is a key area of concern. Any business that connects its operational technology to the internet becomes exposed. A lack of security may undermine all of IIoT's potential, exposing a business to significant danger. In key areas, security design is required:

A Security Operations Center that is Central

A security operations centre (SOC) is required for every company interested in using the IIoT. This will allow for proactive monitoring and defence against a wide range of threats. A centralised setup helps businesses respond quickly to a variety of red signals. These centres are ideal for organisations that need constant monitoring and analysis of their security posture. The key to avoiding any breach is to anticipate security problems. Companies may improve and update outdated systems using this approach.

Low system visibility and sluggish reaction times may also be addressed by a SOC. Alerts are prioritized and handled in a security operations center, which makes the best use of IT and operational technology.

Recruiting Security Experts to Mitigate Threats

The IT security environment, like any other technology-based industry, is always evolving, and organizations must adapt to emerging threats. Companies that implement IIoT must build a dedicated security team that is always working to secure operational technology environments in order to be prepared. This is a specialist area that needs the use of professionals. These IT security experts can work to predict the next possible attack and put safeguards in place to stop it, so they can keep your company safe.

Full-Stack Protection

Security must be built into all of the important parts of the IIoT.

The device: This layer comprises of IIoT devices and the accompanying applications offered by different manufacturers and service providers. IIoT adopters should know how manufacturers and service providers use, transmit, and keep data. If there is a security breach, manufacturers and service providers need to be able to quickly tell businesses about it, so they don't lose money.

The network area has a gateway via which data is collected from devices. Organizations need to have state-of-the-art intrusion prevention systems (IPSs) to screen for potential assaults. The control centre houses this gateway, which sends orders to numerous devices. To protect the control center from malware and hackers, security measures must be put in place.

Cloud security: Providers must have server-based protection against hackers who try to take control of systems and data from them.

In a nutshell, IIoT security necessitates interconnected threat defences and end-to-end protection systems. These systems should be able to:

Keep an eye out for malware infestations at all times. Identify threats and abnormalities as soon as possible.

- ❖ Shield IT and operational technologies from proactive threats and attacks.
- ❖ Ensure that data is sent in a secure manner.
- ❖ Use a sophisticated intrusion prevention system (IPS) to guard against security flaws.
- ❖ Safeguard servers and applications in the data center and cloud.

When combined with digital transformation efforts, the IIoT is recognised for its potential to allow faster and better decision-making. By safeguarding real-time, precise data, the IIoT helps businesses better understand business processes. IIoT analyses data from a variety of sensors, improving process efficiency and generating new income sources. Companies may learn more about all parts of their supply chain, which will help them work together and coordinate better.

Despite all of these benefits, the IIoT is still a relatively young technology. Several IIoT-based technologies, such as 5G networks, are too costly to be used just once. The deployment of the IIoT in its entirety will be gradual and steady. Organizations that look for a digital future with care and sense will be rewarded, on the other hand.

7.10 Industrial Internet of Things Use Cases

People often use the term "Internet of Things IIoT" when talking about Internet of Things applications that aren't related to the Consumer Internet of Things. This is despite the fact that the IIoT is often used to refer to applications that aren't related to the Consumer Internet of Things.

As a result, it is de facto employed in industries such as agriculture, linked logistics, finance, government (including smart cities), healthcare (hospitals and healthcare facilities), and cross-industry IoT use cases such as facility management. This means that there may be fewer or more IIoT use cases depending on the perspective and industries that are known in an Industrial Internet of Things environment.

If we widen IIoT beyond manufacturing and the like, below are a few common IIoT application cases and business scenarios.

- ❖ Intelligent industrial and warehouse applications
- ❖ Predictive as well as remote maintenance.
- ❖ Freight, product, and transportation monitoring

- ❖ Interconnected logistics
- ❖ Smart metering and smart grid
- ❖ Eco-friendly alternatives
- ❖ Smart city applications
- ❖ Intelligent farming and livestock monitoring.
- ❖ Industrial facility security systems
- ❖ energy consumption optimization
- ❖ Workplace HVAC (heating, ventilation, and air conditioning)
- ❖ manufacturing equipment monitoring
- ❖ Asset tracking and smart logistics are two of the most important aspects of asset management.
- ❖ Ozone, gas, and temperature monitoring in industrial settings.
- ❖ monitoring of employees' safety and health conditions.
- ❖ Asset performance management
- ❖ Remote service, field service, remote maintenance, and control use cases
- ❖ The Industrial Internet of Things market's size, growth, and economic impact

The IIoT presents a massive market potential. According to IndustryARC research (June 2016), the industrial IoT market is expected to reach \$123.89 billion by 2021.

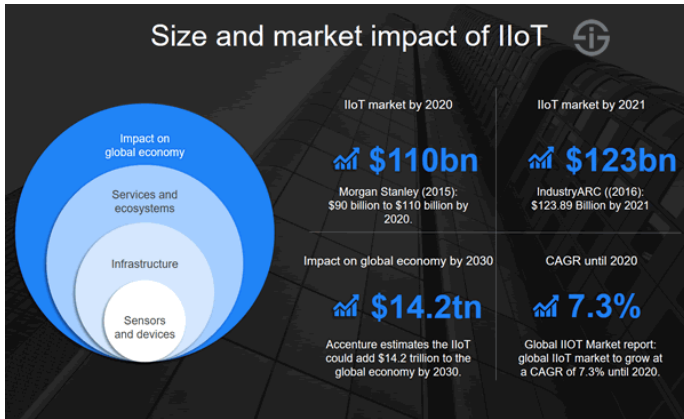
In the picture below, you can also see Morgan Stanley's estimates, Accenture's data on the impact of IIoT on the global economy, and Research and Markets' forecast. Leaders in the IIoT field, including GE, have similarly high projections. Again, it depends on how you define IIoT and what you measure.

There was a lot of information about the size and impact of the Industrial Internet of Things that Morgan Stanley, IndustryARC, Accenture, and Research and Markets gave to us.

The primary challenges in implementing the Industrial Internet of Things

Even though the Industrial Internet of Things is expected to grow, there are still problems.

Visual Capitalist's infographic at the bottom of this page, as well as Morgan Stanley's and other research, illustrates a few. An overview of the IIoT challenges as seen by executives

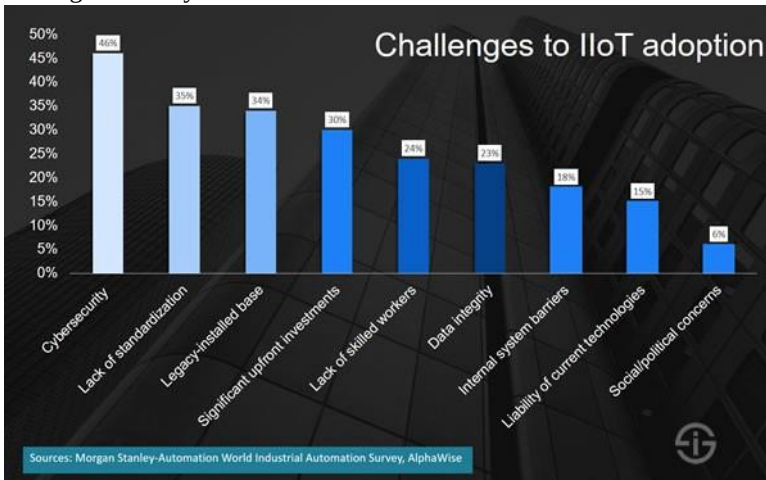


Size and market impact of the Industrial Internet of Things – source: Morgan Stanley, IndustryARC, Accenture and Research and Markets.

Industrial Internet of Things adoption barriers: the major challenges

For the reasons mentioned in the infographic, industrial data is difficult to come by (based upon IDC 2016 data). Most are also among our time's (big) data challenges.

Consider different data source types, large data quantities (especially in "heavy" industrial applications), varying data frequencies, and complicated data relationships. The solution, as in the larger picture of big data "chaos," is intelligent data systems.



Challenges and barriers to Industrial Internet of Things adoption

Data integration challenges (and data is key in IIoT).

According to the study, data integration is the most significant hurdle, with 64 percent of respondents citing it as such. In the context of the IIoT, the perennial issue of transferring data to business value becomes clear. Future income potential, on the other hand, is found in sharing ecosystems where data, insights, and expertise are shared.

Lack of skills (and access to skills)

Another main cause of organizations' lack of readiness for the Industrial Internet of Things, according to the survey, is a skills shortage.

For 36% of respondents, a lack of access to the required skills and knowledge is a problem. This lack of skills doesn't just apply to data integration. It also applies to a lot of other skills that are needed for the IIoT.

A shortage of qualified workers was also mentioned in the Morgan Stanley- Automation World Industrial Automation Survey, with 24 percent of respondents citing it. There is a general shortage of highly specialised skills, but it may be required to seek farther "outside" to get the essential skills. Let's assume one thing is clear in this era of digital transformation and the Industrial Internet of Things. No organization can flourish on its own, so networks, ecosystems, and platforms of partners are crucial.

Cybersecurity and Data Security

The main problems for executives in the Morgan Stanley and Automation

World magazine surveys were cybersecurity and data security.

When Morgan Stanley released some of its findings in April 2016, it stated that data security was becoming a greater concern for organisations that rely on universal connectivity, which is true for the Internet of Things, which in industrial applications often requires a mix of IoT connectivity solutions depending on the use case.

There are a lot of hybrid IoT connectivity solutions for industrial applications out there, from cellular IoT and low-power wide area networks (LPWAN) to industrial connectivity solutions that are both fixed and moving.

It is important to note that before the end of 2016, severe IoT security concerns and cyberattacks, even if they were not all employing the sorts of devices and connectivity one thinks of in an IIoT context, emerged as the first IIoT adoption obstacle in the Morgan Stanley research. We've already discussed these Internet of Things security concerns.

However, the Industrial Internet of Things poses unique security challenges, such as merging IT and OT, and the need to rethink security architectures.

Other obstacles to IIoT adoption

A lack of standardisation is also a concern, according to the Morgan Stanley survey, and there is more.

Survey: The top five problems with adopting the IIoT are cybersecurity, lack of standardization, lack of experience, and high upfront costs.

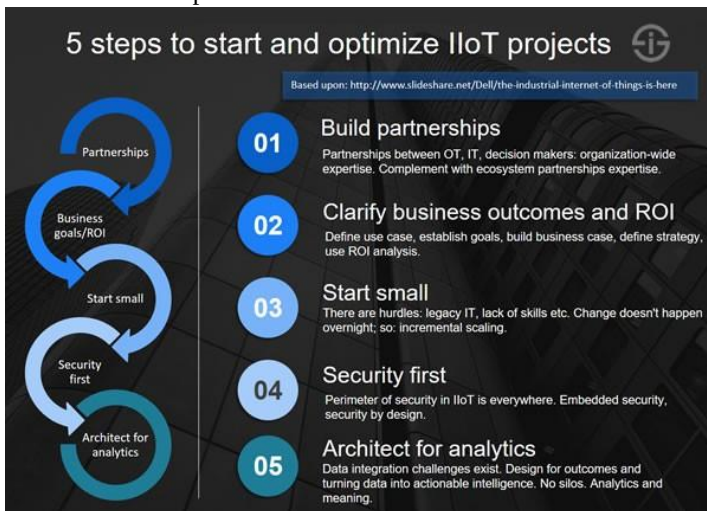
As seen in the diagram below, data integrity came in sixth place (23 percent). Several of these challenges have been noted by others and seem to be universal.

The Industrial Internet of Things (IIoT) adoption: the driving benefits

According to Morgan Stanley research, the following are the main reasons why the Industrial Internet of Things is becoming more popular:

- ❖ Improving operational efficiency.
- ❖ Improving efficiency.
- ❖ Launching new business ventures
- ❖ Minimizing downtime
- ❖ Making the most of your assets

More drivers in the report.



steps to start and optimize IIoT projects – based upon Dell IIoT presentation Based on a Dell IIoT presentation, here are five steps to get started and optimise IIoT initiatives.

Create Partnerships

As previously mentioned, partnerships between OT and IT are crucial. However, business decision-makers must be engaged as well. IDC says that for every digital transformation project, you should build partnerships and work with people who aren't obvious.

Determine the business consequences and return on investment.

This may seem self-clear, but as almost every IoT expert will tell you, the business benefits aren't always evident.

For IIoT initiatives, as for others, unclear business benefits are just fatal. An IIoT project begins with the discovery of an idea, a need, or an opportunity. However, the business case must be clear.

Begin with a Modest Project.

This is frequently the de facto approach in Internet of Things and Industrial IoT initiatives.

It's best to start with pilots and grow slowly over time. You should start small and fail, then try again and try again. You should also know how to grow quickly if you have the right ecosystem in place.

First and foremost, there is security.

Security is just as essential as business objectives and partnerships with key stakeholders and others.

Keep an eye on the IoT providers you work with and consider security from beginning to finish. There are several components involved, ranging from connectivity to linked devices and applications. Security through design, as well as inherent security, is essential. And, like with every revolutionary initiative, start with security.

Analytics Architect

Big data, and what you do with it, is everything: insight, action, and automation.

It's all about turning data into insights, actions, and automation in your Industrial Internet of Things project, and the necessity to employ analytics to do so, even if you already have data.

Real-life examples or cases of the Industrial Internet of Things

As previously mentioned, there are several IoT application cases. Some are industry-specific, while others are cross-industry.

Numerous actual Internet of Things implementation examples demonstrate the real-world benefits, ranging from streamlining manufacturing operations to predictive and remote maintenance. Case

studies are excellent for helping you see the benefits without relying on too much theory.

Manufacturing: ABB Robotics' IIoT-connected services offering

Thanks to the Industrial Internet of Things, ABB Robotics was able to increase productivity, improve customer service, and create a new linked services approach.

ABB Robotics, a branch of the automation giant ABB, creates industrial robots for its customers. Because of the Internet of Things, the company was able to go from fixing problems with robots after the fact to being able to fix them remotely and completely change the company's service offerings in this technologically dense environment.

Engineers may analyze difficulties without having to travel to the client by creating a linked ecosystem of the industrial robots that the company has deployed for them, often at difficult-to-reach customers. In an environment where uptime is critical, this results in actual benefits for customers. The Internet of Things (and the platforms that exploit the data and insights gained from it) opened new opportunities for ABB Robotics to build new services and online applications that allow monitoring, asset optimization, and more.

CHAPTER 8

8.1 IoT Applications

The IoT has a wide range of applications, but today we'll look at the top 11 IoT applications and their purposes. So, let's have a look at each one separately:

The smart home When we think about IoT systems, the most essential and efficient application that stands the test of time is Smart Home, which consistently ranks as the most popular IoT application across all channels. Every month, the number of people looking for smart homes climbs by roughly 60,000 people.

Another intriguing fact is that 256 organizations and startups are represented in the smart home database for IoT Analytics. Smart homes are currently being aggressively pursued by more firms than related IoT applications.

Smart Home firms have received more than \$2.5 billion in financing, and the amount is increasing all the time. The list of startups includes well-known startup names like AlertMe or Nest, as well as global firms like Philips, Haier, and Belkin, among others.



FIG 8.1 internet of things applications – smart home

Wearable

Wearables, like smart houses, are a popular subject among potential IOT applications. Every year, Apple's smartwatch is eagerly anticipated by consumers all around the world.

Other wearable devices that make our lives easier are the Sony Smart B Trainer, LookSee bracelet, and Myo gesture control.



FIG 8.2 internet of Things Applications – Wearable Smart City

The smart city, as the name implies, is a huge invention. It can be used for a wide range of things, like water distribution, traffic management, waste management, environmental monitoring, and city security.

It is so popular because it aims to alleviate the pain and problems that city people face. IoT solutions in the Smart City area assist in tackling a variety of city-related problems such as traffic congestion, decreasing air and noise pollution, and making cities safer.



FIG 8.3 IoT Applications – Smart City

Another application area that stands out is smart grids. A smart grid allows for the automatic collection of information on the behavior of users and electrical providers.

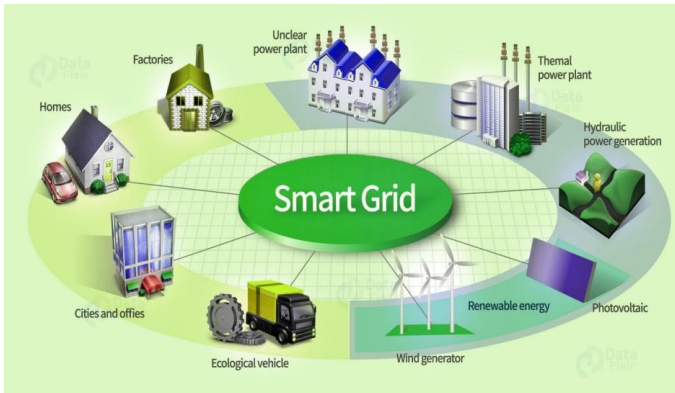


FIG 8.4 Uses Of IoT – Smart Grid Industrial Internet

Connecting machinery and devices in sectors like power production, oil and gas, and healthcare is one way to conceive of the Industrial Internet. It's also employed in circumstances when unforeseen downtime or system breakdowns might result in people's lives being in danger.

Devices such as fitness bands for heart monitoring or smart home appliances are examples of IoT-enabled systems. These systems are functional and may provide ease of use, yet they are reliable since they seldom cause crises if there is an outage.

Connected Car



FIG 8.5 IoT Applications – Connected Car

Connected automobile technology is a huge and broad network of many sensors, antennas, integrated software, and technologies that assist with communication in our complicated environment. It is in charge of making judgments that are consistent, accurate, and timely.

It must also be reliable. When humans hand over complete control of the steering wheel and brakes to the autonomous or automated cars that are now being tested on our roadways, these needs will become even more crucial.

Connected Health (Digital Health/Telehealth/Telemedicine)

From remote monitoring equipment to advanced and smart sensors to equipment integration, the Internet of Things offers a wide range of healthcare applications. It has the potential to improve how doctors provide care while also keeping patients safe and well.

Patients may be able to spend more time communicating with their physicians thanks to the Internet of Things in healthcare. It has the potential to increase patient happiness and involvement.

From personal fitness sensors to surgical robots, IoT in healthcare delivers new tools updated with the newest technology in the ecosystem that helps in producing better healthcare.

The Internet of Things (IoT) is helping to improve healthcare by giving patients and healthcare professionals cost-effective ways to get things done.



**FIG 8.6 Internet of Things Applications
– Connected Health**

Smart Retail

Retailers have begun to use IoT solutions and embedded systems in a variety of applications to improve shop operations. These include raising sales, minimizing theft, improving inventory management, and improving the shopping experience for customers.

Physical stores can compete more effectively against internet adversaries thanks to the Internet of Things. They may regain lost market share by attracting consumers inside the shop, making them spend more while saving money.

For a couple of us

Future research with integrated longitudinal data about

Figure 1. The effect of the concentration of the solution on the adsorption of the dye. The concentration of the solution was 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 15.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0, 90.0, 100.0, 150.0, 200.0, 300.0, 400.0, 500.0, 600.0, 700.0, 800.0, 900.0, 1000.0, 1500.0, 2000.0, 3000.0, 4000.0, 5000.0, 6000.0, 7000.0, 8000.0, 9000.0, 10000.0, 15000.0, 20000.0, 30000.0, 40000.0, 50000.0, 60000.0, 70000.0, 80000.0, 90000.0, 100000.0, 150000.0, 200000.0, 300000.0, 400000.0, 500000.0, 600000.0, 700000.0, 800000.0, 900000.0, 1000000.0, 1500000.0, 2000000.0, 3000000.0, 4000000.0, 5000000.0, 6000000.0, 7000000.0, 8000000.0, 9000000.0, 10000000.0, 15000000.0, 20000000.0, 30000000.0, 40000000.0, 50000000.0, 60000000.0, 70000000.0, 80000000.0, 90000000.0, 100000000.0, 150000000.0, 200000000.0, 300000000.0, 400000000.0, 500000000.0, 600000000.0, 700000000.0, 800000000.0, 900000000.0, 1000000000.0, 1500000000.0, 2000000000.0, 3000000000.0, 4000000000.0, 5000000000.0, 6000000000.0, 7000000000.0, 8000000000.0, 9000000000.0, 10000000000.0, 15000000000.0, 20000000000.0, 30000000000.0, 40000000000.0, 50000000000.0, 60000000000.0, 70000000000.0, 80000000000.0, 90000000000.0, 100000000000.0, 150000000000.0, 200000000000.0, 300000000000.0, 400000000000.0, 500000000000.0, 600000000000.0, 700000000000.0, 800000000000.0, 900000000000.0, 1000000000000.0, 1500000000000.0, 2000000000000.0, 3000000000000.0, 4000000000000.0, 5000000000000.0, 6000000000000.0, 7000000000000.0, 8000000000000.0, 9000000000000.0, 10000000000000.0, 15000000000000.0, 20000000000000.0, 30000000000000.0, 40000000000000.0, 50000000000000.0, 60000000000000.0, 70000000000000.0, 80000000000000.0, 90000000000000.0, 100000000000000.0, 150000000000000.0, 200000000000000.0, 300000000000000.0, 400000000000000.0, 500000000000000.0, 600000000000000.0, 700000000000000.0, 800000000000000.0, 900000000000000.0, 1000000000000000.0, 1500000000000000.0, 2000000000000000.0, 3000000000000000.0, 4000000000000000.0, 5000000000000000.0, 6000000000000000.0, 7000000000000000.0, 8000000000000000.0, 9000000000000000.0, 10000000000000000.0, 15000000000000000.0, 20000000000000000.0, 30000000000000000.0, 40000000000000000.0, 50000000000000000.0, 60000000000000000.0, 70000000000000000.0, 80000000000000000.0, 90000000000000000.0, 100000000000000000.0, 150000000000000000.0, 200000000000000000.0, 300000000000000000.0, 400000000000000000.0, 500000000000000000.0, 600000000000000000.0, 700000000000000000.0, 800000000000000000.0, 900000000000000000.0, 1000000000000000000.0, 1500000000000000000.0, 2000000000000000000.0, 3000000000000000000.0, 4000000000000000000.0, 5000000000000000000.0, 6000000000000000000.0, 7000000000000000000.0, 8000000000000000000.0, 9000000000000000000.0, 10000000000000000000.0, 15000000000000000000.0, 20000000000000000000.0, 30000000000000000000.0, 40000000000000000000.0, 50000000000000000000.0, 60000000000000000000.0, 70000000000000000000.0, 80000000000000000000.0, 90000000000000000000.0, 100000000000000000000.0, 150000000000000000000.0, 200000000000000000000.0, 300000000000000000000.0, 400000000000000000000.0, 500000000000000000000.0, 600000000000000000000.0, 700000000000000000000.0, 800000000000000000000.0, 900000000000000000000.0, 1000000000000000000000.0, 1500000000000000000000.0, 2000000000000000000000.0, 3000000000000000000000.0, 4000000000000000000000.0, 5000000000000000000000.0, 6000000000000000000000.0, 7000000000000000000000.0, 8000000000000000000000.0, 9000000000000000000000.0, 10000000000000000000000.0, 15000000000000000000000.0, 20000000000000000000000.0, 30000000000000000000000.0, 40000000000000000000000.0, 50000000000000000000000.0, 60000000000000000000000.0, 70000000000000000000000.0, 80000000000000000000000.0, 90000000000000000000000.0, 100000000000000000000000.0, 150000000000000000000000.0, 200000000000000000000000.0, 300000000000000000000000.0, 400000000000000000000000.0, 500000000000000000000000.0, 600000000000000000000000.0, 700000000000000000000000.0, 800000000000000000000000.0, 900000000000000000000000.0, 10000000



Consequently, the

II. THE PROPOSED APPROACH TO THE PROBLEM OF IDENTIFICATION

Smart farming could become a big business, especially in countries that export agricultural goods.

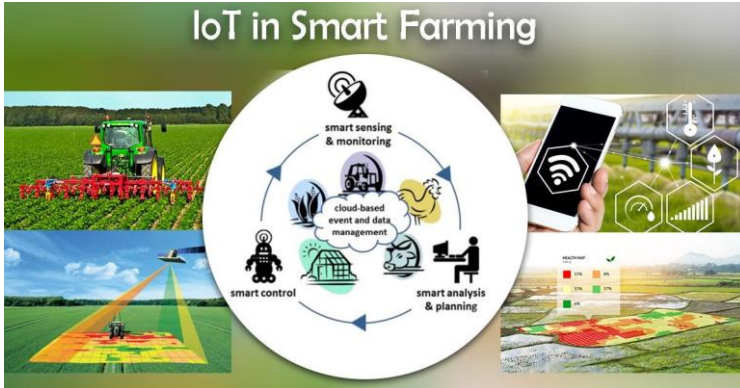


Fig 8.9 Internet of Things Applications – Smart Farming

8.2 IoT Consumer Applications

Customers gain from the IoT's information analysis and optimization.

The technology of things functions as a team of advisers, helpers, and security. Consumer Internet of Things applications range from basic and low-cost devices such as personal health monitors to high- quality smart home automation systems.

As a result, there are a lot of different IoT use cases, devices, and packages out there for people to choose from.

a. Home security and smart domestic

Domestic safety is one of the most important areas where the internet of things is becoming an increasingly important intrusion. Unauthorized entries may now be monitored and stopped before things get out of hand.

One of the smart domestic's concerns is home security. We see linked devices becoming more popular in family home equipment, for example. Other smart home packages include room management devices and recreational structures.

These applications range in price from low-cost to high-cost smart home automation solutions.

The Internet of Things (IoT) tracks the location of a whole team of workers.

- ❖ Butler IoT anticipates your arrival and ensures that your home is fully prepared. Its visual displays bring together your assets, family, and the kingdom of your own home. It takes action to resolve any problems that arise.
- ❖ Chef – An IoT kitchen may prepare meals for you or assist you in preparing them.
- ❖ In the event of an emergency, Nanny IoT can act as a mother or father by regulating access, providing supplies, and notifying the appropriate people.
- ❖ Gardner A farm's IoT infrastructure may be used for home landscaping as well.
- ❖ Repairman Smart systems both perform routine maintenance and repairs and make requests for them.
- ❖ The IoT security shield monitors you 24 hours a day, seven days a week. It may detect suspicious people from a long distance and predict the potential of young equipment problems.

b. Personal healthcare, healthcare carriers, and healthcare payers

Personal healthcare, healthcare carriers, and healthcare payers are all included in this category.

Another area where the consumer net of factors has a big impact is private health care.

IoT-powered blood pressure and heart rate bracelets connect us to the medical equipment right away, so we can get help right away if something isn't working right.

In the world of sports, monitoring the vital symptoms of athletes via training and the use of personal wearables is becoming more prevalent.

There are also other uses in the healthcare industry for things like patient monitoring, fall detection, and so on.

c. Wearable technology

Wearables are often used for personal healthcare, but they're also gaining popularity for applications other than smartwatches and health monitors.

Wearables, for example, are used to keep factory workers safe. This means they are part of the commercial web of things.

The majority of wearables, on the other hand, are consumer electronics. The market is on the lookout for smart wearables that are less reliant on cellphone technology.

d. Asset Tracking – Tracking your Valuable Assets

Consumer IoT is well-known for anything from smart smartphone tracking to GPS pet monitoring and tracking any item you choose. Puppy monitoring is becoming more popular as a way to provide pet owners with total peace of mind.

They could keep an eye on their dogs' movements. This is possible with asset monitoring solutions, which enable you to listen to anything. IoT coverage is also available across greater distances for low-energy wide-area networks.

e. Workplace

Have fun with it.

For my part, the IoT learns just as much about you as it does professionally. This enables the next generation to direct their delight.

- ❖ Entertainment and nightlifeIoT can look at what you do and how you react in the real world to help you find more things and places you like, like suggesting restaurants and events based on your preferences and past events.
- ❖ ## Holidays For some, making plans and budgeting for vacations is tough, and many firms that use IoT can be replaced.
- ❖ Goods and servicesDue to its deeper access, IoT provides a more accurate assessment of the products you desire and need than existing analytics. It uses data from important areas of your life, such as your money, to make answers.
- ❖ IIoT (Industrial Internet of Things) Applications

The Industrial Internet of Things (IIoT) is a fast-rising market that accounts for the majority of worldwide IoT investment.

Industrialists and manufacturers in practically every industry have a fantastic chance to not only monitor but also to act. However, many complicated manufacturing processes may be automated.

Sensors and systems have been used to measure development in businesses and facilities for a long time. IoT, on the other hand, goes a step farther and adds complexity to even the tiniest problems.

Let's have a look at some of the IoT's applications in manufacturing and industrial operations.

8.3 IoT Applications in Agriculture

The Internet of Things (IoT) has the potential to transform the ways we live in the world; we have more-efficient industries, more connected cars, and smarter cities, all these as components of an integrated IoT system.

The ever-growing global population would touch around 8.6 billion by 2050. So, to feed this immense population, the agriculture industry needs to embrace IoT.

The demand for more food has to meet overcoming challenges such as, rising climate change, extreme weather conditions and environmental impact that results from intensive farming practices.

Smart farming through the use of IoT technologies will help farmers to reduce generated wastes and enhance productivity. That can come from the quantity of fertilizer that has been utilized to the number of journeys the farm vehicles have made.

So, smart farming is a hi-tech system of growing clean and sustainable food for the masses. It is the induction and the application of modern ICT (Information and Communication Technologies) into agriculture.

At below, we discussed some important IoT Applications in Agriculture, let's discuss them one by one:

a. Precision Farming

Precision farming is a process or a practice that makes the farming procedure more accurate and controlled for raising livestock and growing crops. In this approach, the use of IT and items like sensors, autonomous vehicles, automated hardware, control systems, robotics, etc in this approach are key components.

Precision agriculture in the recent years has become one of the most famous applications of IoT in agricultural sector and a vast number of organizations have started using this technique around the world.

The products and services offered by IoT systems include soil moisture probes, VRI optimization, virtual optimizer PRO, etc.

VRI (Variable Rate Irrigation) optimization is a process that maximizes the profitability on irrigated crop fields with soil variability, thereby improving yields and increasing water use efficiency.

b. Agriculture Drones

Agricultural drones are a very good example of IoT applications in Agriculture. Today, agriculture industries have become one of the major industries that drones can incorporate.

Two types of drones, that is, *ground-based* and aerial-based drones are being incorporated in agriculture in many ways such as, for crop health assessment, irrigation, planting, and soil & field analysis.

The benefits that the usage of drones brings to the table include, ease of use, time-saving, crop health imaging, integrated GIS mapping, and the ability to increase yields.

The drone technology will give a high-tech makeover to the agriculture industry by using strategy and planning based on real-time data collection and processing.

The farmers through drones can enter the details of what field they want to survey. Select an altitude or ground resolution from which they want fields' data.

From the data collected by the drone, useful insights can be drawn on various factors such as plant counting and yield prediction, plant health indices, plant height measurement, canopy cover mapping, nitrogen content in wheat, drainage mapping, and so on.

The drone collects data and images that are thermal, multispectral and visual during the flight and then lands at the same location it took off initially.

c. Livestock Monitoring

IoT applications help farmers collect data regarding the location, well-being, and health of their cattle. This information helps them in identifying the condition of their livestock.

Such as, finding sick animals so that they can separate from the herd, preventing the spread of the disease to the entire cattle. The feasibility of ranchers locating their cattle with the help of IoT-based sensors helps bring down labor costs by a substantial amount.

One example of an IoT system used by a company is JMB North America. Which is an organization that provides cow monitoring solutions to cattle producers? One of the many solutions provided is to help the cattle owners observe their cows that are pregnant and about to give birth.

From them, a battery that is sensor powered is expelled when its water breaks. An information is then sent to the herd manager or the rancher. The sensor thus enables farmers will more focus.

d. Smart Greenhouses

Greenhouse farming is a technique that enhances the yield of crops, vegetables, fruits etc. Greenhouses control environmental parameters in two ways; either through manual intervention or a proportional control mechanism.

However, since manual intervention has disadvantages such as production loss, energy loss, and labor cost, these methods are less effective. A smart greenhouse through IoT embedded systems not only monitors intelligently but also controls the climate. Thereby eliminating any need for human intervention.

Different sensors that measure the environmental parameters according to the plant requirement are used for controlling the environment in a smart greenhouse. Then, a cloud server create for remotely accessing the system when it connects using IoT.

Inside the greenhouse, the cloud server helps in processing data and applies a control action. This design provides optimal and cost- effective solutions to the farmers with minimal and almost no manual intervention.

One example of this is Illuminum Greenhouses, an Agri-Tech greenhouse organization that uses technologies and IoT to provide services. It builds modern and affordable greenhouses by using IoT sensors that are solar powered.

The greenhouse state and water consumption can supervise with these sensors through sending SMS alerts to the farmer with an online portal.

The sensors in the IoT system in the greenhouse provide information on temperature, pressure, humidity, light levels.

8.4 Iot Applications in Media, Marketing, and Advertising

The IoT gadget that is in use in media, marketing, and advertising consists of a custom designed enjoyment wherein the device analyzes and responds to the desires and pursuits of each client.

This includes their famous conduct styles, shopping for conduct, choices, tradition, and

a. IoT Applications in Media & leisure

A leisure layout and manufacturing firm uses sensors in turnstiles of venues to recognize the foot traffic of people at activities.

Their IoT utility visualizes the attendee site visitors patterns in real time to help sponsors understand the great locations to market it and ensure the attendee remember stays inside the fire code compliance of the venue.

b. IoT Applications in Marketing

The conventional sample-based content intake metrics is most effective display who watches a program, when and – optimistically!

– how. Sensor facts accumulated through wearable devices can also help

content cloth companies and digital marketers determine how many human beings noticed the identical advert on a couple of structures and how many impressions precipitated conversions.

27% of agencies that leverage IoT-driven records for advertising purposes record a regular revenue increase.

c. IoT Applications in Advertising

IoT's abilities are comparably and more deeply to contemporary generation, analytics, and big facts. Gift technology collects particular information to supply associated metrics and styles over time, however, that information regularly lacks depth and accuracy.

IoT improves this through looking greater behaviors and reading them otherwise.

This leads to more facts an element, which could provide extra reliable metrics and styles. It lets businesses examine and respond to consumer goals or opportunities.

It improves business productiveness and strategy, and improves the client enjoyment through handiest turning in applicable content material and answers.

d. Advanced Advertising

Modern advertising and marketing is poorly concentrated on. No matter modern day analytics, contemporary-day advertising fails. IoT guarantees exceptional and personalized advertising and marketing in preference to one-size-fits-all strategies.

It transforms marketing from noise to a realistic part of life because of the truth purchasers engage with advertising and marketing through IoT in preference to in reality receiving it.

This makes advertising and marketing extra realistic and beneficial to humans searching the market for answers or questioning if those solutions exist.

8.5 Application of Iot in Education

One of the most innovative features of today's universities and classrooms is that the Internet of Things enhances education by bringing modern technology to the physical environment and systems. An intelligent institution offers facilities that are simple to use and allow for a greater level of personal interaction.

The wi-fi network is used by the smart devices on campus to send facts and receive orders. A computational internet of things device can be used by teachers and learning facilities to make better lesson plans, keep track of important resources, improve admission records, make campuses safer, and more.

These are some key areas in which the Internet of Things may be used in education.

a. Transforming Poster Boards into IoT-Enabled Boards

It's impossible to make a comparison between earlier presentation boards and today's multimedia poster boards. Glogster and other internet-based tools have altered this ease. It makes it easy for us to make digital posters that include photos, music, video, text, and links.

This enables us to share them electronically with others while also revealing the scholar's activities without difficulty. These virtual posters may then be sent to students and instructors, accessible through the poster's URL address, and posted on elegant blogs.

b. Knowledge Acquisition Through Interaction

Nowadays, learning is not limited to a combination of words and images, but extends much beyond that. The majority of textbooks are accompanied by web-based websites that provide additional materials, videos, examinations, animations, and other materials to aid in learning.

This provides students with a wider perspective, allowing them to analyze new things with greater knowledge and interact with instructors and their peers. Educational experts bring real-world problems into the classroom and allow students to discover answers.

C. Learning can Take Place at any Time and from any Location.

The Internet of Things (IoT) plays a vital role in the construction of a network using unique internet-based solutions. Academics may use advanced technology to show how their scholars have progressed. Edmodo is a fantastic tool for trainer-pupil communication.

Edmodo enables newcomers to have access to information from any place. Using the Internet of Things, students and teachers can connect in new ways. For example, they can keep an eye on messages and upcoming events while they're not in the classroom, or even respond to social media posts.

It's a powerful program that ensures a secure network and total privacy. It also enables you to preserve your individual ideas and class assignments without fear of losing secrecy.

Extraordinary safety features

IoT's application in education is critical, since applying advanced technology solutions in classrooms and training areas may be quite beneficial. It features emergency indications, audio augmentation, wi-fi clocks, and hearing-impaired warnings, all of which provide a sense of security to the scholars and workers.

It may also help to mitigate the destruction and save lives that can occur in the aftermath of a catastrophe. Colleges and educational institutions are using particular security measures to help calm down their campuses.

Special emergency tones, live bulletins, a couple of bell schedules, and pre-recorded instructional messages may all be used with the IoT-enabled communications system to lead a group of workers and students in an emergency.

Chalkboards Say Goodbye

In recent years, students have been using a highly sophisticated platform that incorporates smart boards. With the use of online displays and videos, it allows instructors to explain the lectures more thoroughly. Students are vocal in their support for interactive gaming as a viable medium.

Web-based tools and programs may help scholars learn more effectively than paper-based or chalkboard-based methods. Because of clever generation, instructors and students can access the internet, edit videos, and exchange homework.

Monitoring System for Attendance

A reliable faculty attendance system ensures the safety of a learning institution. It may help universities and educational institutions in a variety of ways. It enables academics to rapidly enter essential records into the device.

This might assist the agency in cutting down on the time it takes to publish attendance facts and enable school authorities to deliver a message to the parents through email.

It may also be used to keep track of how many times a student has spoken to the doctor and to conduct a test on the student's clinical desires

and the medication they will be taking. It also gives students the option of verifying their daily meal.

8.6 Government Applications

The Internet of Things (IoT) aids in the development of smart international locations and smart cities. This includes improvements to previously mentioned infrastructure (e.g., healthcare, electricity, transportation, etc.), security, and community engineering and maintenance.

Government Applications in IoT

IoT helps the improvement of clever international locations and clever cities. This includes enhancement of infrastructure previously discussed (e.g. healthcare, power, transportation, etc.), protection, and the engineering and keeping up of communities.

a. National Defense

Threats to a country's national security vary in severity and complexity. IoT enhances and supports militia systems and services, as well as provides the technology required to control the national defense landscape.

It helps to increase border security by providing more affordable, high-performance devices that are both controllable and outstanding.

IoT automates the safety obligations that are often distributed across various departments and people. This is accomplished while improving accuracy and speed.



**FIG 8.11 Government Applications In Iot –
Future Scope of Internet of Things**

b. Smart Cities

Smart towns are authorities that use the time of the internet to modify physical buildings and services in ways that benefit residents and businesses while also increasing area efficiency.

It's more than just process automation; it also connects diverse structures and networks to acquire and analyse facts. These facts are then used to describe whole systems and effect change.

While the notion of smart cities has been around for a long time, it has taken on new importance as more people flock to city facilities. Sustainability is becoming more and more important.

To some degree, technology has progressed. Real-time and meaningful contact between communities, residents, and government agencies is possible.



**FIG 8.12 Government Applications In Iot –
Future Scope of Internet of Thing**

City Control and Planning Governments and engineers may utilise the Internet of Things to analyse the very complicated factors that go into city planning and control. The Internet of Things makes this easier by looking at a variety of factors such as population growth, zoning, mapping, water distribution, transportation patterns, food delivery, social offers, and land usage.

Having the ability to "live" with people in a metropolis allows it to get more valuable and accurate data than other analytics.

The IoT helps cities by incorporating key services and infrastructure, such as transportation and healthcare, into the management realm. It also helps with important things like water management, waste management, and emergency management.

d. Creating Jobs

This is one of the most Important government applications in IoT. IoT offers thorough economic evaluation. It makes previous blind spots seen and helps better monetary tracking and modeling. It analyzes the industry and the market to spot possibilities for increase and obstacles.

e. Building an Ecosystem for Water Safety

IoT era can resolve the complicated challenges surrounding water security, allowing governments to better define priorities for water supply, consumer call for, and governance.

Like other problems driven using multiple and various factors, improving results for water management will require contributions from a surrounding of companions, many of whom are not even privy to the role they play in water conservation.



**FIG 8.13 Benefits of Iot In Government Sector –
Future Scope of Internet of Things**

E. Creating a Water Safety Ecosystem

The Internet of Things era may help governments better figure out what to do about water supply, consumer demand, and governance. This could help governments figure out what to do about water supply, consumer demand, and governance.

Improving water management results, like other problems that involve many different factors, would require help from a lot of people, many of whom don't know they play a role in water conservation.



**FIG 8.14 internet of things government applications –
Future scope of iot**

8.7 IoT Energy Applications

The benefits of IoT in other sectors apply to energy usage as well. The Internet of Things (IoT) can be used to monitor and control a wide range of things that use energy, like businesses and homes, as well as devices and the energy supply.

IoT optimization is based on a deep study that most organisations and people do not have access to.

a. Dependability

The IoT system ensures system dependability by providing actionable data and analytics. In addition to efficient utilisation, the IoT prevents a system from being throttled or overwhelmed.

By paying attention to system performance and stability issues, the system is protected from things like broken equipment, down time, and injuries.

B. Industrial Power

Small firms' high energy organizations, which result in increased energy waste, may have a significant influence on enterprises. Smaller organisations deliver products with lower profit margins. They struggle to keep company expenditures in check while working with limited cash and technology.

Larger organisations are in charge of monitoring a vast, complicated ecology of energy consumption with few effective and easy energy management solutions.

The IoT streamlines the process of energy monitoring and management while maintaining a low cost and high degree of precision. Every aspect of a company's usage across devices is carefully considered.

Through the IoT system's depth of analysis and control, organizations are given a powerful tool for controlling their consumption through cost savings and output optimization.

IoT systems deal with energy issues in the same way that problems in a complicated corporate network are dealt with and solutions are found.

Energy Applications of the Internet of Things: Commercial Energy

C. Domestic Energy

The growing technology of energy is a result of technological advancements. Consumers are always looking for strategies to control or minimise their consumption. IoT has a wide range of ways to look at and improve use across the whole home system as well as at the level of each individual device.

This could be as simple as dimming the lights, turning them off, or changing different home settings or gadget settings to use less energy.

IoT also has the capability of detecting problematic consumption patterns that are caused by problems like defective appliances, ageing appliances, or faulty system components. Environmental Monitoring using the Internet of Things We'll go through four key IoT environmental monitoring applications that are good for the environment.

8.7 IoT Environmental Monitoring

Vehicle Tracking:

Things-on-the-internet IoT Environmental Monitoring – Vehicle Tracking

The vehicle tracking facility makes use of road sensors and clever display technologies to assist vehicles in locating the best parking spots in the city.

There are several advantages to using this service. The less time it takes a vehicle to find a parking spot, the less CO₂ it emits, the fewer traffic issues it causes, and the happier its residents are. The vehicle parking facility may be immediately integrated into the IoT infrastructure.

Furthermore, as previously discussed, we can understand the electronic confirmation system of parking and locate slots reserved for residents or disabled people by using communication technologies such as Near Field Communication (NFC) or Radio Frequency Identifiers (RFID), thus

providing a better service to residents who can use those slots and as an efficient tool to spot any violations quickly.

Things-on-the-internet IoT Environmental Monitoring – Vehicle Tracking

Manual labour, complex instrumentation, and lab processing procedures are mostly used in the present monitoring technology for air and water safety. The use of IoT systems reduces the requirement for physical labour.

As a result, more frequent sampling is allowed, broadening the range of monitoring and sampling, enabling comprehensive on-site testing and reacting to detecting systems. In this way, there is less risk that more water bodies and other natural resources will become polluted and that there will be more natural disasters, too.c. Severe Weather

Deep monitoring is possible because of today's powerful, modern weather forecasting technologies. They do, however, suffer from the use of widespread equipment like radar and satellites. These small- scale gadgets lack the precision targeting capabilities of smart technology.

Benefits of IoT Environmental Monitoring-Extreme Weather

Now, thanks to recent IoT advancements, the IoT system can get more detailed data, be more flexible, and be more accurate now, too.

Weather forecasting processes that are effective need a high level of detail and flexibility in terms of equipment kind, range, and deployment. As a result, early interventions are possible, preventing the loss of life and property.

8.8 Transportation and the Internet of Things (IoT)

At each stage of transportation, the Internet of Things allows for improved verbal conversation and information sharing. This comprises personal automobiles, commercial automobiles, railways, unmanned aerial applications, and other devices.

Let's discuss more IoT – Internet of Things in detail

a. Vehicle

Most automotive firms have begun to imagine a future for automobiles in which IoT makes vehicles "smart," providing enticing options similar to MRT, yet IoT provides few significant advantages to private automobiles.

Better control over connected infrastructure and the inherent weaknesses in vehicle transportation provide the greatest advantages. On the other hand, IoT improves private motors in non- public areas. IoT

allows for the same upgrades and customizations in a car as it does in a home.

IoT in Transportation: Car Applications IoT Transportation Applications – Car Avenue b.

Congestion control, accident reduction, and parking are the three top traffic problems. Using the Internet of Things (IoT), we can better track and measure the flow of visitors through devices in every traffic comment section.

If you have trouble parking, this helps by making the storage glide more visible. Current systems don't give you very many or no statistics.

Accidents are caused by a variety of circumstances, but visitor management has an influence on how often they result. Production sites, bad rerouting and not knowing where traffic is good or bad are all factors that lead to an accident.

Improved data exchange with the general public and between different events that have an impact on road users is one way IoT can help.

Avenue of Transportation IoT Applications

Public Transportation and Rails Modern buildings provide advanced integration and overall performance, but they deliver MRT to a bygone period and practice. This leads to better results when it comes to managing things like general performance, maintenance, and new features.

Transportation IoT Applications-Rail

Mass transportation options outside of the preferred MRT may not have the right combination to turn them from a choice to a long- term provider.

The IoT provides a low-cost, high-tech way to improve overall performance and transfer MRT characteristics to other modes of transportation, such as buses.

Increasing service and offer delivery in areas like scheduling, shipping time optimization, reliability and responding to customer requests is a result of this.

D. Industrial Transportation

The benefits of managing the transportation arm of enterprises extend to commercial enterprises and manufacturing.

Analytical tools make it easier to control and monitor things like idling and fuel use, as well as idling and travel time. This lowers and eliminates problems caused by poor fleet control.

As a result, transportation of goods works more like a single company than a group of separate services.

8.8 IoT Applications in Healthcare

A. Conduct research

Current medical research does not have access to vital real-world data. For medical examinations, it usually employs leftovers, controlled conditions, and volunteers. Through analysis, real-time field data, and testing, the Internet of Things opens doors to a sea of valuable data and information.

By deploying equipment capable of undertaking future research, IoT may deliver considerably better data than typical analytics.

As a result, IoT aids healthcare by giving more practical and trustworthy data, resulting in improved solutions and the identification of previously undisclosed concerns; research is therefore one of the most significant IoT applications in healthcare.

Healthcare Applications of the Internet of Things (IoT) – Research

B. Devices

Even if existing devices are advancing in terms of power, precision, and availability, they still fall short of the advantages and attributes provided by an IoT system. The Internet of Things has the potential to unlock current technologies and lead to improved healthcare and medical device solutions.

By developing a system rather than merely tools, the Internet of Things seeks to bridge the gap between how we deliver healthcare and the equipment. It then looks for weaknesses, trends, and missing aspects in healthcare, and makes recommendations for changes.

Benefits of IoT in Healthcare – Devices

Benefits of the Internet of Things in Healthcare – Devices Benefits of the Internet of Things in Healthcare – Devices C. Take care

The Internet of Things allows healthcare practitioners to better use their expertise and training to address issues. It enables them to make greater use of data and equipment, allowing them to take more accurate and quick actions.

IoT empowers healthcare professionals to advance their careers by allowing them to put their skills to use rather than wasting time on administrative activities.

Applications of the Internet of Things in Healthcare - Care

D. Dissemination of Medical Information

The dissemination of correct and updated information to patients remains one of the most problematic aspects of medical care, and this is a most notable innovation of IoT applications in healthcare.

Individuals' everyday life, as well as facilities and professional practise, are improved by IoT devices.

IoT solutions enable invasive care at the workplace, home, or social environment, removing healthcare from facilities such as hospitals. Individuals are empowered and enabled to take care of their health, and healthcare practitioners are able to give better care to patients.

As a result, fewer accidents caused by misunderstanding have occurred, patient satisfaction has improved, and preventative care has improved. Applications of the Internet of Things in Healthcare - Medical Information Distribution

E. Immediate Care

Limited resources and disconnections from the base facility have traditionally plagued emergency response services. IoT's sophisticated automation and analytics address this issue in the healthcare industry.

An emergency may be assessed from a long way away, even kilometres distant. Providers have access to patient profiles well in advance of their arrival. They have the ability to provide critical care to patients on schedule. Associated losses are decreased as a result, and emergency health care is improved.

The Internet of applications in Healthcare - Emergency Care Applications of the Internet of Things in Law Enforcement Regulation enforcement in an area that plays a significant role in securing people and ensuring that they are protected is one way to enhance human life. Because criminal charges are on the rise all across the world, policing is vital.

Law enforcement agencies are using technology to make that its staff are up to date on technology in order to better do their responsibilities as part of their mission to reduce crime rates.

It's no surprise that bundles of IoT in law enforcement are being sought to improve the government's present reputation.

The Internet of Things (IoT), a dynamic age that has delivered significant improvements in a variety of sectors, is now altering the realm of law enforcement. Here are several IoT applications in law enforcement that might help people and authorities boost law enforcement's cutting-edge reputation.

8.9 IoT Law Enforcement Applications

A. Single traffic sensors for reduced congestion

Smart site visitor signals are becoming fashionable in a number of countries throughout the world. They've shown to be effective in decreasing injuries and better managing site visitors. According to a research, the loss of parking places is responsible for 30% of all accidents.

This congestion not only results delays in people getting to their destinations, but it also causes fuel waste.

With the Internet of Things, traffic signs become sophisticated indicators that may assist people and criminal authorities in a variety of ways, such as warning authorities about congestion in a certain area and what preventative actions are necessary.

Parking problems are reduced thanks to sophisticated sensors. Those devices no longer need batteries that must be replaced after a few months and may operate for many years on the energy they already have. These sensors may also alert clients to any open parking spaces.

B. Law Enforcement wearables

Smart wearables are now regarded the most up-to-date innovations among humans. With the increased use of smart devices, law enforcement authorities understood how important they are in monitoring crimes and reducing their frequency.

A government may immediately communicate with its server rooms thanks to the availability of smartwatches. A pedometer, a cardiac charge sensor, and other equipment are included in the smartwatches to determine whether an officer has gotten enough sleep and maintained stress levels.

c. Surveillance Vehicles that aren't Armed

There are times when physically reaching an area is difficult because the area may pose a threat to government officials' lives.

IoT becomes effective in combating such situations and gaining more monitoring. Authorities may use drones to reach a faraway location without having to be personally there.

When mounted with different types of cameras, drones may hover over a certain area that needs to be monitored and act as an extra pair of eyes. The use of cameras makes it easier for authorities to monitor criminals and assures that their region's crime cost is reduced.

Aside from cameras, drones are equipped with a variety of sensors that aid law enforcement authorities in their monitoring efforts.

Applications of the Internet of Things in Law Enforcement Regulation enforcement agencies must intentionally seek out ways to train their people to understand and use new technology as part of their mission to make things safe and productive.

d. The Judicial System

Traditional generation and assets are used in today's courtrooms. With the exception of minimal legal responsibilities, they seldom use modern analytics or automation.

Advanced analytics, improved evidence, and streamlined procedures are brought to judicial systems by IoT, which enhance tactics, eliminate inefficient processes, control corruption, save costs, and increase pride.

This might result to a more effective and trustworthy mechanism within the criminal justice system. It adds automation comparable to that of typical authorities office solutions in traditional court products; for example, IoT may automate the formation of an LLC.

Internet of Things Security – IoT Law Enforcement Applications

IoT combined with new legislation has the potential to remove or decrease the need for attorneys in many typical criminal duties. This lowers expenses and expedites numerous processes, which might take months to complete because to the complexity of criminal procedures and documents.

e. Surveillance

It's not always easy to enforce the law. Through greater information, data sharing, and superior automation, the Internet of Things serves as a regulatory enforcement instrument that helps people expend less effort and make better judgments.

IoT technologies save costs by reducing human labour in certain regions while also reducing positive traffic offences.

IoT contributes in the development of better solutions to challenges via the use of generation in the pressure area.

A far-flung statement, documented film of infractions, and digital ticketing, for example, may be replaced by moderate in-person or women examinations of suspicious activities. It also decreases corruption by removing human control and opinion in the case of a few minor infractions.

Acronyms

HVAC	Heating Ventilation and Air Conditioning
MWSN	Mobile Wireless Sensor Network
WSN	Wireless Sensor Network
IoT	Internet of Things
IoE	Internet of Everything AWS
Amazon Web Services	
MitM	Man in the Middle (attack)
MEMSMicro	Electro Mechanical Systems
RFID	Radio Frequency Identification
FCC	Federal Communications Commision
M2M	Machine to Machine
IP	Internet Protocol
IPv6	Internet Protocol version 6
BPME	Business Process Management Everywhere
EIoT	Enterprise Internet of Things
A2DP	Advanced Audio Distribution Profile
ACL	Asynchronous Connection-Less
ASB	Active Slave Broadcast
ATT	ATtribute protocol
AVCTP	Audio/Video Control Transport Protocol
AVDTP	Audio/Video Distribution Transport Protocol
AVRCP	Audio/Video Remote Control Profile
BLE/LE	Bluetooth Low Energy / Low Engery (Both terms are used to alternately) Bluetooth SIG Bluetooth Special Interest Group
BNEP	Bluetooth Network Encapsulation Protocol BR/EDR Basic
Rate /	Enhanced Data Rate
DUN	Dial-Up Networking
GATT	Generic ATtribute profile
HCI	Host Controller Interface
HCRP	Hard Copy Replacement Profile
HFP	Hands-Free Profile
HID	Human Interface Device
HOGP	HID over GATT Profile
IPSP	Internet Protocol Support Profile L2CAPLogical Link Control and Adaptation Protocol LC Link Control (protocol)
LELL	Low Energy Link Layer
LL	Link Layer

LMP	Link Management Protocol LMP Link Manager Protocol OBEX Object exchange
OPP	Object Push Profile
PANU	Personal Area Network User
PPP	Point-to-Point Protocol PSB Parked Slave Broadcast RFCOMM Radio frequency communication RTLS Real Time Location System
SCO	Synchronous Connection-Oriented
SDP	Service Discovery Protocol
SPP	Serial Port Profile
TCS	Telephony Control Protocol Specification
UUID	Universally Unique Identifier

References

1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547
2. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014,ISBN: 9789350239759
Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On
3. Approach
4. Waltenegus Dargie,Christian Poellabauer, “Fundamentals of Wireless Sensor Networks: Theory and Practice”
5. Beginning Sensor networks with Arduino and Raspberry Pi – Charles Bell, Apress, 2013